Analog Electronics

Detailed Solutions

GATE ELECTRONICS AND COMMUNICATION ENGINEERING

(EC)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in Analog Electronics (EC), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○○○ ●○○ Easy ●●○ Medium ●●● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

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SOLUTIONS

Diode Circuits and applications

Topics

1. Diode Models with open and short circuit tests
2. Operating Point and Small Signal Analysis of a Diode
3. Zener Diode as a Voltage Regulator
4. Series and Shunt Clipper Circuits
5. Two-Level Clipper Circuits
6. Clamper Circuits and their examples
7. Peak Detector and Average Diode Current
8. Voltage Multiplier Circuits
9. Half Wave Rectifier and Centre Tapped Full Wave Rectifier
10. Full Wave Bridge Rectifier and Capacitive Filters

Unit Snapshot*

Stream: EC	**Questions:** 39	**MCQ:** 30	**MSQ:** 0	**NAT:** 9
1M: 25	**2M:** 14	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 0M(2015)–4M(2021)

**See preface for how Avg Weightage and Range are calculated.*

Q1.26.1 NAT 2026 1M Ans: 32.5 ● ○ ○

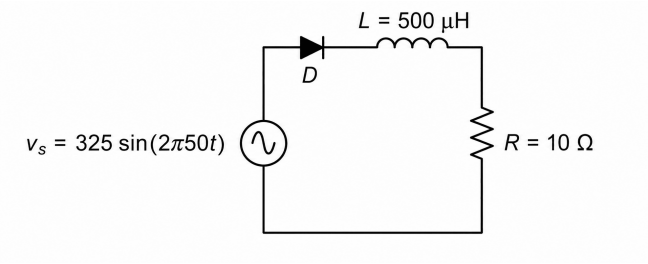


Fig. Solution Q1.26.1

During the positive half-cycle, the ideal diode conducts and the circuit behaves as a series $R-L$ circuit.

$$L\frac{di}{dt} + Ri = V_m \sin(\omega t)$$

where

$$V_m = 325\,\text{V}, \qquad \omega = 2\pi(50) = 100\pi\,\text{rad/s},$$
$$R = 10\,\Omega, \qquad L = 500\,\mu\text{H} = 0.0005\,\text{H}.$$

The current during conduction is

$$i(t) = I_m \sin(\omega t - \phi) + I_m \sin\phi\, e^{-t/\tau},$$

where

$$I_m = \frac{V_m}{\sqrt{R^2 + (\omega L)^2}}, \qquad \phi = \tan^{-1}\left(\frac{\omega L}{R}\right), \qquad \tau = \frac{L}{R}.$$

Now,

$$\omega L = 100\pi \times 0.0005 = 0.157\,\Omega,$$
$$I_m = \frac{325}{\sqrt{10^2 + 0.157^2}} \approx 32.5\,\text{A}.$$

Since

$$\tau = \frac{0.0005}{10} = 50\,\mu\text{s},$$

the exponential term decays much faster than one AC cycle, so the peak current is essentially

$$I_{\text{peak}} \approx I_m \approx 32.5\,\text{A}.$$

$$\boxed{32.5}$$

Q1.26.2 NAT 2026 2M Ans: 1.09 ● ○ ○

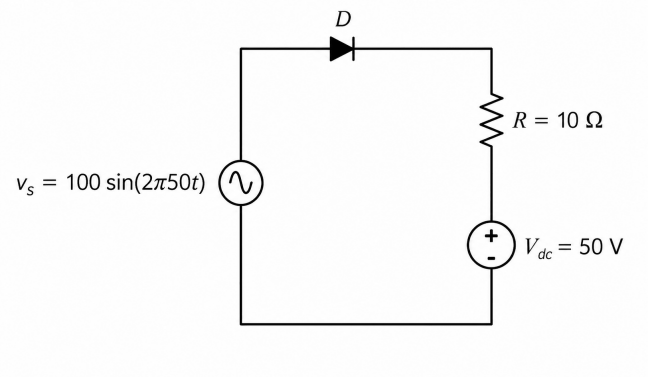


Fig. Solution Q1.26.2

The diode conducts only when the source voltage exceeds the DC source.

$$v_s = 100\sin\theta, \qquad V_{dc} = 50\,\text{V}$$

Conduction starts when

$$100\sin\theta = 50$$
$$\Rightarrow \sin\theta = \frac{1}{2}$$
$$\therefore \theta = \frac{\pi}{6} \text{ to } \frac{5\pi}{6}.$$

During conduction,

$$i(\theta) = \frac{v_s - V_{dc}}{R} = \frac{100\sin\theta - 50}{10} = 10\sin\theta - 5.$$

The average diode current over one complete cycle is

$$I_{\text{avg}} = \frac{1}{2\pi}\int_{\pi/6}^{5\pi/6} (10\sin\theta - 5)\, d\theta.$$

Evaluating,

$$I_{\text{avg}} = \frac{1}{2\pi}\left[-10\cos\theta - 5\theta\right]_{\pi/6}^{5\pi/6}$$
$$= \frac{1}{2\pi}\left(10\sqrt{3} - \frac{10\pi}{3}\right) \approx 1.09\,\text{A}.$$

1.09

Q1.24.1 NAT 2024 2M Ans: 1.67 ● ○ ○

Let's assume
$D_1, D_2 = ON$
The right node is clamped to 3 V

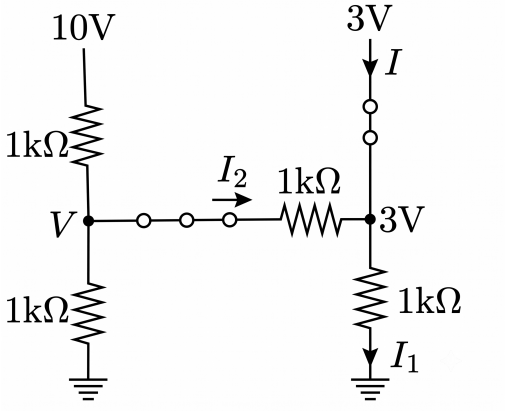


Fig. Solution Q1.24.1

Let the left junction voltage be V. Applying KCL at this node,

$$\frac{V}{1\,\text{k}\Omega} + \frac{V-10}{1\,\text{k}\Omega} + \frac{V-3}{1\,\text{k}\Omega} = 0$$
$$3V - 13 = 0$$

$$V = \frac{13}{3} = 4.33\,\text{V}.$$

Also,

$$I_1 = \frac{3}{1\,\text{k}\Omega} = 3\,\text{mA}.$$

and

$$I_2 = \frac{V-3}{1\,\text{k}\Omega} = \frac{4.33-3}{1\,\text{k}\Omega} = 1.33\,\text{mA}.$$

Applying KCL at the 3 V node,

$$I = I_1 - I_2 = 3 - 1.33 = 1.67\,\text{mA}.$$

Since $I_2 > 0$ and $I_{D1} > 0$, the assumed state is valid and both D_1 and D_2 are ON.

$$\boxed{1.67}$$

Q1.23.1 MCQ 2023 2M Ans: B,C

When S is ON, diode D_3 is reverse biased. Assume D_1 is ON and D_2 is OFF.

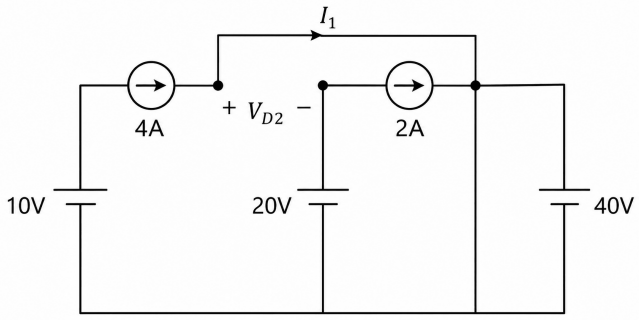


Fig. Solution Q1.23.1

The current through D_1 is

$$I_1 = 4\,\text{A}.$$

Since

$$I_1 = 4\,\text{A} > 0,$$

the assumption that D_1 is ON is valid.
The voltage across D_2 is

$$V_{D2} = -20\,\text{V}.$$

As

$$V_{D2} < 0,$$

D_2 is reverse biased, validating the assumption that D_2 is OFF.
Hence, the assumed diode states are consistent.
When S is OFF, diode D_3 becomes ON due to the 2 A current source. Assume D_1 is OFF and D_2 is ON.

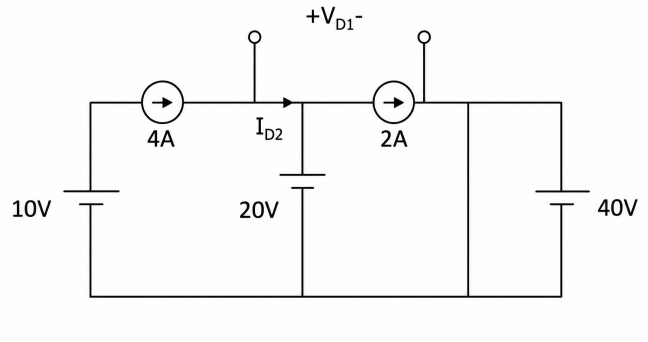


Fig. Solution Q1.23.1

The current through D_2 is

$$I_{D2} = 4\,\text{A}.$$

Since $I_{D2} = 4\,\text{A} > 0, \Rightarrow D_2$ is ON.
The voltage across D_1 is

$$V_{D1} = 20 - 40 = -20\,\text{V}.$$

Since $V_{D1} < 0, \Rightarrow D_1$ is OFF.
Hence, the assumed states are consistent in both cases.

Option B and option C are correct

Q1.22.1 MCQ 2022 1M Ans: A ● ○ ○

For positive half cycle, D_1 and D_2 will be ON

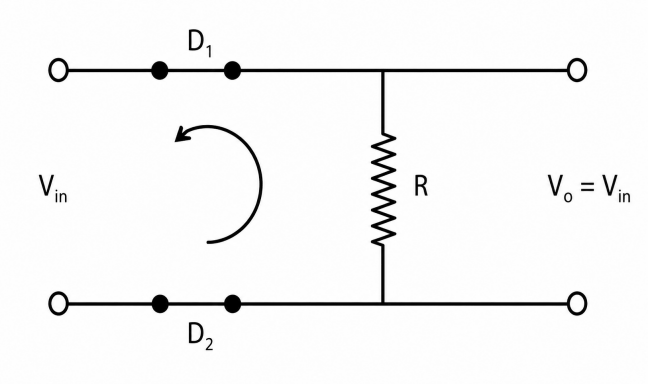


Fig. Solution Q1.22.1

For negative half cycle, D_1 and D_2 will be OFF

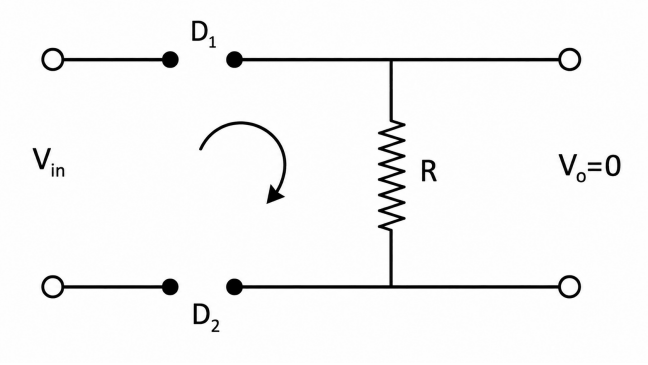


Fig. Solution Q1.22.1

Therefore,

$$\boxed{\textbf{A. } V_{\text{out}} = V_{\text{in}} \text{ for } V_{\text{in}} > 0}$$

Q1.21.1 NAT 2021 2M Ans: 1.21 ● ○ ○

Write $\theta = \omega t$ and use one period $0 \le \theta \le \pi$. The clipping level is $V_m/\sqrt{2}$, reached at $\theta = \pi/4$ and $3\pi/4$. By symmetry,

$$V_{c,\text{rms}}^2 = \frac{V_m^2}{\pi}\left(2\int_0^{\pi/4} \sin^2\theta\, d\theta + \frac{1}{2}\frac{\pi}{2}\right).$$

Since $\int_0^{\pi/4} \sin^2\theta\, d\theta = \pi/8 - 1/4$, this reduces to

$$V_{c,\text{rms}}^2 = V_m^2 \frac{\pi - 1}{2\pi}.$$

The unclipped full-wave RMS is $V_{f,\text{rms}} = V_m/\sqrt{2}$. The original GATE question asks for full-wave RMS divided by clipped RMS:

$$\frac{V_{f,\text{rms}}}{V_{c,\text{rms}}} = \sqrt{\frac{\pi}{\pi - 1}} = 1.21117\ldots.$$

Thus the requested two-decimal answer is $\boxed{1.21}$, within the official interval 1.20 to 1.23. The imported wording had reversed this ratio; clipped/full-wave would instead be approximately 0.82565, rounding to 0.83.

Q1.21.2 NAT 2021 1M Ans: 30 ● ● ○

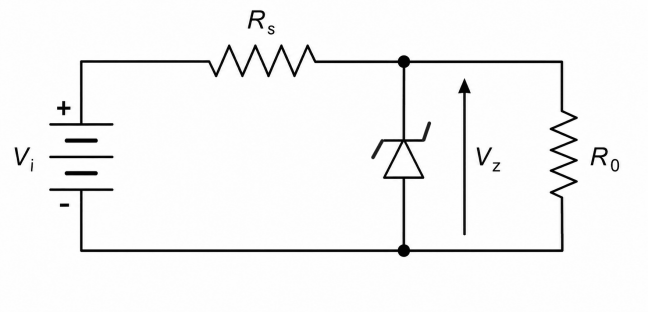


Fig. Solution Q1.21.2

For the minimum value of R_0, the load current is maximum.
Applying KCL at the Zener node,

$$I_{s_{\min}} = I_{z_{\min}} + I_{L_{\max}}$$

where

$$I_{s_{\min}} = \frac{V_{i_{\min}} - V_z}{R_s}.$$

Since the Zener diode is ideal,

$$I_{z_{\min}} = 0.$$

Hence,

$$\frac{V_{i_{\min}} - V_z}{R_s} = I_{L_{\max}} = \frac{6-5}{6} = \frac{1}{6}\,\text{A}.$$

Therefore,

$$R_{0_{\min}} = \frac{V_z}{I_{L_{\max}}} = \frac{5}{1/6} = 30\,\Omega.$$

30

Key Points

- Minimum R_0 corresponds to maximum load current.
- For an ideal Zener diode at the regulation boundary, $I_{z_{\min}} = 0$.
- Use $R_0 = V_z / I_L$ after finding the maximum load current.

Mistakes to be avoided

- Do not use $V_{i_{\max}}$ while calculating $R_{0_{\min}}$.
- The Zener power rating is not required since the limiting condition occurs at $V_{i_{\min}}$.
- Use the minimum input voltage to determine the maximum permissible load current.

Q1.21.3 NAT | 2021 | 1M | Ans: 390.32 | ● ○ ○

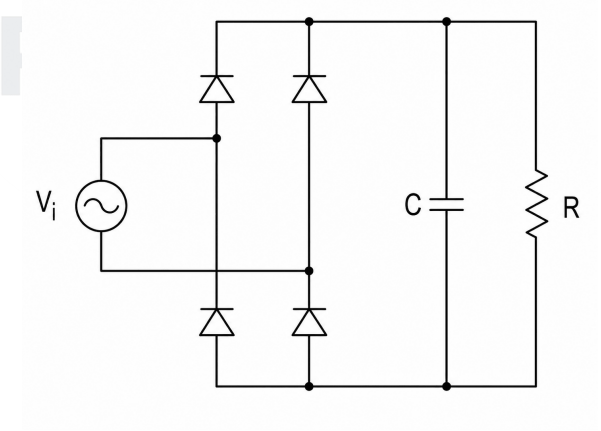


Fig. Solution Q1.21.3

The given circuit is a full-wave bridge rectifier with a capacitive filter.
Since $RC \gg T$, the capacitor charges to the peak value of the input voltage and discharges negligibly through the load. Therefore,

$$V_o = V_c = V_{i_{\text{peak(max)}}} = \sqrt{2} \times 230 \times (1 + 0.2).$$

$$V_o = \sqrt{2} \times 230 \times 1.2 = 390.323\,\text{V}.$$

For a bridge rectifier with a capacitor filter, the peak inverse voltage (PIV) across each diode is equal to the peak output voltage.

$$\therefore \text{ PIV} = 390.323\,\text{V}.$$

$$\boxed{390.32}$$

Q1.20.1 MCQ 2020 2M Ans: B ● ○ ○

Using the diode current equation,

$$I_D = I_0\left(e^{\frac{V_D}{\eta V_T}} - 1\right),$$

where

$$V_T = 26\,\text{mV} = 0.026\,\text{V}, \qquad \eta = \frac{15}{13}.$$

At $V_D = -0.03\,\text{V}$,

$$I_1 = I_0\left(e^{\frac{-0.03}{(15/13)\times 0.026}} - 1\right) = I_0\left(e^{-1} - 1\right).$$

When the current becomes $1.5I_1$,

$$1.5I_1 = I_0\left(e^{\frac{V_D}{(15/13)\times 0.026}} - 1\right).$$

Taking the ratio,

$$1.5 = \frac{e^{\frac{V_D}{0.03}} - 1}{e^{-1} - 1}.$$

Solving,

$$V_D \approx -0.088\,\text{V} \approx -0.09\,\text{V}.$$

$$\boxed{\textbf{B. } -0.09\,\text{V}}$$

Q1.20.2 MCQ 2020 2M Ans: D ● ○ ○

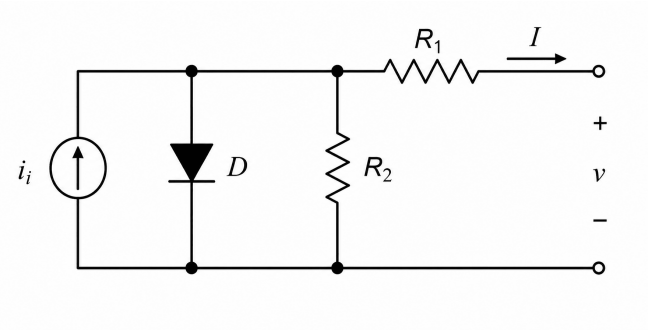


Fig. Solution Q1.20.2

To transfer maximum power to the load, the output voltage should be as high as possible while minimizing the voltage drop across R_1.

The output voltage is given by the voltage divider relation,

$$V_D = V\left(\frac{R_2}{R_1 + R_2}\right).$$

For a larger diode voltage,

$$R_2 \gg R_1 \quad \Rightarrow \quad V_D \approx V.$$

Hence,

- R_1 should be small so that the series voltage drop is minimum.
- R_2 should be large so that very little current is wasted through it, allowing maximum power to be delivered to the load.

D. Small R_1 and large R_2

Q1.17.1 MCQ 2017 1M Ans: B

During the positive half cycle, D_1 is forward biased, while D_2 and D_3 are OFF. The two resistors form a voltage divider, hence

$$v_1 = v_2 = \frac{\pi}{2}\sin(100\pi t).$$

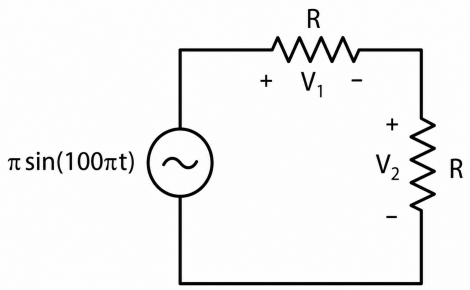


Fig. Solution Q1.17.1

During the negative half cycle, D_2 and D_3 are forward biased, which shorts the output node to ground. Therefore,

$$v_1 = \pi\sin(100\pi t), \qquad v_2 = 0.$$

The DC component is the average value over one complete cycle.

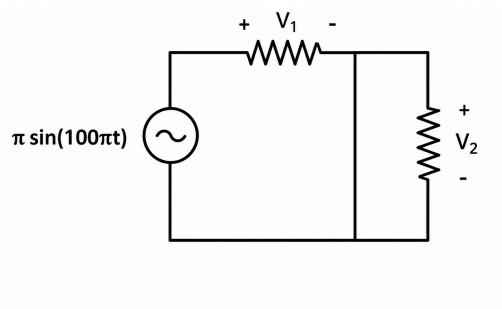


Fig. Solution Q1.17.1

For v_1,

$$V_{1(\mathrm{DC})} = \frac{1}{2\pi}\left[\int_0^{\pi}\frac{\pi}{2}\sin\theta\,d\theta + \int_{\pi}^{2\pi}\pi\sin\theta\,d\theta\right] = \frac{1}{2}\left[\frac{1}{2}\times 2 + 1\times(-2)\right] = -\frac{1}{2}\,\mathrm{V}.$$

For v_2,

$$V_{2(\text{DC})} = \frac{1}{2\pi}\left[\int_0^{\pi} \frac{\pi}{2}\sin\theta\, d\theta + \int_{\pi}^{2\pi} 0\, d\theta\right] = \frac{1}{2}\left[\frac{1}{2}\times 2\right] = \frac{1}{2}\text{ V}.$$

B. -0.5 V and 0.5 V

Q1.17.2 MCQ 2017 1M Ans: A ● ○ ○

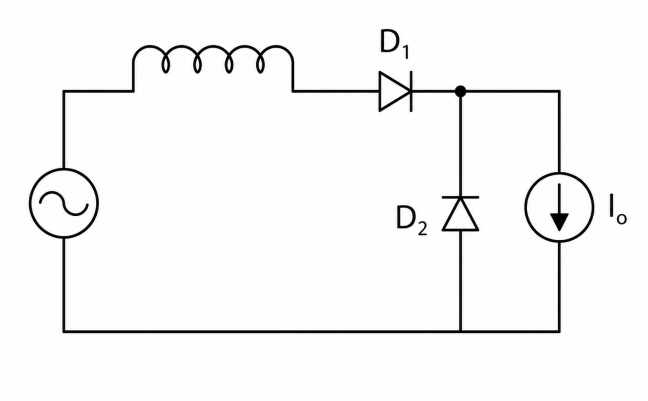


Fig. Solution Q1.17.2

Since the inductance is small but non-zero, the inductor current cannot change instantaneously. Hence, the current through the inductor is continuous.

During the positive half cycle, D_1 conducts and supplies the load current. Due to the inductance, the current continues even after the source voltage crosses zero. Therefore, D_1 conducts for slightly more than 180°.

When the source voltage becomes sufficiently negative, D_1 turns OFF and the inductor current freewheels through D_2. Thus, D_2 conducts only during the remaining interval of the negative half cycle.

Hence,

A. D_1 conducts for more than 180° and D_2 conducts for greater than 180°

Q1.15.1 MCQ 2015 1M Ans: C ● ● ○

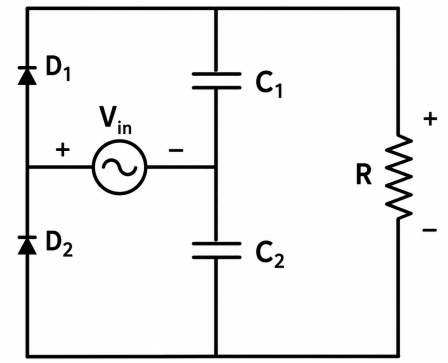


Fig. Solution Q1.15.1

During the positive half cycle, diode D_1 conducts and the upper capacitor is charged to the positive peak value of the input, $\Rightarrow V_{C_1} = V_m = 100$ V

Similarly, during the negative half cycle, diode D_2 conducts and the lower capacitor is charged to the negative peak value of the input $\Rightarrow V_{C_2} = V_m = 100$ V

Given,

$$100\pi RC = 50$$

$$\Rightarrow RC = \frac{50}{100\pi} = \frac{1}{2\pi}$$

Since,

$$T = \frac{1}{f} = \frac{1}{50}\text{ s}$$

we get

$$RC = \frac{5T}{2}$$

Thus, the time constant is much larger than the time period of the input. Hence, the discharge of the capacitors through resistors is negligible and the output voltage remains approximately constant.
Therefore, the average voltage across the resistor is

$$V_R = V_{C_1} + V_{C_2} = 100 + 100 = 200\text{ V}$$

Hence, the correct answer is

C. 200 V

Q1.14.1 MCQ 2014 2M Ans: B ●●○

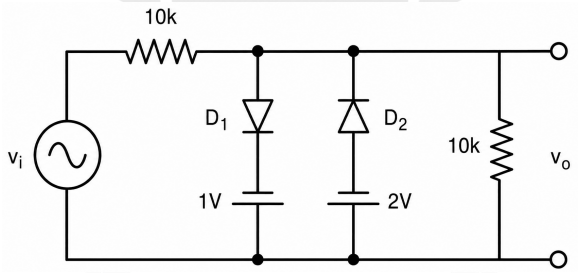


Fig. Solution Q1.14.1

Assume both diodes are OFF.
The voltage across the diode branches is obtained using voltage division,

$$V_o = \frac{10k}{10k + 10k}V_i = \frac{V_i}{2}$$

The output will be clipped if either diode conducts.
For the diode D1 connected to the 1 V source, it conducts if

$$Vo > -1 \Rightarrow \frac{V_i}{2} > -1$$

$$V_i > -2\text{ V}$$

For the diode D2 connected to the 2 V source, it conducts if

$$V_0 < -2 \Rightarrow \frac{V_i}{2} < -2$$

$$V_i < -4\text{ V}$$

Therefore, the output is unclipped for

$$-4\text{ V} < V_i < -2\text{ V}$$

Hence, the input voltage must lie outside this range for the output to be clipped.
Hence, the correct answer is

B. − 2 V to − 4 V

Q1.14.2 NAT | 2014 | 1M | Ans: 300 | ●●○

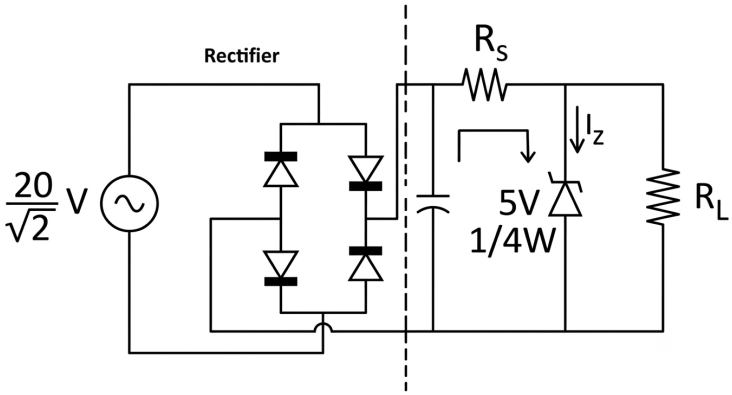


Fig. Solution Q1.14.2

The bridge rectifier converts the input AC into a full-wave rectified voltage.

Since, $V_{\text{rms}} = \dfrac{20}{\sqrt{2}}$ V

The peak value of the rectified voltage is, $V_m = 20$ V

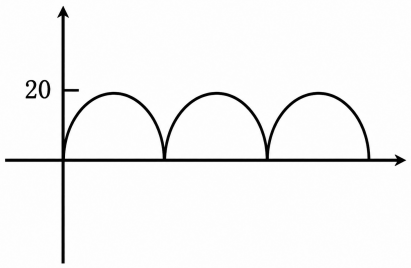


Fig. Solution Q1.14.2

To obtain the minimum value of R_s, the maximum current through the Zener diode must be considered.

This occurs when, $R_L = \infty$

So that the entire current through R_s flows through the Zener diode.

The Zener diode is rated at

$$V_Z = 5 \text{ V}, \qquad P_Z = \frac{1}{4} \text{ W}$$

Hence, the maximum permissible Zener current i

$$I_Z = \frac{P_Z}{V_Z} = \frac{1/4}{5} = \frac{1}{20} \text{ A}$$

Applying KVL,

$$20 = 5 + I_Z R_s$$

$$20 = 5 + \frac{R_s}{20}$$

$$15 = \frac{R_s}{20} \Rightarrow R_s = 300\ \Omega$$

Hence,

$$\boxed{300\ \Omega}$$

Q1.14.3 NAT | 2014 | 1M | Ans: 23 | ●●○

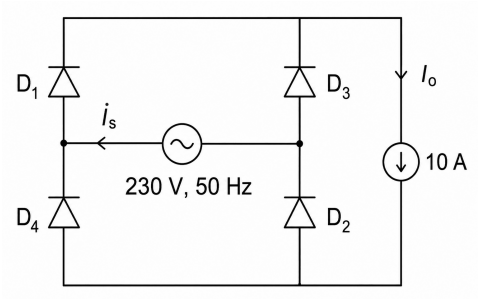


Fig. Solution Q1.14.3

The load is a constant current source of $I_0 = 10$ A.

$$\Rightarrow I_{0,\text{rms}} = I_{0,\text{avg}} = 10 \text{ A}.$$

During the positive half cycle, diodes D_1 and D_2 conduct $\Rightarrow i_s = I_0$
During the negative half cycle, diodes D_3 and D_4 conduct $\Rightarrow i_s = -I_0$

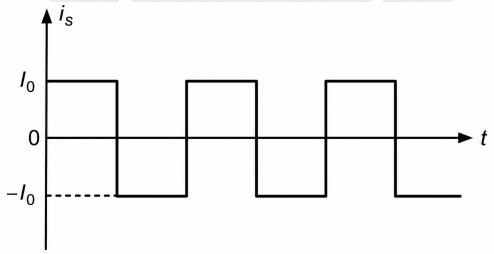


Fig. Solution Q1.14.3

$$\Rightarrow i_{s,\text{rms}} = 10 \text{ A}.$$

To obtain the same RMS current after replacing the current source with a resistor, the equivalent circuit is

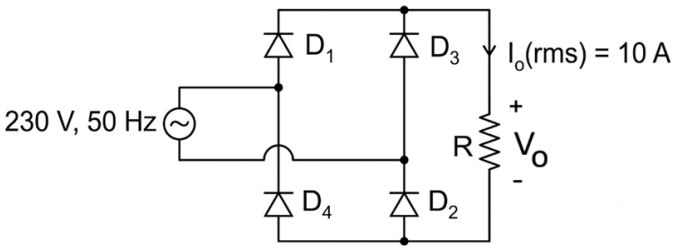


Fig. Solution Q1.14.3

The bridge rectifier produces a full-wave rectified output.
Hence, the RMS output voltage is equal to the RMS input voltage, $V_{o,\text{rms}} = 230$ V.
The RMS load current is

$$I_{o,\text{rms}} = \frac{V_{o,\text{rms}}}{R} = \frac{230}{R}.$$

Since the RMS source current must remain unchanged

$$I_{o,\text{rms}} = i_{s,\text{rms}} = 10 \text{ A}.$$

$$\therefore\ 10 = \frac{230}{R} \Rightarrow R = 23\ \Omega$$

Hence,

23

Q1.13.1 MCQ | 2013 | 2M | Ans: B | ● ● ○

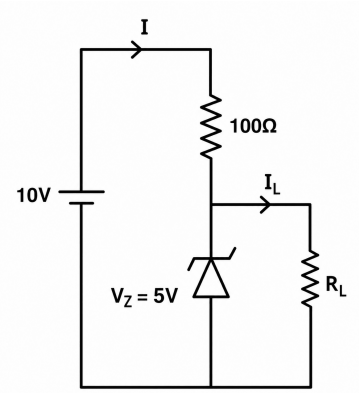


Fig. Solution Q1.13.1

The minimum Zener current required for regulation is

$$I_{Z(\min)} = I_{Z(\text{knee})} = 10\ \text{mA}$$

The current through the 100 Ω resistor is

$$I = \frac{10 - 5}{100} = 50\ \text{mA}$$

Applying KCL, $I = I_Z + I_L$
For the minimum value of R_L, the load current is maximum while the Zener current is at its knee value.
Hence,

$$I_{L(\max)} = 50 - 10 = 40\ \text{mA}$$

Therefore,

$$R_{L(\min)} = \frac{V_Z}{I_{L(\max)}} = \frac{5}{40 \times 10^{-3}} = 125\ \Omega$$

The maximum Zener current occurs when the load is open-circuited.

$$I_{Z(\max)} = 50\ \text{mA}$$

Hence, the maximum power dissipated in the Zener diode is

$$P_{Z(\max)} = V_Z I_{Z(\max)} = 5 \times 50\ \text{mA} = 250\ \text{mW}$$

Hence, the correct answer is

B. 125 *and* 250

Q1.13.2 MCQ 2013 1M Ans: D ● ○ ○

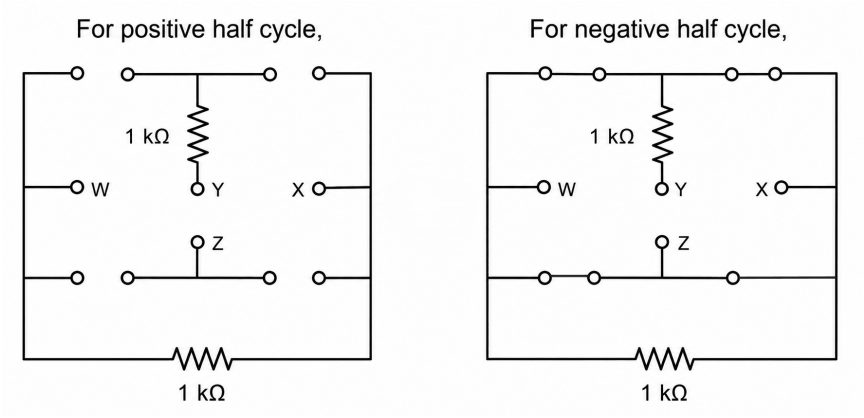


Fig. Solution Q1.13.2

Consider the two half cycles separately. Irrespective of the polarity of the input, $V_{WX} = 0$
Hence, the correct answer is

D. 0 for all t

Q1.12.1 MCQ 2012 1M Ans: D ● ● ○

Method 1

From the given i-v characteristic,

$$i = \frac{v - 0.7}{500}, \qquad v \geq 0.7 \text{ V}$$

Hence, when the diode is ON, it can be modeled as a 0.7 V voltage source in series with a 500 Ω resistance.

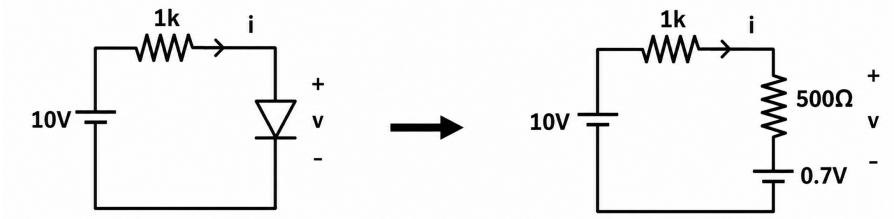


Fig. Solution Q1.12.1

Applying KVL,

$$10 = 0.7 + i(1000 + 500)$$

$$i = \frac{10 - 0.7}{1500} = \frac{9.3}{1500} = 6.2 \text{ mA}$$

Hence, the correct answer is

D. 6.2 mA

Method 2

Applying KVL to the given circuit,

$$i = \frac{10 - v}{1000}$$

From the diode characteristic,

$$i = \frac{v - 0.7}{500}$$

Equating the two expressions,

$$\frac{10 - v}{1000} = \frac{v - 0.7}{500}$$

$$10 - v = 2v - 1.4 \Rightarrow v = 3.8 \text{ V}$$

Therefore,

$$i = \frac{3.8 - 0.7}{500} = 6.2 \text{ mA}$$

Hence, the correct answer is

D. 6.2 mA

Q1.11.1 MCQ 2011 2M Ans: C ● ● ○

Consider the following cases.
Case I: Diode D is forward biased.

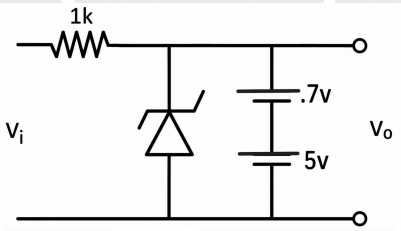


Fig. Solution Q1.11.1

The output diode conducts and clamps the output voltage to $V_o = 5 + 0.7 = 5.7$ V
The Zener diode is reverse biased but does not enter breakdown.
This condition occurs when $V_i > 5.7$ V
Case II: Diode D is reverse biased and the Zener diode is also reverse biased.

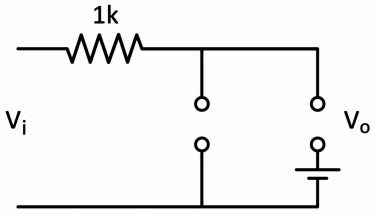


Fig. Solution Q1.11.1

Both diodes remain OFF.
Hence, $V_o = V_i$
This occurs for $-0.7 < V_i < 5.7$ V
Case III: Diode D is reverse biased and the Zener diode is forward biased.

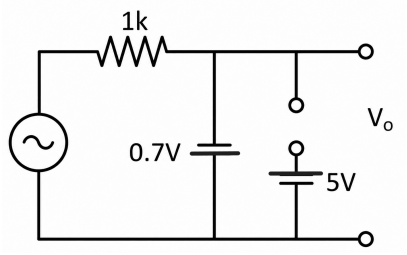


Fig. Solution Q1.11.1

The Zener diode conducts in the forward direction and clamps the output voltage to $V_o = -0.7$ V
This condition occurs when $V_i < -0.7$ V
Thus, the input-output transfer characteristic is shown below.

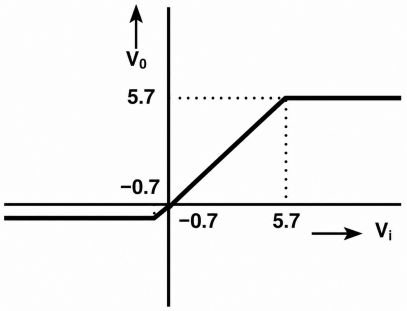


Fig. Solution Q1.11.1

Hence, the correct answer is

C

Q1.10.1 MCQ 2010 1M Ans: B ●●○

Using the open-circuit test, Assume both diodes are OFF.

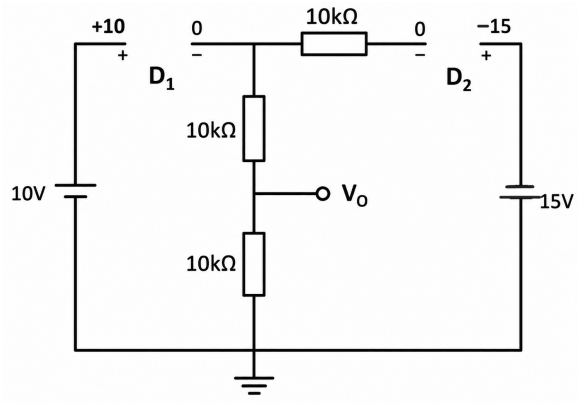


Fig. Solution Q1.10.1

For diode D_1,

$$V_A = 10 \text{ V}, \qquad V_K = 0 \text{ V}$$

Therefore, D_1 is forward biased and turns ON.
Now assume D_1 is ON and D_2 is OFF.

The equivalent circuit is shown below.

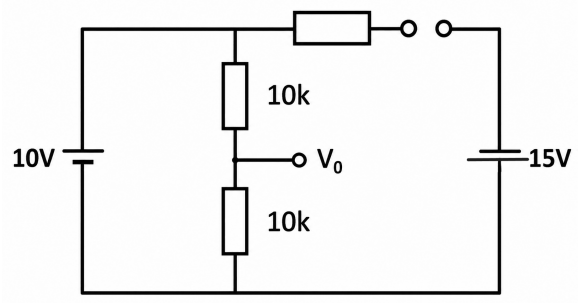


Fig. Solution Q1.10.1

The output voltage is obtained using the potential divider,

$$V_o = 10\left(\frac{10k}{10k + 10k}\right) = 5 \text{ V}$$

And our assumption is valid since the cathode voltage of D_2 is 10 V and its anode voltage is -15 V
Thus, D_2 is reverse biased.
Hence, the correct answer is

B. 5 V

Q1.09.1 MCQ 2009 1M Ans: A ● ● ○

For different intervals of time,
For $0 < t < 100$ ms,

$$V_s = 0 \Rightarrow i = 0$$

For $100 < t < 200$ ms,

$$V_s = 10 \text{ V}$$

The current through the resistor is

$$i = \frac{10}{10 \text{ k}\Omega} = 1 \text{ mA}$$

Thus, the element conducts during the positive voltage.
For $200 < t < 300$ ms,

$$V_s = -10 \text{ V}$$

From the given current waveform, a negative current flows initially and then decays to zero.
This is the reverse recovery current exhibited by a diode when it is suddenly reverse biased.
After the reverse recovery time, the diode becomes reverse biased and blocks the current.
For $300 < t < 400$ ms,

$$V_s = 0 \Rightarrow i = 0$$

Therefore, the element connected between terminals a and b is a diode with anode at a and cathode at b.
Hence, the correct answer is

A

Q1.08.1 MCQ 2008 1M Ans: D ●●○

This is a voltage doubler circuit. During the positive half cycle, diode D_1 is ON and diode D_2 is OFF.
Hence, capacitor C_1 charges to the peak value of the input voltage. $V_{C_1} = 5$ V

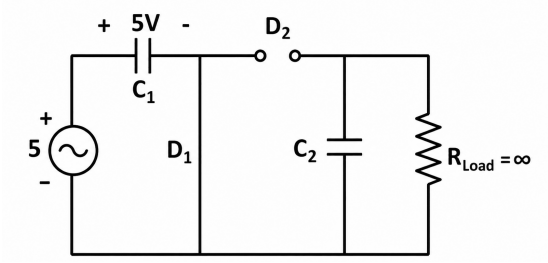


Fig. Solution Q1.08.1

During the negative half cycle, diode D_1 is OFF and diode D_2 is ON.
Capacitor C_2 is charged by the source and the previously charged capacitor C_1.
Thus,

$$V_{C_2} = 5 + 5 = 10 \text{ V}$$

With the polarity shown across C_2, $V_{C_2} = -10$ V

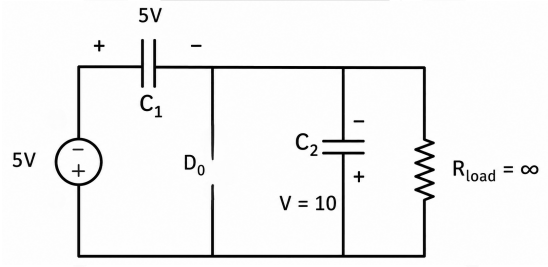


Fig. Solution Q1.08.1

Hence, the correct answer is

D. $V_{C_1} = 5$ V, $V_{C_2} = -10$ V

Key Points

- A **voltage doubler** can be understood as a combination of two basic circuits:

  Voltage Doubler = Clamper + Peak Detector

  The **clamper** shifts the input waveform by charging one capacitor, while the **peak detector** captures the shifted peak using the second capacitor. Consequently, the output voltage becomes approximately twice the input peak value (neglecting diode drops).

Q1.08.2 MCQ 2008 1M Ans: A ●○○

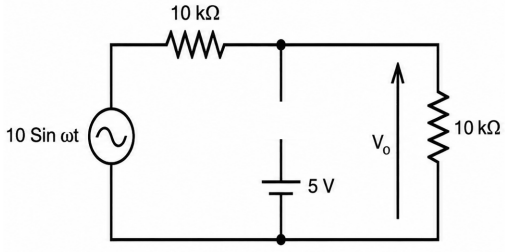


Fig. Solution Q1.08.2

Using the open-circuit test, assume the diode is OFF.
The output voltage is obtained using voltage division,

$$V_o = V_i\left(\frac{10k}{10k + 10k}\right) = \frac{V_i}{2}$$

Since, $V_i = 10\sin\omega t$
The maximum output voltage is,

$$V_{o(\max)} = \frac{10}{2} = 5\text{ V}$$

For the diode to turn ON,

$$V_o > 5 + 0.7 = 5.7\text{ V}$$

However,

$$V_{o(\max)} = 5\text{ V} < 5.7\text{ V}$$

Therefore, the diode never turns ON.
Hence, $V_o = \frac{V_i}{2} = 5\sin\omega t$
Hence, the correct answer is

A

Q1.06.1 MCQ 2006 2M Ans: A ●●○

Consider the following cases.
Case I: Assume D_1 is ON and D_2 is OFF.

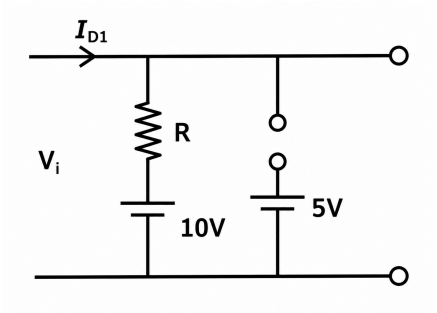


Fig. Solution Q1.06.1

The current through D_1 is

$$I_{D_1} = \frac{V_i - 10}{2}$$

For D_1 to remain ON,

$$I_{D_1} \geq 0$$

$$V_i \geq 10 \text{ V}$$

Hence, $V_o = V_i$

Case II: Assume D_1 is OFF and D_2 is OFF.

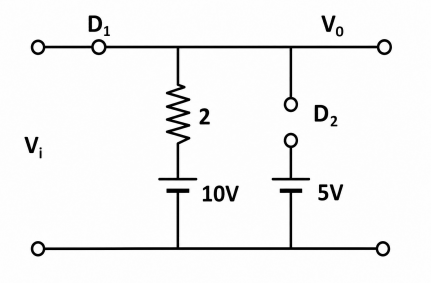


Fig. Solution Q1.06.1

For

$$V_i < 10 \text{ V}$$

$$I_{D_1} = \frac{V_i - 10}{2} < 0$$

Hence, D_1 remains OFF.

Since no current flows through the 2 Ω resistor, $V_o = 10$ V

Also, diode D_2 remains OFF because its anode voltage is less than its cathode voltage.

Therefore,

$$V_o = \begin{cases} 10 \text{ V}, & V_i < 10 \text{ V} \\ V_i, & V_i \geq 10 \text{ V} \end{cases}$$

Hence, the transfer characteristic is shown below.

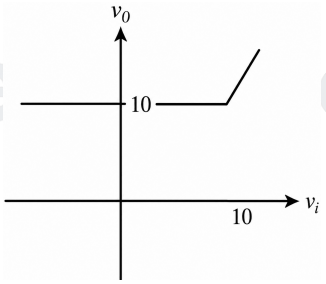


Fig. Solution Q1.06.1

Hence, the correct answer is

A

Q1.06.2 MCQ 2006 1M Ans: A ●●○

Using the short-circuit test, assume D_1, D_2 and D_3 are ON.

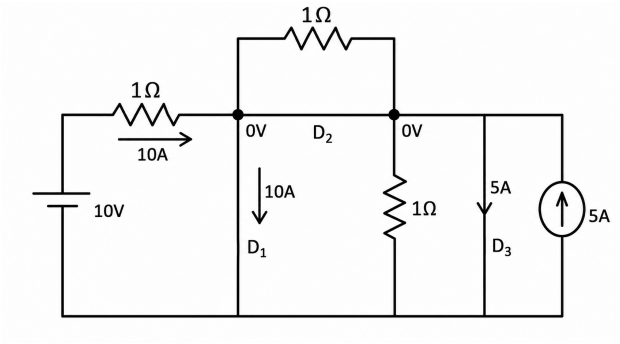


Fig. Solution Q1.06.2

It is observed that the current through D_1 flows from anode to cathode.
Hence, D_1 is ON.
However, the current through D_3 flows from cathode to anode.
Hence, D_3 is OFF.
Now, consider D_1 ON and D_3 OFF.
The modified circuit is shown below.

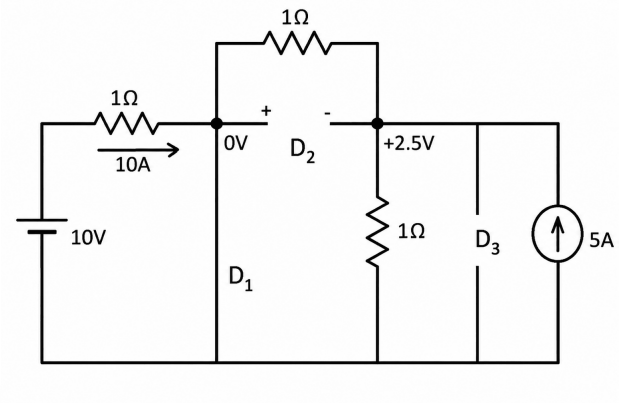


Fig. Solution Q1.06.2

The voltage at the anode of D_2 is $V_A = 0$ V
The voltage at the cathode of D_2 is $V_K = 2.5$ V
Since, $V_A < V_K$
Diode D_2 is reverse biased.
Therefore,

$$D_1 \text{ ON}, \qquad D_2 \text{ OFF}, \qquad D_3 \text{ OFF}$$

Hence, the correct answer is

A. D1 ON, D2 OFF, D3 OFF

Key Points

- In exam you can solve the question by verifying the options directly.

Mistakes to be avoided

- After determining the ON/OFF state of each diode, redraw the circuit with the assumed states and verify that every diode state is consistent with the resulting voltages and currents.

Q1.05.1 MCQ 2005 1M Ans: A ● ○ ○

Since the diodes are ideal, current can flow only from anode to cathode.
Thus, D_1 is ON and D_2 is OFF.
The equivalent circuit is shown below.

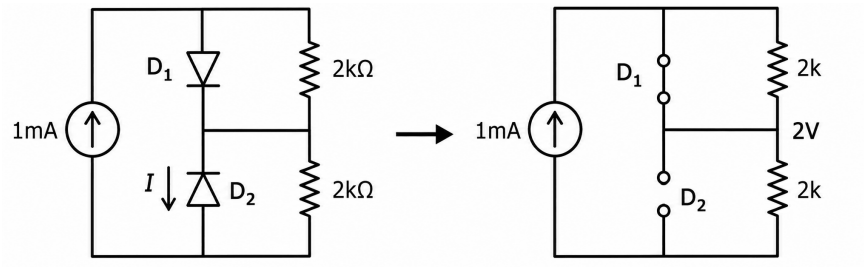


Fig. Solution Q1.05.1

Hence, no current flows through diode D_2.
Hence, the correct answer is

A. 0 mA

Q1.04.1 MCQ 2004 2M Ans: C ● ● ○

Using the short-circuit test, assume both diodes are ON.
The current directions are shown below.

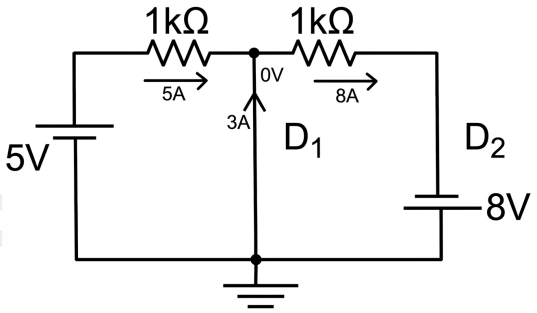


Fig. Solution Q1.04.1

For diode D_1, the current flows from cathode to anode.
Hence, D_1 is OFF.
For diode D_2, the current flows from anode to cathode.
Hence, D_2 is ON.
Therefore, no current flows through diode D_1.
Hence, the correct answer is

C. 0 mA

Q1.04.2 MCQ 2004 1M Ans: A ● ● ○

The transformer secondary is rated as 50 – 0 – 50 V
Hence, the RMS voltage across each half of the secondary winding is, $V_{rms} = 50$ V
Therefore, the peak voltage across each half winding is $V_m = 50\sqrt{2}$ V
When diode D_1 is ON, diode D_2 is OFF.

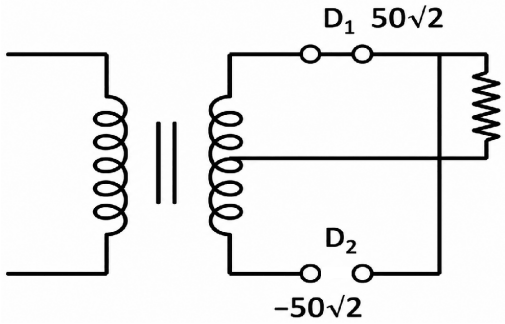


Fig. Solution Q1.04.2

The anode of D_2 is at $-50\sqrt{2}$ V
while its cathode is at $+50\sqrt{2}$ V
Hence, the peak inverse voltage across D_2 is

$$PIV = 50\sqrt{2} - (-50\sqrt{2}) = 100\sqrt{2} \text{ V}$$

Similarly,

$$PIV_{D_1} = 100\sqrt{2} \text{ V}$$

Hence, the correct answer is

A. $100\sqrt{2}$ V

Q1.04.3 MCQ 2004 1M Ans: C ● ● ○

The Zener diode is operating in the breakdown region as $V_Z < 3.5V$
Hence, it can be replaced by its equivalent model consisting of an ideal voltage source V_Z in series with the Zener resistance R_Z.

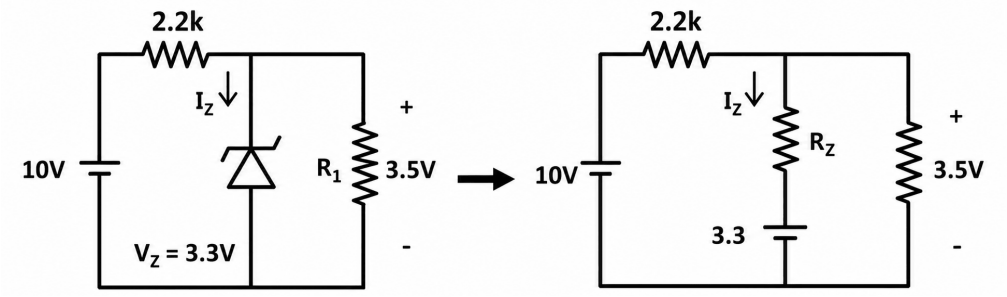


Fig. Solution Q1.04.3

From the figure, $V_o = 3.5$ V
Applying KVL,

$$3.5 = 3.3 + I_Z R_Z$$

$$\Rightarrow I_Z = \frac{3.5 - 3.3}{100} = 2 \times 10^{-3} \text{ A} = 2 \text{ mA}$$

Hence, the correct answer is

C. 2 mA

Q1.03.1 MCQ | 2003 | 2M | Ans: D | ● ● ○

Consider the following cases.

Case I: Positive half cycle

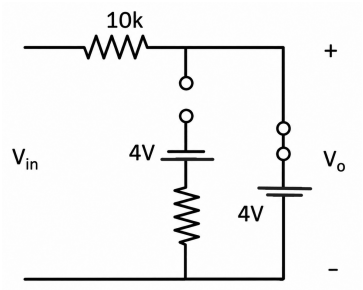


Fig. Solution Q1.03.1

Diode D_2 becomes forward biased when $V_i \geq 4$ V $\Rightarrow$ $V_o = 4$ V

For $0 \leq V_i \leq 4$ V, both diodes remain OFF $\Rightarrow$ $V_o = V_i$

Case II: Negative half cycle

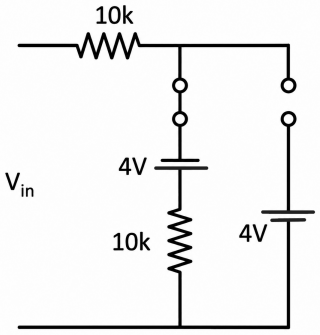


Fig. Solution Q1.03.1

For D_1 to conduct, $V_i \leq -4$ V

In this case,

$$V_o = V_i - I_{D_1}(10k)$$

$$V_o = V_i - \frac{V_i + 4}{20k} \times 10k = \frac{V_i}{2} - 2$$

At the negative peak,

$$V_i = -10 \text{ V}$$

$$V_o = \frac{-10}{2} - 2 = -7 \text{ V}$$

For $-4 \leq V_i \leq 0$ V, both diodes remain OFF $\Rightarrow$ $V_o = V_i$

The output characterstics is as;

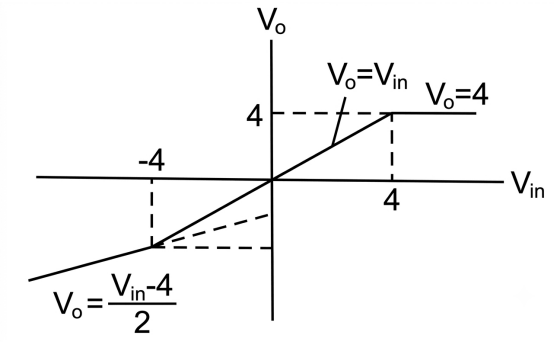


Fig. Solution Q1.03.1

Hence, the correct answer is

D. $+4$ V and -7 V

Q1.02.1 MCQ 2002 1M Ans: B

Determine the Thevenin equivalent across the diode and Zener diode combination.
The Thevenin voltage is

$$V_{th} = V_i\left(\frac{1k}{1k + 1k}\right) = \frac{V_i}{2} = 5\sin\omega t$$

The Thevenin resistance is, $R_{th} = 1k \parallel 1k = 500\ \Omega$
The equivalent circuit is shown below.

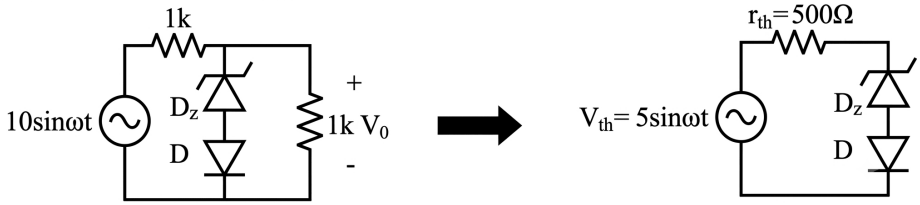


Fig. Solution Q1.02.1

For the positive half cycle, $V_{th} > 0$

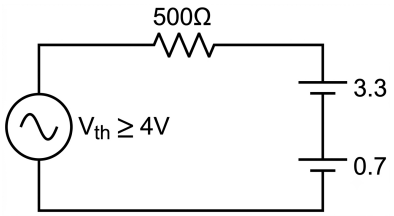


Fig. Solution Q1.02.1

The Zener is reverse biased The Zener diode conducts only after entering breakdown. When Zener enters breakdown, then the diode D is forward biased with a voltage drop of 0.7 V.
This happens when; $V_{th} \geq 3.3 + 0.7 = 4$ V
After conduction begins, $V_o = 4$ V
For the negative half cycle, $V_{th} < 0$
The Zener diode becomes forward biased and the diode D enters reverse breakdown only when

$$|V_{th}| \geq 50 + 0.7 = 50.7\ \text{V}$$

However, $V_{th(\min)} = -5$ V,
Which is insufficient to turn ON the diode combination.
Hence, the diode branch remains open and $V_o = V_{th}$
Therefore, the negative peak output voltage is $V_{o(\min)} = -5$ V
Hence, the correct answer is

$$\boxed{V_{o,\max} = 4 \text{ V}} \qquad V_{o,\min} = -5 \text{ V} \quad \text{(B)}$$

Q1.02.2 MCQ 2002 1M Ans: A ●●○

When the diode is forward biased, it is replaced by its forward resistance.
Hence, the equivalent circuit is shown below.

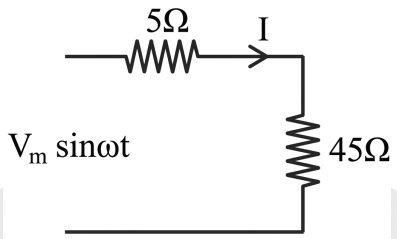


Fig. Solution Q1.02.2

During the positive half cycle,

$$i(t) = \frac{V_m \sin \omega t}{50}$$

During the negative half cycle, the diode is reverse biased $\Rightarrow i(t) = 0$
The current waveform is shown below.

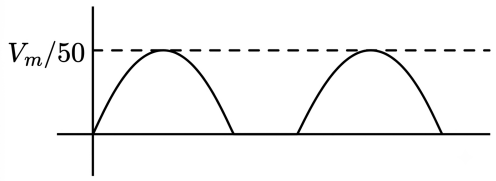


Fig. Solution Q1.02.2

The dc component of the current is

$$I_{dc} = \frac{1}{2\pi} \int_0^{\pi} \frac{V_m}{50} \sin\theta \, d\theta$$

$$= \frac{1}{2\pi} \cdot \frac{V_m}{50} \cdot 2 = \frac{V_m}{50\pi}$$

Hence, the correct answer is

$$\boxed{\textbf{A.} \ \frac{V_m}{50\pi}}$$

Q1.00.1 MCQ 2000 2M Ans: B ● ○ ○

The diode current is given by

$$i_D = I_s e^{\frac{v}{V_T}}$$

Differentiating with respect to v,

$$\frac{di_D}{dv} = I_s e^{\frac{v}{V_T}} \cdot \frac{1}{V_T}$$

$$\Rightarrow \frac{di_D}{dv} = \frac{i_D}{V_T} \qquad [\because i_D = I_s e^{\frac{v}{V_T}}]$$

The dynamic resistance is

$$r_d = \left(\frac{di_D}{dv}\right)^{-1} = \frac{V_T}{i_D}$$

Given,

$$V_T = 25 \text{ mV}, \qquad i_D = 2 \text{ mA}$$

$$r_d = \frac{25 \times 10^{-3}}{2 \times 10^{-3}} = 12.5\ \Omega$$

Hence, the correct answer is

B. 12.5 Ω

Q1.99.1 MCQ 1999 1M Ans: B ● ○ ○

The diode current is given by

$$I_D = I_S \left(e^{\frac{V_D}{\eta V_T}} - 1 \right)$$

For a constant diode current, I_D = constant
As temperature increases, $I_S \uparrow$ and $V_T \uparrow$
To maintain the same current, the exponential term must decrease.
Hence, $V_D \downarrow$
Practically, the forward voltage of a silicon diode decreases at a rate of approximately

$$-2.5 \text{ mV/°C}$$

Therefore, the voltage across the diode decreases with increase in temperature.
Hence, the correct answer is

B. Decreases

Key Points

- The reverse saturation current of a diode is approximately

$$I_S \propto T^3 e^{-E_g/(kT)}$$

  where E_g is the semiconductor bandgap.
- As temperature increases, more covalent bonds break, generating more electron-hole pairs. This increases the minority carrier concentration, causing I_S to increase exponentially.
- **Rule of Thumb:** The reverse saturation current approximately **doubles for every** 10°**C** rise in temperature.

Q1.99.2 MCQ 1999 1M Ans: A ● ○ ○

The mobility of a charge carrier is defined as the ratio of drift velocity to the applied electric field.

$$\mu = \frac{v_d}{E}$$

where,

v_d = Drift velocity, E = Electric field

The unit of drift velocity is

$$[v_d] = \frac{\text{cm}}{\text{s}}$$

The unit of electric field is

$$[E] = \frac{\text{V}}{\text{cm}}$$

Therefore,

$$[\mu] = \frac{\frac{\text{cm}}{\text{s}}}{\frac{\text{V}}{\text{cm}}} = \frac{\text{cm}^2}{\text{V} \cdot \text{s}}$$

Hence, the correct answer is

A. $\text{cm}^2/(\text{V} \cdot \text{s})$

Q1.96.1 MCQ 1996 1M Ans: C ● ○ ○

In a *p-n* junction, electrons and holes near the junction recombine after diffusion.
As a result, the depletion region is left with only immobile ionized impurity atoms:

Donor atoms $\rightarrow$ Positive ions Acceptor atoms $\rightarrow$ Negative ions

Hence, the depletion region contains only positive and negative ions, with no free electrons or holes.
Therefore, the correct answer is

C. Positive ions and negative ions

Q1.92.1 MCQ 1992 1M Ans: A ● ● ○

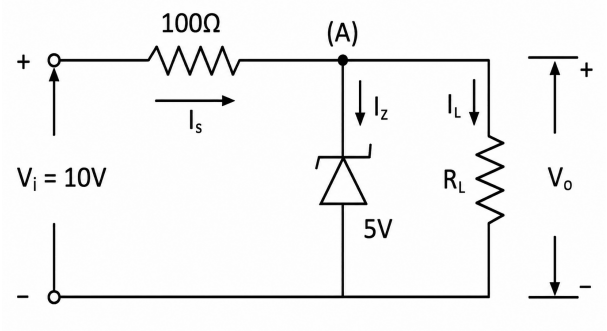


Fig. Solution Q1.92.1

The output voltage is regulated at

$$V_o = V_Z = 5 \text{ V}.$$

Hence, the source current through the 100 Ω resistor is

$$I_s = \frac{V_i - V_o}{100} = \frac{10 - 5}{100} = 50 \text{ mA}.$$

Applying KCL at node A,

$$I_s = I_Z + I_L.$$

The maximum load current occurs when the Zener current is at its minimum value required for regulation,

$$I_{Z,\min} = 10 \text{ mA}.$$

Therefore,

$$I_{L,\max} = I_s - I_{Z,\min} = 50 - 10 = 40 \text{ mA}.$$

The maximum Zener current occurs when the load is disconnected, i.e.,

$$I_L = 0.$$

Hence,

$$I_{Z,\max} = I_s = 50 \text{ mA}.$$

The minimum power rating of the Zener diode is

$$P_Z = V_Z I_{Z,\max} = 5 \times 50 \times 10^{-3} = 0.25 \text{ W}.$$

Hence, the correct answer is

A. 40 mA and 0.25 W

Q1.91.1 NAT 1991 2M Ans: 80 ●●○

For satisfactory voltage regulation, the Zener diode must conduct at least its minimum current.

$$I_{Z(\min)} = 25 \text{ mA}$$

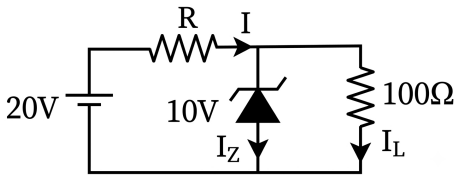


Fig. Solution Q1.91.1

The load current is

$$I_L = \frac{V_Z}{R_L} = \frac{10}{100} = 100 \text{ mA}$$

Hence, the current through the series resistor must satisfy

$$I \geq I_{Z(\min)} + I_L = 25 + 100 = 125 \text{ mA}$$

Applying Ohm's law,

$$I = \frac{20 - 10}{R} = \frac{10}{R}$$

Therefore,

$$\frac{10}{R} \geq 125 \text{ mA}$$

$$R \leq \frac{10}{125 \times 10^{-3}} = 80 \ \Omega$$

Hence, the required value of resistance is

80

SOLUTIONS

II

MOSFETs

Topics

1. Introduction to MOS - Depletion and Enhancement Type MOSFETs
2. Important Concepts of MOS Physics
3. Biasing a MOSFET - Transconductance and Output Impedance
4. Biasing a MOSFET - Amplification & Small Signal Analysis
5. Small Signal Model of MOSFET and Port Impedance
6. Common Source Amplifier - With and Without Degeneration
7. Common Source Amplifier with Different Biasing
8. Miller's Theorem
9. Common Gate and Common Drain Amplifiers
10. GmRout Method for Gain
11. PMOS Amplifiers
12. Cascode Current Source and Amplifiers
13. Current Mirror Circuits
14. Multistage Amplifiers and Pass Transistor Logic
15. Frequency Response of MOSFET Amplifiers
16. Frequency Response of Common Gate Amplifiers
17. Introduction to Differential Amplifiers, Current Source, Voltage Source as Tail
18. Large Signal Analysis of Differential Amplifiers, Concept of Half Circuit
19. Building a Two-Stage Op Amp
20. Types and Effects of Feedback
21. Feedback Topologies - Voltage-Voltage, Voltage-Current, Current-Voltage, Current-Current
22. MOS Amplifiers with Body Effect

Unit Snapshot*

Stream: EC	**Questions:** 15	**MCQ:** 14	**MSQ:** 0	**NAT:** 1
1M: 8	**2M:** 7	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2022)–3M(2025)

**See preface for how Avg Weightage and Range are calculated.*

PrepFusion

Q2.25.1 MCQ 2025 1M Ans: C ● ○ ○

Parameters	Common Source (CS)	Common Drain (CD)	Common Gate (CG)
Input resistance	$R_i = \infty$	$R_i = \infty$	$R_i = \frac{1}{g_m}$
Output resistance	$R_0 = R_0 \parallel R_L$	$R_0 = \frac{1}{g_m}$	$R_0 = R_0 \parallel R_L$
Voltage gain	$A_v = -g_m R'_L \quad (R'_L = R_L \parallel R_D)$	$A_v = \frac{g_m R'_L}{1+g_m R'_L} \approx 1$	$A_v = g_m R'_L$
Phase shift	180°	0°	0°

From the above table.
(i) In common source amplifier the phase shift between input and output voltage is 180°. Hence, it is an inverting amplifier.
(ii) In common drain amplifier the phase shift between input and output voltage is 0°. Hence, it is a non inverting amplifier.
(iii) In common gate amplifier the phase shift between input and output voltage is 0°. Thus, it is a non inverting amplifier.
Hence, the correct option is (C).

C. Common source amplifier is inverting and common gate amplifier is non-inverting amplifier

Key Points

- The Common Source (CS) amplifier is an inverting amplifier with a 180° phase shift.
- The Common Drain (CD) and Common Gate (CG) amplifiers are non-inverting amplifiers with a 0° phase shift.

Q2.25.2 MCQ 2025 2M Ans: C ● ● ○

For small-signal analysis,

- The ideal DC current source is replaced by an open circuit.
- V_{DC} and V_{in} are AC grounds.

The resulting small-signal equivalent circuit is

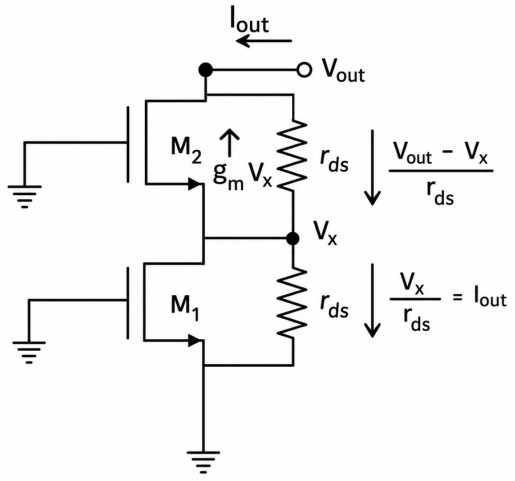


Fig. Solution Q2.25.2

Applying KCL at node V_x,

$$\frac{V_x}{r_{ds}} = I_{\text{out}}.$$

Applying KCL at the output node,

$$I_{\text{out}} + g_m V_x = \frac{V_{\text{out}} - V_x}{r_{ds}}.$$

Substituting

$$V_x = I_{\text{out}} r_{ds},$$

we get

$$I_{\text{out}} + g_m I_{\text{out}} r_{ds} = \frac{V_{\text{out}}}{r_{ds}} - I_{\text{out}}.$$

Hence,

$$\frac{V_{\text{out}}}{I_{\text{out}}} = g_m r_{ds}^2 + 2r_{ds}.$$

Therefore, the small-signal output impedance is

$$R_{\text{out}} = g_m r_{ds}^2 + 2r_{ds}.$$

Hence, the correct answer is

C. $g_m r_{ds}^2 + 2r_{ds}$

Q2.22.1 MCQ 2022 1M Ans: D ● ○ ○

An ideal MOSFET has insulated gate and zero small-signal gate current. A common-drain stage can supply a nonzero source/load current while drawing no input gate current, so its ideal current-gain magnitude is unbounded: (D) infinite.

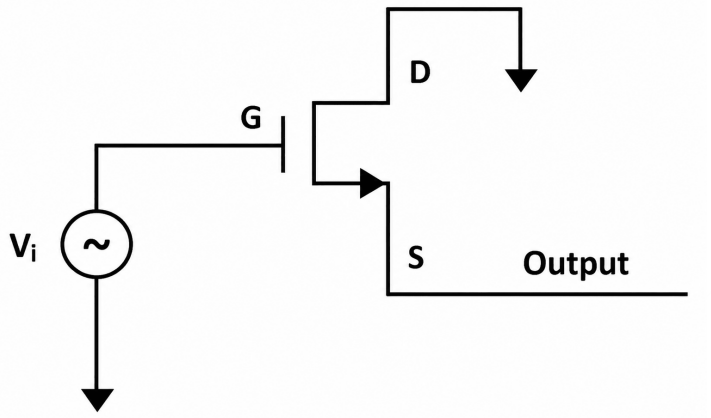


Fig. Solution Q2.22.1

Saturation does not require shorting gate to drain. For an ON n-channel device the condition is $V_{DS} \geq V_{GS} - V_{th}$. A diode connection is one way to satisfy that inequality, but is not a necessary condition for a common-drain amplifier. Gate leakage and capacitance limit real-device current gain; the question explicitly uses an ideal device.

Q2.20.1 NAT 2020 1M Ans: 0 ● ● ○

Let the magnitude of each induced charge be Q.
When a positive voltage is applied between the gate and the body,

$$\text{Gate} \rightarrow +Q, \qquad \text{Body} \rightarrow -Q.$$

As given in the question, the charge inside the silicon dioxide layer is +Q.
Hence, the silicon region must contain an equal and opposite charge, -Q.

The charge distribution is shown below.

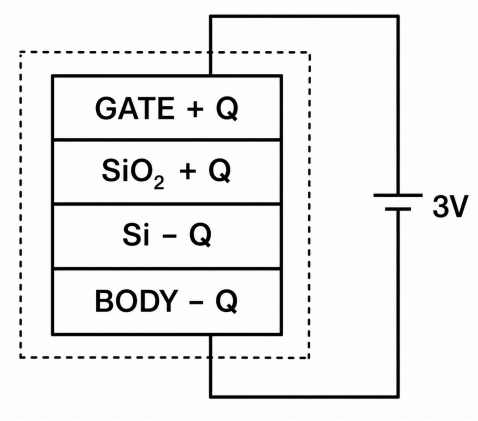


Fig. Solution Q2.20.1

The total charge enclosed within the dashed box is

$$(+Q) + (+Q) + (-Q) + (-Q) = 0.$$

Hence,

$$\boxed{0}$$

Q2.20.2 MCQ 2020 1M Ans: D

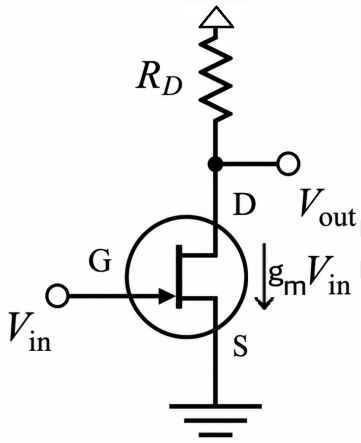


Fig. Solution Q2.20.2

For a common-source amplifier, neglecting the output resistance of the MOSFET,

$$-g_m V_{in} = \frac{V_o}{R_D}$$

$$\Rightarrow A_v = -g_m R_D.$$

Given,

$$g_m = 520\ \mu\text{A/V}, \qquad R_D = 4.7\ \text{k}\Omega.$$

Hence

$$A_v = -(520 \times 10^{-6})(4.7 \times 10^3) = -2.44.$$

Hence, the correct answer is

$$\boxed{\textbf{D. } -2.44}$$

Q2.19.1 MCQ | 2019 | 2M | Ans: B ● ● ○

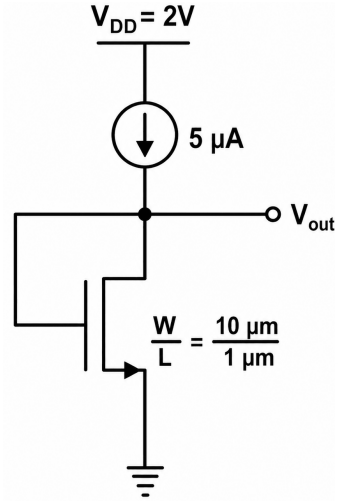


Fig. Solution Q2.19.1

The drain current is given by

$$I_D = \frac{\mu_n C_{ox}}{2}\frac{W}{L}(V_{GS} - V_T)^2.$$

Given,

$$I_D = 5\ \mu\text{A},$$
$$\mu_n C_{ox} = 100\ \mu\text{A/V}^2,$$
$$\frac{W}{L} = \frac{10}{1} = 10,$$

and

$$V_T = 0.5\ \text{V}.$$

Substituting,

$$5 = \frac{100}{2} \times 10(V_{GS} - 0.5)^2$$
$$5 = 500(V_{GS} - 0.5)^2$$
$$(V_{GS} - 0.5)^2 = \frac{5}{500} = 0.01.$$

Taking the positive root,

$$V_{GS} - 0.5 = 0.1$$
$$V_{GS} = 0.6\ \text{V}.$$

Since the source is grounded,

$$V_o = V_D = V_{GS} = 0.6\ \text{V} = 600\ \text{mV}.$$

Hence, the correct answer is

B. 600 mV

Q2.19.2 MCQ 2019 1M Ans: C ● ○ ○

For an enhancement-type NMOS transistor, the operating regions are summarized below.

Region	V_{GS}	V_{DS}
Cut-off	$V_{GS} < V_{th}$	Any value
Triode (Linear)	$V_{GS} > V_{th}$	$V_{DS} < V_{GS} - V_{th}$
Saturation	$V_{GS} > V_{th}$	$V_{DS} \geq V_{GS} - V_{th}$

Hence, for saturation,

$$V_{GS} > V_{th}$$

and

$$V_{DS} \geq V_{GS} - V_{th}.$$

Therefore, the correct answer is

C. $V_{GS} > V_{th}, \quad V_{DS} \geq V_{GS} - V_{th}$

Q2.05.1 MCQ 2005 1M Ans: D ● ○ ○

From the given circuit, the gate voltage (V_G) is connected to a 2 V source, and the source terminal (V_S) is connected to ground (0 V). The gate-to-source voltage is calculated as:

$$V_{GS} = V_G - V_S = 2\,\text{V} - 0\,\text{V} = 2\,\text{V}$$

The threshold voltage of the N-channel MOSFET is given as $V_T = 1$ V. Since $V_{GS} > V_T$ (2 V > 1 V), the MOSFET is turned ON. Because the MOSFET is specified as "ideal" and no conduction parameters (such as $\mu_n C_{ox}$ or W/L) are provided, it is modeled as an ideal switch. When turned ON, an ideal MOSFET acts as a short circuit between the drain and source terminals.

$$V_D = V_S = 0\,\text{V}$$

Node 'a' is connected to the drain terminal through a 1 kΩ resistor, and it is connected to ground (node 'b') through a 2 kΩ resistor. Since the drain terminal is effectively at ground potential ($V_D = 0$ V) and node 'b' is also at ground (0 V), there is no potential difference to drive current through this rightmost branch. Therefore, the voltage at node 'a' is 0 V.

$$V_{ab} = V_a - V_b = 0\,\text{V} - 0\,\text{V} = 0\,\text{V}$$

D. 0V

Key Points

- In the absence of specific transconductance parameters, an "ideal" MOSFET operates as a perfect switch.
- When $V_{GS} > V_T$, the ideal MOSFET is in the ON state and behaves as a short circuit ($V_{DS} = 0$ V).
- When the drain is shorted to ground, any parallel resistive branches connected to the drain will have 0 V across them, resulting in zero current and zero voltage drop across those components.

Q2.05.2 MCQ 2005 2M Ans: A ● ● ○

In a metal, all free charges reside on the surface.
Hence, the electric field inside the metal is, E=0.
The dielectric contains no free charge. Therefore, the electric field remains constant throughout the dielectric region.
In the semiconductor, negative space charge is distributed throughout the depletion region.
Consider a Gaussian surface starting at the oxide-semiconductor interface and extending a distance x into the semiconductor.
As the Gaussian surface moves deeper into the semiconductor, it encloses more negative charge.

From Gauss's law,

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enc}}}{\varepsilon},$$

the enclosed negative charge reduces the net electric flux. Therefore, the electric field decreases with increasing distance inside the semiconductor.

Since the charge density decreases linearly, the slope of the electric field also decreases linearly, and the electric field finally becomes zero at the edge of the depletion region. Hence, the electric field distribution is

Region	Electric Field
Metal	$E = 0$
Dielectric	Constant
Semiconductor	Decreases linearly to zero

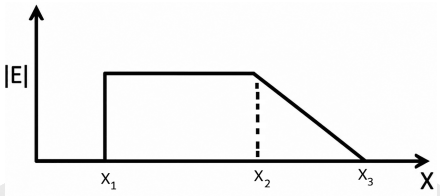


Fig. Solution Q2.05.2

Therefore, the correct electric field variation is shown in

A

Q2.05.3 MCQ 2005 2M Ans: B ● ○ ○

Trans-conductance of MOSFET is given by

$$g_m = \frac{\partial i_D}{\partial V_{GS}}$$

$$= \frac{(2-1)\text{ mA}}{(2-1)\text{ V}} = 1\text{ mS}$$

Hence (B) is correct option.

B. 1 ms

Q2.05.4 MCQ 2005 2M Ans: D ● ○ ○

Voltage gain can be obtain by small signal equivalent circuit of given amplifier.

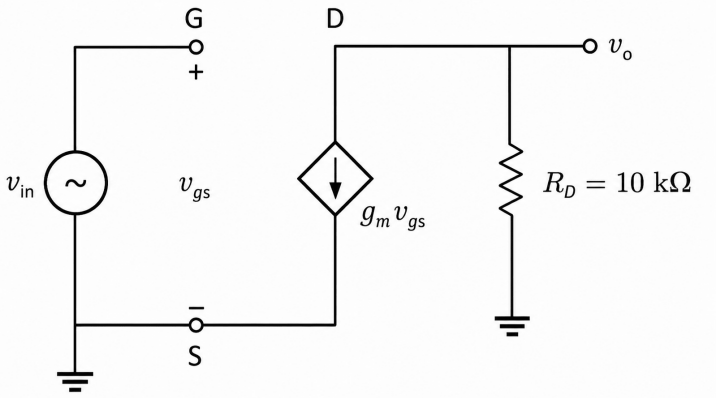


Fig. Solution Q2.05.4

$$v_o = -g_m v_{gs} R_D \quad \text{and,} \quad v_{gs} = v_{in}$$

So,

$$v_o = -g_m R_D v_{in}$$

Voltage gain

$$\begin{aligned} A_v = \frac{v_o}{v_i} &= -g_m R_D \\ &= -(1\,\text{mS})(10\,\text{k}\Omega) \\ &= -10 \end{aligned}$$

Hence (D) is correct option.

D. -10

Key Points

- The negative sign indicates a 180° phase inversion between the input and output signals.

Q2.04.1 MCQ | 2004 | 2M | Ans: D | ● ○ ○

Given,

$$V_S = 4\text{ V}, \qquad V_G = 0\text{ V}, \qquad V_T = -1\text{ V}, \qquad I_D = 1\text{ mA}.$$

For a PMOS transistor to operate in the linear (triode) region,

$$V_{SD} < V_{SG} - |V_T|.$$

Now,

$$V_{SG} = V_S - V_G = 4 - 0 = 4\text{ V}.$$

Hence,

$$V_S - V_D < 4 - 1 = 3.$$

$$4 - V_D < 3 \Rightarrow V_D > 1\text{ V}$$

Since,

$$V_D = I_D R = (1\text{ mA})R,$$

we get

$$(1\text{ mA})R > 1\text{ V}$$

$$R > \frac{1}{1 \times 10^{-3}} = 1000\ \Omega.$$

Among the given options, the only value satisfying this condition is

$$R = 1200\ \Omega.$$

Hence, the correct answer is

D. 1200 Ω

Key Points

- A PMOS transistor enters the linear (triode) region when the source-to-drain voltage is less than the overdrive voltage: $V_{SD} < V_{SG} - |V_T|$.
- Conversely, it operates in the saturation region when $V_{SD} \geq V_{SG} - |V_T|$.

- In this configuration, increasing the resistor R increases the drain voltage V_D, which decreases V_{SD}, eventually pushing the transistor from saturation into the linear region.

Q2.03.1 MCQ 2003 2M Ans: C ● ● ○

The gate is connected to the drain, so an ON device satisfies $V_{DS} = V_{GS} \geq V_{GS} - V_{th}$ and is in saturation. The initial 4 mA through 1 kOhm gives $V_{GS} = 10 - 4 = 6$ V. Thus

$$K = \frac{4}{(6-2)^2} = 0.25 \text{ mA/V}^2.$$

After increasing R_D to 4 kOhm, let I be the drain current in mA. Then $V_{GS} = 10 - 4I$ and

$$I = 0.25(8 - 4I)^2,$$
$$0 = 4I^2 - 17I + 16.$$

The roots are $I = (17 \pm \sqrt{33})/8$. The larger root, 2.843 mA, gives $V_{GS} < 2$ V and violates the ON condition. The physical root is

$$I = 1.40693\ldots \text{ mA}.$$

This is closest to 1.4 mA, option (C).

The "4 Ohm" text was missing its kilo prefix. Treating it literally would decrease the resistance, contradict "increased," and would not yield any supplied option.

Key Points

- When the gate is tied to the drain ($V_{GS} = V_{DS}$), an enhancement-mode MOSFET is always biased in the saturation region.
- The conduction parameter K_n is a constant physical property of the transistor and remains unchanged when the external bias resistors are modified.
- When quadratic equations yield two roots for currents or voltages in transistor circuits, always verify the region of operation conditions ($V_{GS} > V_{th}$) to reject the unphysical mathematical root.

Q2.03.2 MCQ 2003 1M Ans: C ● ● ○

From the given I_D-V_{GS} characteristic,

$$I_D \neq 0 \quad \text{at} \quad V_{GS} = 0.$$

This indicates that a conducting channel already exists at zero gate bias.
Hence, the MOSFET is a depletion-mode device.

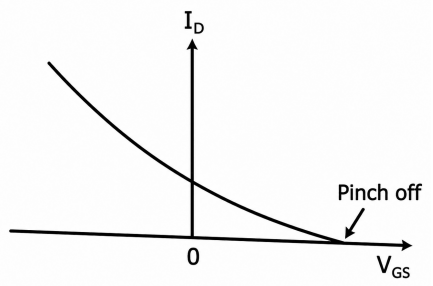


Fig. Solution Q2.03.2

Also, the drain current becomes zero at a positive value of V_{GS}, i.e., the pinch-off (threshold) voltage is positive.
A positive threshold (pinch-off) voltage corresponds to a p-channel depletion-mode MOSFET.
Hence, the correct answer is

C. A p-channel depletion mode device

Q2.99.1 MCQ 1999 1M Ans: A ● ○ ○

The symbols shown correspond to the following MOSFET types.

Symbol	Device
A	Enhancement-type n-channel MOSFET
B	Enhancement-type p-channel MOSFET
C	Depletion-type n-channel MOSFET
D	n channel JFET

Hence, the symbol representing an enhancement-type n-channel MOSFET is

A

PrepFusion

SOLUTIONS

III

Operational Amplifier

Topics

1. Basics of Op-Amp, Comparator and Virtual Short
2. Inverting and Non-Inverting Amplifiers
3. Adder, Differential and Instrumentation Amplifiers
4. Log, Anti-Log and Rectifier Using Op-Amp
5. Filters Using Op-Amp
6. Offsets in Op-Amp
7. CMRR, Slew Rate and Frequency Response of Op-Amp
8. Integrator Using Op-Amp
9. Differentiator Using Op-Amp
10. Active Clipper, Clamper and Peak Detector
11. Multivibrators Using Op-Amp
12. Oscillators Using Op-Amp
13. Schmitt Triggers

Unit Snapshot*

Stream: EC	**Questions:** 81	**MCQ:** 53	**MSQ:** 0	**NAT:** 18
1M: 27	**2M:** 44	**Descriptive:** 10	**Avg Weightage** ~3M	**Range:** 1M(2018)–5M(2022)

**See preface for how Avg Weightage and Range are calculated.*

Q3.26.1 MCQ 2026 2M Ans: A ●●○

For an inverting operational amplifier with a finite open-loop voltage gain A_0, the closed-loop voltage gain A_{CL} is modified to account for the finite gain error:

$$A_{CL} = \frac{-\frac{R_f}{R_1}}{1 + \frac{1}{A_0}\left(1 + \frac{R_f}{R_1}\right)}$$

- Input resistor, $R_1 = 5\,\text{k}\Omega$
- Feedback resistor, $R_f = 100\,\text{k}\Omega$
- Open-loop gain, $A_0 = 105$

$$A_{CL} = \frac{-\frac{100\,\text{k}\Omega}{5\,\text{k}\Omega}}{1 + \frac{1}{105}(1 + \frac{100\,\text{k}\Omega}{5\,\text{k}\Omega})} = \frac{-20}{1 + \frac{21}{105}} \approx -16.67$$

$$\boxed{(A) \quad -16.67}$$

Key Points

- **Ideal vs. Practical Op-Amp:** For an ideal op-amp ($A_0 \to \infty$), the gain reduces to the standard form $A_{CL} = -R_f/R_1 = -20$.
- **Feedback Factor (β):** The term $\beta = \frac{R_1}{R_1+R_f}$ represents the feedback fraction. The denominator of the practical gain formula can also be expressed as $1 + \frac{1}{A_0\beta}$.

Q3.25.1 MCQ 2025 1M Ans: D ●○○

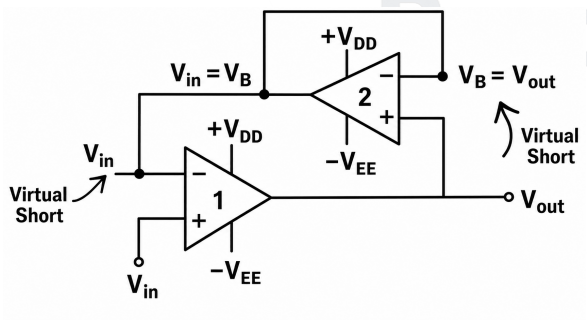


Fig. Solution Q3.25.1

Assuming ideal operational amplifiers under stable negative feedback configuration, we invoke the **virtual short** concept ($V_+ = V_-$) for both elements:

- For Op-Amp 1: $V_{-1} = V_{+1} = V_{in}$

$$V_B = V_{in}$$

- For Op-Amp 2: $V_{-2} = V_{+2}$

$$V_B = V_{out}$$

From both conditions: $V_{out} = V_{in}$

Since the overall closed-loop system possesses unity voltage gain ($A_v = 1$) along with high input impedance and zero output impedance characteristic properties, this block strictly functions as a **buffer**.

$$\boxed{\text{(D) buffer}}$$

Key Points

- **Virtual Short Concept:** For an ideal op-amp with high open-loop gain operating under negative feedback, the differential input voltage approaches zero ($V_+ \approx V_-$).
- **Buffer Characteristics:** A voltage buffer or unity-gain voltage follower satisfies $V_{out} = V_{in}$, performing impedance

transformation to isolate electrical stages.

Q3.25.2 NAT | 2025 | 2M | Ans: 2 | ● ○ ○

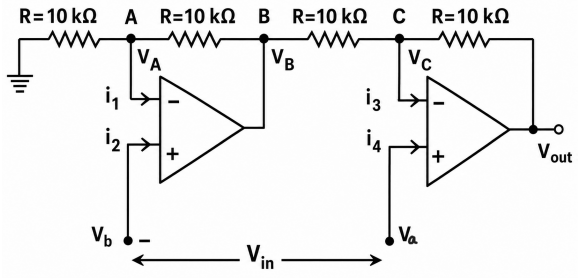


Fig. Solution Q3.25.2

Applying KCL at node A:

$$\frac{0 - V_A}{R} = \frac{V_A - V_B}{R}$$

$$V_B = 2V_A$$

Using the virtual short concept for both ideal op-amps ($i_{in} = 0$):

- For the first op-amp: $V_b = V_A$
- For the second op-amp: $V_a = V_C$

From the differential input definition:

$$V_{in} = V_a - V_b = V_C - V_A$$

$$V_{in} = V_C - \frac{V_B}{2}$$

$$V_{in} = \frac{2V_C - V_B}{2}$$

Applying KCL at node C:

$$\frac{V_B - V_C}{R} = \frac{V_C - V_{out}}{R}$$

$$V_{out} = 2V_C - V_B$$

Substituting this in V_{in} expression:

$$V_{in} = \frac{V_{out}}{2} \implies A_v = \frac{V_{out}}{V_{in}} = 2$$

$\boxed{2}$

Mistakes to be avoided

Input Voltage Reference: Ensure V_{in} is taken correctly as the difference $(V_a - V_b)$ and not referenced directly to the system ground node.

Q3.24.1 NAT | 2024 | 2M | Ans: 39.50 to 41.50 | ● ● ●

By voltage divider rule, the non-inverting input V_+ is:

$$V_+ = v_{in2}\left(\frac{101.5}{101.5+9.5}\right) = 0.9144(v_{in2})$$

By virtual short, $V_- = V_+ = 0.9144(v_{in2})$. Applying KCL at the inverting node:

$$\frac{v_{in1}-V_-}{10.5} = \frac{V_- - V_o}{101}$$

$$V_o = V_- + V_-\left(\frac{101}{10.5}\right) - v_{in1}\left(\frac{101}{10.5}\right)$$

Substituting V_- gives:

$$V_o = -9.619(v_{in1}) + 9.71(v_{in2})$$

From this expression, the single-ended gains are:

$$A_1 = \left.\frac{V_o}{v_{in1}}\right|_{v_{in2}=0} = -9.619; \quad A_2 = \left.\frac{V_o}{v_{in2}}\right|_{v_{in1}=0} = 9.71$$

Using the mode relations:

$$V_d = v_{in1} - v_{in2}, \quad V_{cm} = \frac{v_{in1}+v_{in2}}{2}$$

The differential and common-mode gains are:

$$A_d = \frac{A_1 - A_2}{2} = \frac{-9.619 - 9.71}{2} = -9.6645$$

$$A_{cm} = A_1 + A_2 = -9.619 + 9.71 = 0.091$$

The magnitude ratio for CMRR is:

$$\text{CMRR} = \left|\frac{A_1 - A_2}{2(A_1 + A_2)}\right| = \left|\frac{-9.6645}{0.091}\right| \approx 106.203$$

In decibels:

$$\text{CMRR}_{\text{dB}} = 20\log_{10}(106.203) \approx 40.52\,\text{dB}$$

40.52

Key Points

- **CMRR Definition:** Common Mode Rejection Ratio is a measure of the capability to reject common signals: CMRR $= 20\log_{10}(|A_d/A_c|)$.
- **Mode Components:** Inputs are decomposed into differential and common modes via $V_d = v_{in2} - v_{in1}$ and $V_c = \frac{v_{in1}+v_{in2}}{2}$ where V_d represents the total differential input voltage span, whereas V_{cm} represents the average common-mode bias level applied.

Q3.23.1 MCQ | 2023 | 2M | Ans: D ●●○

For $v_{in} > 0$, the output is negative. Diode D_1 conducts, providing negative feedback through the 1 kOhm resistor; D_2 blocks the other branch. The equal input and feedback resistors give $v_o = -v_{in}$, down to -10 V.

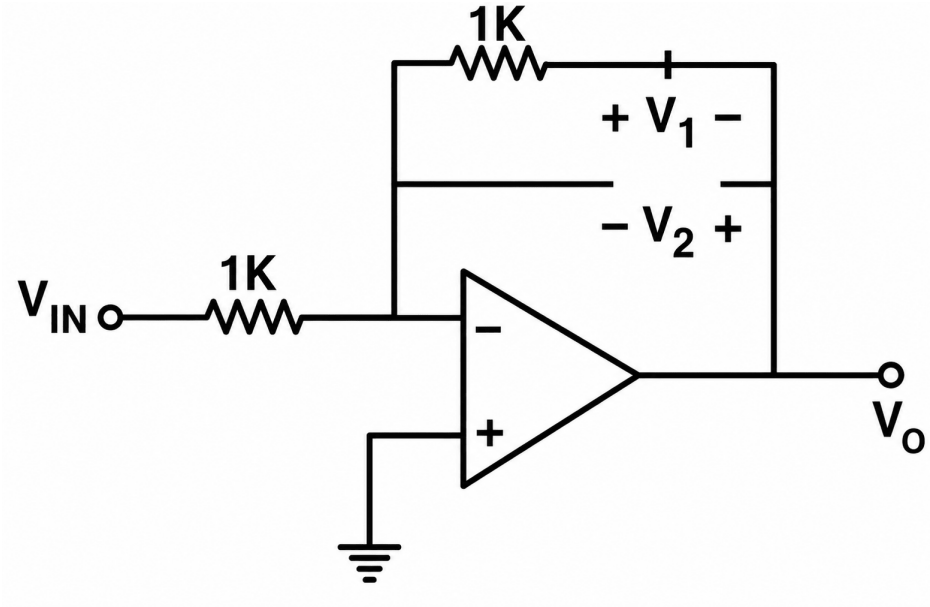


Fig. Solution Q3.23.1

For $v_{in} < 0$, the output becomes positive. D_1 is off. The branch containing D_2 and the reverse-biased Zener conducts once $v_o - v_- = 5$ V, providing feedback and clamping the positive output to 5 V. With $v_- = 0$, the maximum is therefore +5 V. Thus $\boxed{v_{o,\max} = +5\text{ V}}$ and $\boxed{v_{o,\min} = -10\text{ V}}$, option (D). Both limits lie inside the 15 V supply rails.

Q3.22.1 MCQ 2022 1M Ans: B ●○○

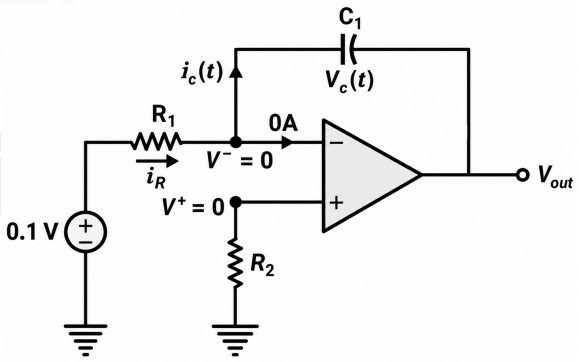
Applying KCL :

$$i_R = i_c(t)$$

$$\frac{V_i - 0}{R_1} = C_1 \times \frac{d}{dt}(0 - V_{\text{out}})$$

$$\frac{dV_{\text{out}}}{dt} = -\frac{V_i}{R_1 C_1}$$

$$V_{\text{out}} = -\frac{1}{R_1 C_1}\int V_i\, dt$$



Given $V_i = 0.1$ V:

$$V_{\text{out}} = -\frac{1}{R_1 C_1}\int_t 0.1 \times dt = -\frac{0.1}{R_1 C_1} \times t$$

As time approaches steady-state ($t \to \infty$), V_{out} decreases linearly towards $-\infty$. However, the output is bounded by the negative power supply rail. Thus, the op-amp will saturate :

$$V_{\text{out}} = -V_{\text{EE}}$$

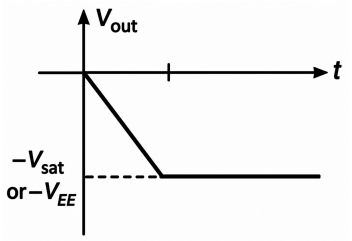


Fig. Solution Q3.22.1

$\boxed{\text{(B) saturate to } -V_{\text{EE}}}$

Mistakes to be avoided

Open Circuiting Capacitor at DC: Do not blindly replace the capacitor with an open circuit during steady-state evaluation. Because a constant DC input produces a continuous current charging the capacitor, it operates as a ramp

generator until hitting saturation.

Q3.22.2 MCQ 2022 2M Ans: C ●●○

The given non-ideal op-amp configuration forms a voltage-series (series-shunt) negative feedback amplifier with a feedback factor $\beta = 1$. The output impedance with feedback is given by:

$$Z_{\text{out}} = \frac{Z_0}{1 + A_{\text{OL}}\beta} = \frac{Z_0}{1 + A_{\text{OL}}}$$

- **Low Frequencies (0 to f_α):** A_{OL} remains constant at its maximum value. Thus, $Z_{out} = \dfrac{Z_0}{1 + A_{\text{OL}}}$ is constant.
- **Mid Frequencies ($f_\alpha < f < f_\beta$):** As frequency increases, A_{OL} begins to decrease. Consequently, the denominator decreases, causing Z_{out} to increase:

$$\uparrow Z_{out} = \frac{Z_0}{1 + A_{\text{OL}} \downarrow}$$

- **High Frequencies ($f > f_\beta$):** At very high frequencies, the open-loop gain drops down significantly ($A_{\text{OL}} \to 0$). Hence, the impedance reaches a maximum constant level:

$$Z_{out} = \frac{Z_0}{1 + 0} = Z_0$$

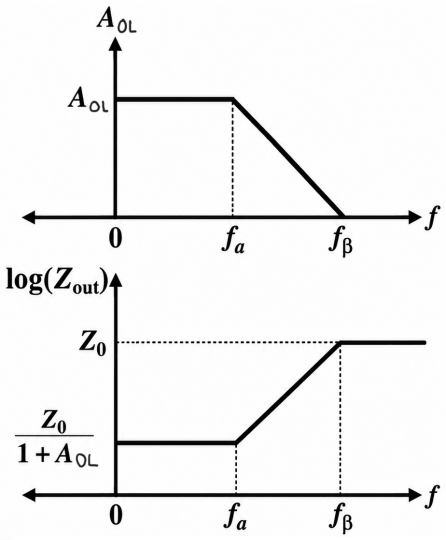


Fig. Solution Q3.22.2

(C)

Q3.22.3 MCQ 2022 2M Ans: C ●○○

From the virtual ground concept, the node voltage at A is zero: $V_A = 0$
The output node voltage V_x is developed due to the input current I_{in} charging the feedback capacitor C_f:

$$V_x = \frac{1}{C_f}\int I_{\text{in}} \times dt = \frac{I_{\text{in}}}{C_f} \times t$$

The output current I_{out} flowing through the capacitor C_c is determined by:

$$I_{\text{out}} = C_c \times \frac{dV_x}{dt} = C_c \times \frac{d}{dt}\left(\frac{I_{\text{in}}}{C_f} \times t\right) = C_c \times \frac{I_{\text{in}}}{C_f}$$

Rearranging the equation to find the current gain:

$$\frac{I_{\text{out}}}{I_{\text{in}}} = \frac{C_c}{C_f}$$

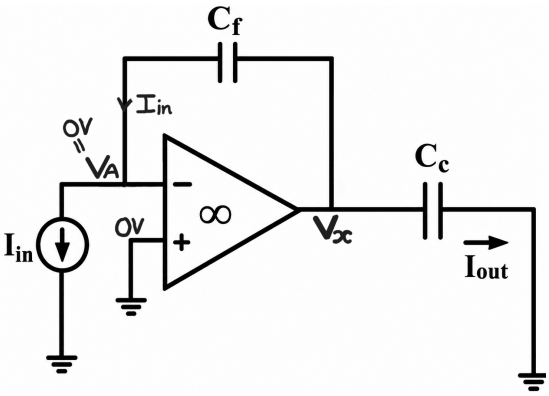


Fig. Solution Q3.22.3

$$(C)\ \frac{C_c}{C_f}$$

Q3.21.1 NAT | 2021 | 2M | Ans: 3150 to 3170 | ● ● ○

The capacitor voltage $v_c(t)$ oscillates continuously between the threshold limits:

$$V_{LT} = 1.6\,\text{V} \quad \text{and} \quad V_{UT} = 2.4\,\text{V}$$

The time constant of the network is:

$$\tau = RC = 10\,\text{k}\Omega \times 47\,\text{nF} = 470\,\mu\text{s}$$

1. Charging Phase (t_1): Here, the output is at high level $V_o = 5\,\text{V}$. The capacitor charges from $v_c(0) = 1.6\,\text{V}$ towards $v_c(\infty) = 5\,\text{V}$:

$$v_c(t) = v_c(\infty) + [v_c(0) - v_c(\infty)]e^{-t/\tau}$$

$$v_c(t) = 5 - 3.4e^{-t/\tau}$$

At $t = t_1$, the voltage reaches the upper threshold $v_c(t_1) = 2.4\,\text{V}$:

$$2.4 = 5 - 3.4e^{-t_1/470\,\mu\text{s}} \implies t_1 \approx 126.08\,\mu\text{s}$$

2. Discharging Phase (t_2): The output drops to $V_o = 0\,\text{V}$. The capacitor discharges from $v_c(0) = 2.4\,\text{V}$ towards $v_c(\infty) = 0\,\text{V}$:

$$v_c(t) = 2.4e^{-t/\tau}$$

At $t = t_2$, the voltage reaches the lower threshold $v_c(t_2) = 1.6\,\text{V}$:

$$1.6 = 2.4e^{-t_2/\tau} \implies t_2 = -\tau \ln\left(\frac{1.6}{2.4}\right) \approx 190.57\,\mu\text{s}$$

Total Time Period (T):

$$T = t_1 + t_2 = 126.08\,\mu\text{s} + 190.57\,\mu\text{s} = 316.65\,\mu\text{s}$$

The frequency of the oscillator is:

$$f = \frac{1}{T} = \frac{1}{316.65 \times 10^{-6}\,\text{s}} \approx 3158.06\,\text{Hz}$$

3158.06

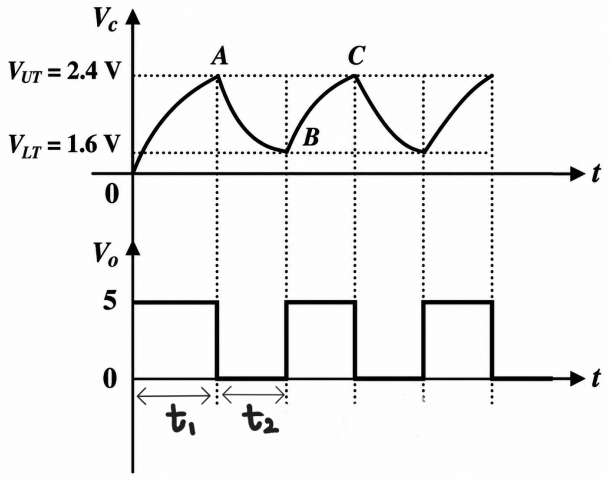


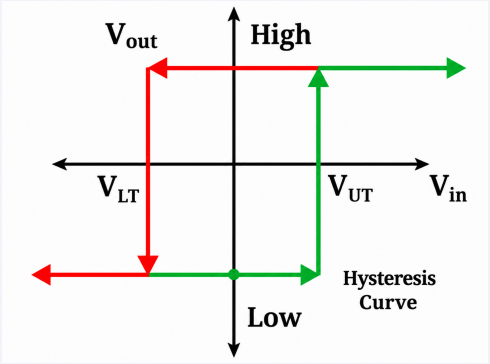
Fig. Solution Q3.21.1

Key Points

- **General Transient Equation:** For any charging or discharging phase of an RC circuit:

$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$$

- **Schmitt-Trigger Oscillator:** The output shifts state when the capacitor voltage reaches either the upper threshold (V_{UT}) or the lower threshold (V_{LT}).



Q3.20.1 NAT | 2020 | 2M | Ans: MTA | ● ● ●

Since the input current of the op-amp is 0: $V_+ = 3 \times \dfrac{R_1}{R_{\text{thermistor}} + R_1}$

Using the superposition principle at the output:

$$V_{\text{out}} = -0.1\frac{R_3}{R_2} + V_+\left(1 + \frac{R_3}{R_2}\right)$$

Substituting the given values $R_1 = 1\,\text{k}\Omega$, $R_2 = 1.3\,\text{k}\Omega$, and $R_3 = 2.6\,\text{k}\Omega$:

$$V_{\text{out}} = -0.2 + \frac{9}{R_{\text{th}} + 1}$$

Case-1: $\alpha = -(4 + 0.25)\,\%/^\circ\text{C} = -0.0425$

$$R_{\text{th}} = 2\,(1 - 0.0425 \times 150) = -10.75\,\text{k}\Omega$$

$$V_{\text{out}} = -0.2 + \frac{9}{-10.75 + 1} = -1.123\,\text{V}$$

Case-2: $\alpha = -(4 - 0.25)\,\%/^\circ\text{C} = -0.0375$

$$R_{\text{th}} = 2\,(1 - 0.0375 \times 150) = -9.25\,\text{k}\Omega$$

$$V_{\text{out}} = -0.2 + \frac{9}{-9.25 + 1} = -1.291\,\text{V}$$

Error Calculation:

$$\Delta V_{\text{out}} = |-1.123 - (-1.291)| = 0.168\,\text{V}$$

$$\text{error} = \frac{\Delta V_{\text{out}}}{2} = 0.084\,\text{V}$$

$$\boxed{MTA}$$

Key Points

Superposition in Op-Amps: The total output voltage is the linear combination of the responses due to individual non-inverting and inverting input configurations.

Mistakes to be avoided

Why this question is MTA: Substituting $T = 150^\circ\text{C}$ and the given negative value of α yields a negative resistance ($R_{\text{th}} < 0$) for the passive thermistor component. Because a passive component cannot exhibit negative absolute resistance under normal operation, the question contains a physical flaw and mathematical contradiction, resulting in it being marked as **Marks to All (MTA)**.

Q3.19.1 MCQ | 2019 | 2M | Ans: B | ● ○ ○

Method 1

By voltage division at the non-inverting node:

$$V_+ = V_2 \times \frac{100}{110} = \frac{10}{11}V_2$$

By using superposition:

$$V_o = V_+\left(1 + \frac{100}{10}\right) + V_1\left(-\frac{100}{10}\right)$$

$$V_o = \frac{10}{11}V_2(11) - 10V_1 = 10(V_2 - V_1)$$

Substituting $V_1 = 10\,\text{mV}$ and $V_2 = 50\,\text{mV}$:

$$V_o = 10(50\,\text{mV} - 10\,\text{mV}) = 400\,\text{mV}$$

Method 2

The given configuration is a standard differential amplifier circuit. The output voltage is given by the formula:

$$V_o = \frac{R_2}{R_1}(V_2 - V_1)$$

Substituting the value of the resistors and voltages:

$$V_o = \frac{100\,\text{k}\Omega}{10\,\text{k}\Omega}(50\,\text{mV} - 10\,\text{mV})$$

$$V_o = 10(40\,\text{mV}) = 400\,\text{mV}$$

$$\boxed{(B)\ \ 400mV}$$

Key Points

Differential Amplifier Gain: When matched resistor ratios are used, the output voltage scales strictly with the difference between the input signals.

Q3.18.1 MCQ 2018 1M Ans: B ●○○

Let V_A be the voltage at the inverting terminal and V_B be the voltage at the non-inverting terminal. According to the virtual ground concept:

$$V_A = V_B = V_{\text{in}}$$

Applying KCL at node A (non-inverting terminal):

$$\frac{V_{\text{in}} - V_0}{Z} = i_{\text{in}} \quad \ldots (\text{i})$$

Applying KCL at the inverting input terminal:

$$\frac{V_{\text{in}} - 0}{R_2} + \frac{V_{\text{in}} - V_0}{R_1} = 0$$

$$V_{\text{in}}\left[\frac{1}{R_2} + \frac{1}{R_1}\right] = \frac{V_0}{R_1}$$

$$V_{\text{in}}\left[\frac{R_1 + R_2}{R_1 R_2}\right] \times R_1 = V_0$$

$$V_0 = V_{\text{in}} \times \left[\frac{R_1 + R_2}{R_2}\right] \quad \ldots (\text{ii})$$

Substituting equation (ii) into equation (i):

$$\frac{V_{\text{in}} - V_{\text{in}}\left[\frac{R_1 + R_2}{R_2}\right]}{Z} = i_{\text{in}}$$

$$\frac{V_{\text{in}}}{i_{\text{in}}}\left[1 - \frac{R_1 + R_2}{R_2}\right] = Z$$

$$\frac{V_{\text{in}}}{i_{\text{in}}}\left[\frac{R_2 - R_1 - R_2}{R_2}\right] = Z$$

$$\frac{V_{\text{in}}}{i_{\text{in}}} = -Z \cdot \frac{R_2}{R_1}$$

$$\boxed{(B)\ \ -Z \cdot \frac{R_2}{R_1}}$$

Q3.17.1 MCQ 2017 2M Ans: A ● ○ ○

Case 1: For positive input ($V_{in} > 0$)

- The output voltage tends to be positive ($+V_{ss}$), which causes the diode D to become forward biased (ON).
- The circuit completes the negative feedback loop and acts as a standard voltage follower.
- Due to the virtual short condition:

$$V_o = V_{in}$$

Case 2: For negative input ($V_{in} < 0$)

- The op-amp output goes negative ($-V_{ss}$), causing the diode D to become reverse biased (OFF).
- The feedback loop breaks, and no current flows through the resistor R.
- Therefore, the output voltage drops to zero:

$$V_o = 0$$

Hence, the circuit operates as a precision positive half-wave rectifier.

$\boxed{(A)}$

Q3.17.2 MCQ 2017 2M Ans: C ● ● ○

The equal 2R resistors give the non-inverting voltage $V_A = V_s/2$. Negative feedback makes the inverting voltage equal to V_A.

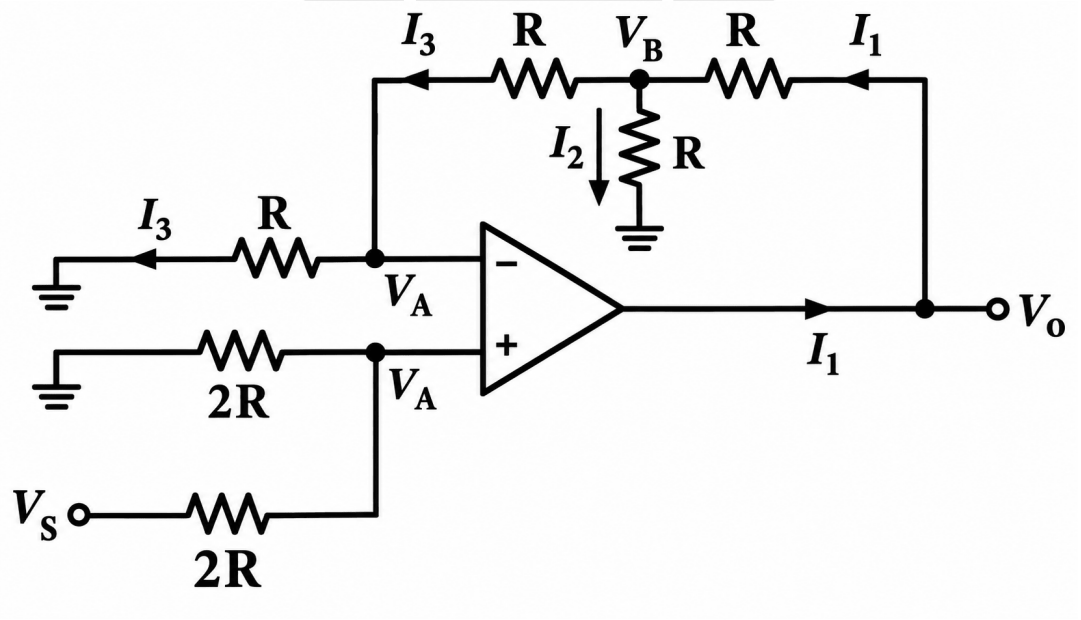


Fig. Solution Q3.17.2

No current enters the op-amp input, so the left R resistor carries $I_3 = V_A/R$. The resistor from V_B to V_A carries the same current. Hence $V_B - V_A = V_A$, or $V_B = 2V_A = V_s$.
At node B, the current toward ground is $I_2 = V_B/R = V_s/R$. Thus $I_1 = I_2 + I_3 = 3V_s/(2R)$, giving

$$V_o = V_B + RI_1 = \frac{5}{2}V_s.$$

Therefore $\boxed{V_o = 2.5V_s}$, option (C).

Q3.16.1 MCQ 2016 1M Ans: A ●○○

Method 1

Transfer Function Approach

The parallel impedance in the feedback loop is:

$$Z_f = R_2 \parallel \frac{1}{j\omega C} = \frac{R_2}{1 + j\omega R_2 C}$$

The gain (transfer function) of the inverting amplifier :

$$\frac{V_{out}}{V_{in}} = -\frac{Z_f}{R_1} = -\frac{R_2}{R_1(1 + j\omega R_2 C)}$$

- At low frequency ($\omega \to 0$): $\left|\frac{V_{out}}{V_{in}}\right| \to \frac{R_2}{R_1}$ (Finite gain)
- At high frequency ($\omega \to \infty$): $\left|\frac{V_{out}}{V_{in}}\right| \to 0$ (Zero gain)

This represents a active low pass filter.

Method 2

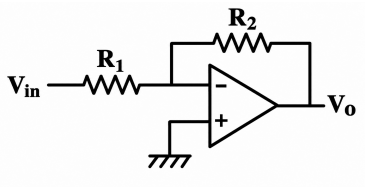
Frequency Limit Inspection

- **At $\omega = 0$ (Capacitor acts as an open circuit.):**

  Circuit acts as an inverting amplifier with gain:

  $$\frac{V_{out}}{V_{in}} = -\frac{R_2}{R_1}$$

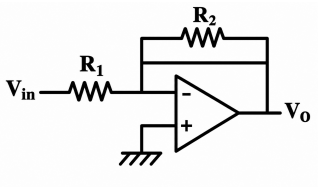
  (Passes low frequencies)



- **At $\omega = \infty$ (Capacitor acts as a short circuit):** Feedback impedance $Z_f = 0$, so:

  $$V_{out} = V_- = 0\ \text{V}$$

  (Blocks high frequencies) The system is a low pass filter.



(A) low pass filter.

Key Points

3-dB cutoff frequency for this first-order active low-pass filter is $\omega_c = \frac{1}{R_2 C}$.

Q3.16.2 MCQ 2016 2M Ans: D ●○○

Op-Amp is ideal, so input impedance = ∞.

Applying KCL at node B:

$$\frac{V - V_1}{1} + \frac{V - V_2}{4} = 0$$

$$4V - 4V_1 + V - V_2 = 0$$

$$V = \frac{4V_1 + V_2}{5}$$

By virtual ground : $V_A = V_B = V$

Applying KCL at node A:

$$\frac{V - V_3}{1} + \frac{V - V_{out}}{9} = 0$$

$$9(V - V_3) + V - V_{out} = 0$$

$$V_{out} = 10\left[\frac{4V_1 + V_2}{5}\right] - 9V_3$$

$$V_{out} = 8V_1 + 2V_2 - 9V_3$$

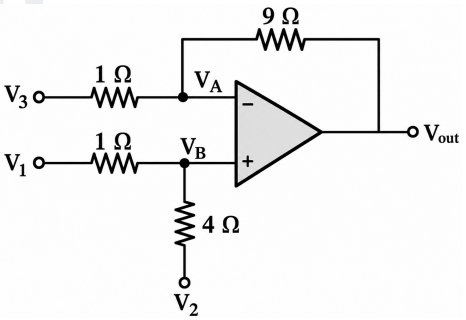


Fig. Solution Q3.16.2

(D) $8V_1 + 2V_2 - 9V_3$

Q3.15.1 MCQ 2015 1M Ans: A ●○○

- An ideal Instrumentation Amplifier features extremely high CMRR, high input impedance, high linearity, and low output impedance to prevent loading effects.
- **Why Low Output Impedance ($Z_{out} \approx 0$)?**
  - **Minimizes Loading Effect**: Prevents voltage attenuation across internal resistance when connected to a load (R_L):

$$V_L = V_{\text{internal}} \times \frac{R_L}{R_{\text{out}} + R_L} \approx V_{\text{internal}} \quad (\text{for } R_{\text{out}} \ll R_L)$$

  - **Ensures Gain Stability**: Maintains linear output voltage regardless of load variations.
  - **Current Drive Capability**: Supplies required load current to downstream stages (ADCs, filters) without signal droop.

(A) P, Q and R only

Q3.15.2 MCQ 2015 1M Ans: D ●●○

Method 1

Using Superposition Principle:

Output due to inverting input:

$$V_{o1} = V_1 \left[\frac{-R}{1/(Cs)}\right] = -sCRV_1$$

Potential at non-inverting node:

$$V_+ = V_2 \left[\frac{R}{R + 1/(Cs)}\right]$$

Output due to non-inverting input:

$$V_{o2} = (1 + RCs) \times \frac{R}{R + 1/(Cs)} \times V_1 = sCRV_2$$

Total output voltage:

$$V_o = V_{o1} + V_{o2} = -sCR(V_1 - V_2) = -sCRV_i$$

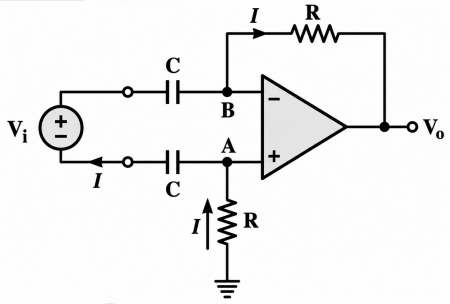
Substitute $s = j\omega = j(100\pi)$:

$$V_o = -j(100\pi)(10^{-6})(10^3)V_i = -j(0.1\pi)V_i$$

$$V_o = 0.1\pi V_i \angle -90^\circ \implies \delta = -90^\circ = -\frac{\pi}{2} \text{ rad}$$

Method 2

Using KVL and Virtual Short Concept:



By virtual short, $V_A = V_B$.

KVL in the input loop from node A to B:

$$I\left(\frac{1}{sC}\right) - V_i + I\left(\frac{1}{sC}\right) = 0 \implies I = \frac{V_i(sC)}{2}$$

$$V_A = -IR = -\frac{sCRV_i}{2} = V_B$$

Current relation at output node:

$$\frac{V_B - V_o}{R} = I \implies V_o = V_B - IR$$

$$V_o = -\frac{V_i(sCR)}{2} - \frac{V_i(sCR)}{2} = -sCRV_i$$

For $s = j\omega$: $V_o = -j\omega RCV_i$, which lags V_i by 90°.

$$\delta = -\frac{\pi}{2} \text{ rad}$$

$(D) \quad -\pi/2$

Key Points

Phasor Angle: Multiplication by $-j = 1\angle -90^\circ$ corresponds to a phase delay of 90° ($-\frac{\pi}{2}$ rad).

Q3.15.3 NAT 2015 1M Ans: 1.1 to 1.4 ● ● ○

Method 1

Given $V_i = 2\sin(2\pi \times 2000t) \implies \omega = 4000\pi$ rad/s.
Capacitive reactance:

$$X_c = \frac{1}{\omega C} = \frac{1}{4000\pi \times 0.1 \times 10^{-6}} = 795.77\,\Omega$$

Equivalent feedback impedance (Z_{eq}):

$$Z_{\text{eq}} = \frac{R \cdot (-jX_c)}{R - jX_c} = 622.67\angle -51.488^\circ\,\Omega$$

Output voltage magnitude:

$$V_o = V_{\text{in}}\left(\frac{|Z_{\text{eq}}|}{R_1}\right) = 2 \times \frac{622.67}{1000} = 1.245\,\text{V}$$

Method 2

Transfer function $H(s)$ of the inverting amplifier:

$$H(s) = \frac{-Z_f}{R_1} = \frac{-\left(\frac{R}{RCs+1}\right)}{R_1} = \frac{-1}{10^{-4}s + 1}$$

For $s = j\omega = j(4000\pi)$:

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (10^{-4} \times 4000\pi)^2}} = \frac{1}{\sqrt{1 + (0.4\pi)^2}} \approx 0.6226$$

Amplitude of output voltage V_o:

$$|V_o| = 2 \times |H(j\omega)| = 2 \times 0.6226 = 1.245\,\text{V}$$

$$\boxed{1.245}$$

Key Points

Output Amplitude: The magnitude of output voltage for a sinusoidal input is given by $|V_o| = V_m \cdot |H(j\omega)|$.

Q3.15.4 MCQ 2015 2M Ans: A ● ● ○

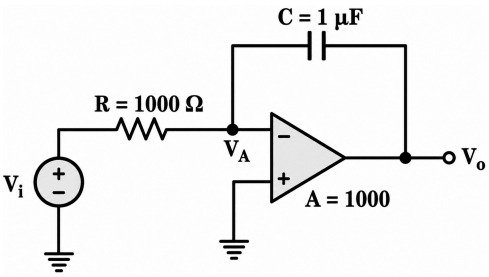


Fig. Solution Q3.15.4

Since $V_+ = 0\,\text{V}$, the inverting node voltage V_A relates to the output as:

$$V_o = A(V_+ - V_A) = -1000V_A \implies V_A = -\frac{V_o}{1000}$$

Applying KCL at node A in the s-domain:

$$\frac{V_A - V_i}{R} + \frac{V_A - V_o}{1/(sC)} = 0$$

$$\frac{V_A - V_i}{1000} = \big(V_A - (-1000V_A)\big)sC$$

$$V_A[1 - 1.001s] = V_i \implies V_A = \frac{V_i}{1 - 1.001s}$$

Substituting $V_A = -\dfrac{V_o}{1000}$ gives the overall transfer relation:

$$V_o(s) = \frac{-1000}{1 - 1.001s}V_i(s)$$

The time constant τ is :

$$\tau = 1.001\,\text{s} = 1001\,\text{ms}$$

$$\boxed{(A)\ \ 1001}$$

Mistakes to be avoided

- **For finite Gain Op-Amp**: Virtual ground $V_A \approx 0$ does not hold strictly; instead, $V_A = -V_o/A$. Avoid assuming ideal virtual ground ($V_A = 0$), which incorrectly yields $\tau = RC = 1\,\text{ms}$.
- Ensure unit conversion from seconds (1.001 s) to milliseconds (1001 ms).

Q3.15.5 MCQ | 2015 | 2M | Ans: A | ● ● ●

Method 1

The given circuit is an astable multivibrator (square wave generator).
Feedback factor β:

$$\beta = \frac{2\,\text{k}\Omega}{2\,\text{k}\Omega + 2\,\text{k}\Omega} = \frac{2}{4} = 0.5$$

Upper and lower threshold voltages:

$$V_{\text{UTP}} = +\beta V_{\text{sat}} = +5\,\text{V}, \quad V_{\text{LTP}} = -\beta V_{\text{sat}} = -5\,\text{V}$$

Capacitor charging equation for $V_o = +10\,\text{V}$:

$$V_c(t) = V_c(\infty) + [V_c(0) - V_c(\infty)]e^{-t/\tau}$$

where $V_c(\infty) = +10\,\text{V}$, $V_c(0) = -5\,\text{V}$, and $\tau = RC = 1\,\text{k}\Omega \times 0.25\,\mu\text{F} = 0.25\,\text{ms}$.

$$V_c(t) = 10 - 15e^{-t/\tau}$$

At $t = T_1$, $V_c(T_1) = 5\,\text{V}$:

$$5 = 10 - 15e^{-T_1/\tau} \implies T_1 = \tau \ln 3 = 0.25 \ln 3 \approx 0.275\,\text{ms}$$

Since UTP and LTP are symmetric, the duty cycle is 50%.
The total time period T is:

$$T = 2T_1 = 2 \times 0.275\,\text{ms} = 0.55\,\text{ms}$$

(A) square wave of period 0.55 ms.

Method 2

Direct formula for time period of an astable multivibrator:

$$T = 2RC \ln\left(\frac{1+\beta}{1-\beta}\right)$$

With $\beta = 0.5$ and $RC = 1\,\text{k}\Omega \times 0.25\,\mu\text{F} = 0.25\,\text{ms}$:

$$T = 2 \times 0.25 \times \ln\left(\frac{1+0.5}{1-0.5}\right) = 0.5\ln(3) \approx 0.55\,\text{ms}$$

Since the output is taken at node V_o of the op-amp, it switches between $+V_{\text{sat}}$ and $-V_{\text{sat}}$, producing a square wave.

Q3.15.6 MCQ 2015 2M Ans: B ●●○

Analyzing gain at extreme frequency limits:
At $\omega \to 0$ (Low Frequencies):

$$H_1(0) = 1, \quad H_2(0) = 0 \implies \left|\frac{V_o}{V_i}\right| \neq 0 \text{ (Passes)}$$

At $\omega \to \infty$ (High Frequencies):

$$H_1(\infty) = 0, \quad H_2(\infty) = 1 \implies \left|\frac{V_o}{V_i}\right| \neq 0 \text{ (Passes)}$$

At mid-band frequencies ($f_1 < f < f_2$):

$$H_1(f) \approx 0, \quad H_2(f) \approx 0 \implies \left|\frac{V_o}{V_i}\right| \approx 0 \text{ (Stops)}$$

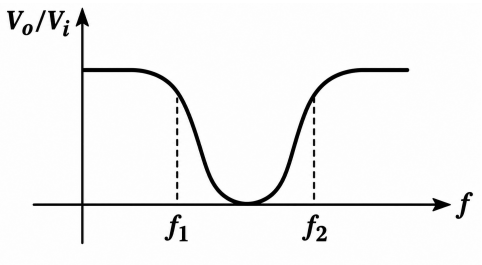


Fig. Solution Q3.15.6

Since low and high frequencies pass while mid-band frequencies are rejected, the overall response represents a **Band-stop filter**.

(B) Band-stop filter

Q3.14.1 MCQ 2014 1M Ans: D ● ○ ○

Stage 1 (First Op-Amp): Since it operates in positive feedback (a comparator/Schmitt trigger circuit), the output is saturated:

$$V_{o1} = +V_{\text{sat}} \quad \text{or} \quad -V_{\text{sat}}$$

Stage 2 (Second Op-Amp): The second stage is configured as a non-inverting amplifier with gain:

$$A_{v2} = 1 + \frac{R}{R} = 2$$

The calculated output for Stage 2 would be:

$$V_o = A_{v2} \cdot V_{o1} = 2(\pm V_{\text{sat}}) = \pm 2V_{\text{sat}}$$

However, the physical output of any op-amp cannot exceed its power supply limits ($\pm V_{\text{sat}}$).
Therefore,

$$V_o = +V_{\text{sat}} \quad \text{or} \quad -V_{\text{sat}}$$

$$\boxed{(D) \quad +V_{\text{sat}} \text{ or } -V_{\text{sat}}}$$

Key Points

- **Positive Feedback**: Causes an op-amp to operate non-linearly, forcing its output to either $+V_{\text{sat}}$ or $-V_{\text{sat}}$.
- **Op-Amp Saturation Limit**: The output voltage magnitude of an op-amp can never exceed its saturation voltage $|V_{\text{sat}}|$.

Q3.14.2 MCQ 2014 2M Ans: B ● ● ○

By virtual ground, the inverting terminals of the first stage op-amps are at potentials V_2 and V_1.
Current flowing downwards through resistor $2R$:

$$I = \frac{V_1 - V_2}{2R}$$

Voltages at the outputs of the input stage:

$$V_- = V_2 - IR = V_2 - \frac{V_1 - V_2}{2} = \frac{3V_2 - V_1}{2}$$

$$V_+ = V_1 + IR = V_1 + \frac{V_1 - V_2}{2} = \frac{3V_1 - V_2}{2}$$

The second stage is a difference amplifier with equal input/feedback resistors (R):

$$V_0 = V_+ - V_-$$

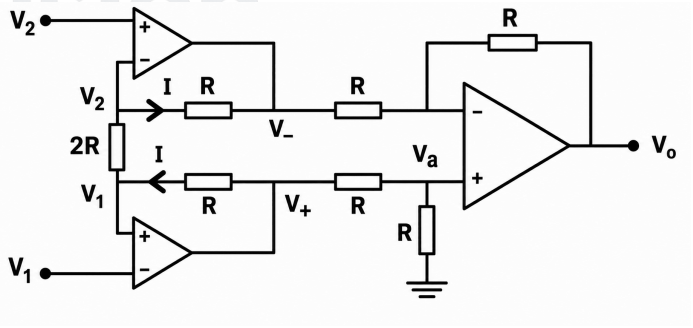


Fig. Solution Q3.14.2

Substituting V_+ and V_-:

$$V_0 = \frac{3V_1 - V_2}{2} - \frac{3V_2 - V_1}{2} = \frac{4V_1 - 4V_2}{2} = 2(V_1 - V_2)$$

$$\boxed{(B) \quad 2(V_1 - V_2)}$$

Q3.14.3 NAT | 2014 | 2M | Ans: 0.8 ●●●

During Δt_1 (positive cycle, $V_0 = +5$ V):

Top diode conducts $\Longrightarrow V_{\text{UTP}} = 5 \times \dfrac{1}{1+3} = 1.25$ V.

Capacitor charges from V_{LTP} towards $V_{\text{final}} = +5$ V:

$$V_c(\Delta t_1) = V_{\text{UTP}} = V_{\text{final}} + (V_{\text{LTP}} - V_{\text{final}})e^{-\Delta t_1/RC}$$

$$1.25 = 5 + (-2.5 - 5)e^{-\Delta t_1/RC}$$

$$\Longrightarrow e^{\Delta t_1/RC} = 2 \quad \text{— (i)}$$

During Δt_2 (negative cycle, $V_0 = -5$ V):

Bottom diode conducts $\Longrightarrow V_{\text{LTP}} = -5 \times \dfrac{1}{1+1} = -2.5$ V.

Capacitor discharges from V_{UTP} towards $V_{\text{final}} = -5$ V:

$$V_c(\Delta t_2) = V_{\text{LTP}} = V_{\text{final}} + (V_{\text{UTP}} - V_{\text{final}})e^{-\Delta t_2/RC}$$

$$-2.5 = -5 + (1.25 - (-5))e^{-\Delta t_2/RC}$$

$$e^{-\Delta t_2/RC} = \frac{2.5}{6.25} = 0.4$$

$$\Longrightarrow e^{\Delta t_2/RC} = \frac{5}{2} = 2.5 \quad \text{— (ii)}$$

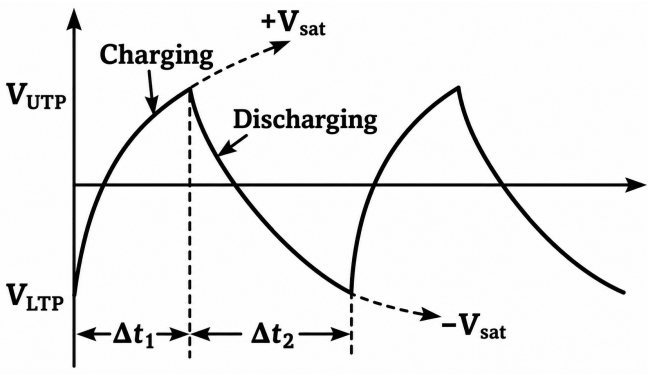


Fig. Solution Q3.14.3

Calculating the required ratio:

$$e^{\frac{\Delta t_1 - \Delta t_2}{RC}} = \frac{e^{\Delta t_1/RC}}{e^{\Delta t_2/RC}} = \frac{2}{2.5} = 0.8$$

$\boxed{0.8}$

Q3.14.4 MCQ | 2014 | 2M | Ans: C ●●○

Case I: When $V_i > 0$

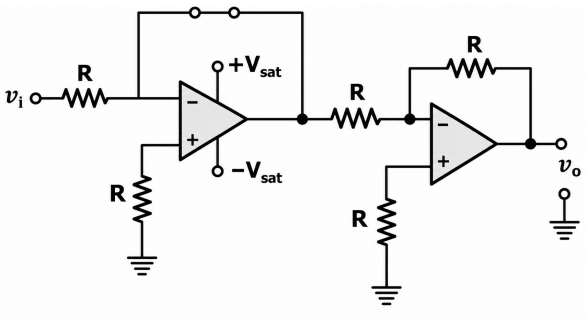


Fig. Solution Q3.14.4

- Output of the first op-amp tries to go negative.
- The feedback diode turns **ON** and the series output diode turns **OFF**.
- No current enters the second stage ($V_{in2} = 0\,V$).
- Therefore, output voltage:

$$V_0 = 0\,V$$

Case II: When $V_i < 0$

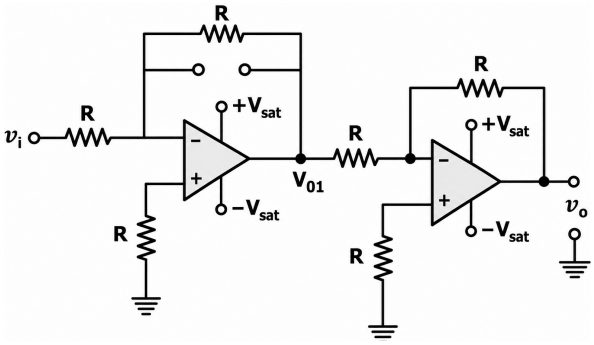


Fig. Solution Q3.14.4

- Output of the first op-amp goes positive.
- The feedback diode turns **OFF** and the series output diode turns **ON**.
- First stage acts as an inverting amplifier:

$$V_{o1} = -\left(\frac{R}{R}\right)V_i = -V_i$$

- Second stage is also an inverting amplifier:

$$V_0 = -\left(\frac{R}{R}\right)V_{o1} = -(-V_i) = V_i$$

Thus, $V_0 = \begin{cases} 0, & V_i > 0 \\ V_i, & V_i < 0 \end{cases}$ (Slope = 1 for $V_i < 0$).

(C)

Key Points

Precision Rectifier: Diode placed inside the feedback loop eliminates the standard diode forward voltage drop ($V_\gamma \approx 0.7\,V$).

Q3.13.1 MCQ | 2013 | 1M | Ans: B | ● ○ ○

Case 1: Assume Q is OFF (Cutoff)

- With Q cutoff, no feedback exists to the op-amp.
- The op-amp acts as an open-loop comparator:

$$V_- = 5\,\text{V}, \quad V_+ = 0\,\text{V}$$

- Since $V_- > V_+$, output saturates at negative supply:

$$V_{\text{out}} = -15\,\text{V}$$

- For transistor Q, base potential $V_B = 0\,\text{V}$ and emitter potential $V_E = V_{\text{out}} = -15\,\text{V}$.
- This gives $V_{BE} = V_B - V_E = 0 - (-15) = +15\,\text{V} > 0.7\,\text{V}$, which forward biases the base-emitter junction.
- Contradiction! Hence, Q cannot be in cutoff.

Case 2: Transistor Q is Active / ON

- With Q conducting, negative feedback path is established.
- Virtual ground concept holds at the inverting terminal:

$$V_- = V_+ = 0\,\text{V}$$

- Collector terminal potential $V_C = V_- = 0\,\text{V}$.
- Base terminal is directly grounded:

$$V_B = 0\,\text{V}$$

- Since $V_{BE} = V_B - V_E = 0.7\,\text{V}$ for a silicon transistor:

$$V_E = V_B - 0.7\,\text{V} = 0 - 0.7\,\text{V} = -0.7\,\text{V}$$

- Since the output V_{out} is connected directly to the emitter of Q:

$$V_{\text{out}} = -0.7\,\text{V}$$

$$\boxed{(B) \quad -0.7}$$

Mistakes to be avoided

Do not assume the op-amp is simply saturated without checking whether the feedback transistor turns ON.

Q3.13.2 MCQ 2013 2M Ans: C ●○○

Since no current enters the ideal op-amp inputs, non-inverting terminal voltage of first stage:

$$V_+ = +1\,\text{V}$$

By virtual short concept, $V_- = V_+ = +1\,\text{V}$.
Nodal analysis at the inverting node V_-:

$$\frac{1-(-2)}{1\,\text{k}\Omega} + \frac{1-V}{1\,\text{k}\Omega} = 0$$

$$3 + 1 - V = 0 \implies V = 4\,\text{V}$$

The second stage acts as a non-inverting amplifier with gain:

$$A_v = 1 + \frac{R_f}{R_1} = 1 + \frac{1\,\text{k}\Omega}{1\,\text{k}\Omega} = 2$$

Therefore, the final output voltage:

$$V_{\text{out}} = V \times 2 = 4 \times 2 = 8\,\text{V}$$

$$\boxed{(C) \quad 8}$$

Q3.12.1 MCQ 2012 2M Ans: B ●●○

- At $\omega = 0$ (DC): Capacitor C acts as an open circuit. No current flows through R_1 or R_2, so $V_{out} = 0\,V$.
- At $\omega = \infty$: C acts as a short circuit. The circuit behaves as an inverting amplifier with gain $-\frac{R_2}{R_1}$.
- Since low frequencies are blocked and high frequencies pass, it is a **High Pass Filter**.

Transfer Function & Cutoff Frequency:

$$\frac{V_{out}}{V_{in}} = -\frac{R_2}{R_1 + \frac{1}{sC}} = -\frac{sR_2C}{1 + sR_1C}$$

Substitute $s = j\omega$:

$$\left|\frac{V_{out}}{V_{in}}\right| = \frac{\omega R_2 C}{\sqrt{1 + \omega^2 R_1^2 C^2}}$$

At the 3 dB cutoff frequency f_{3dB} (or ω_{3dB}):

$$\omega_{3dB} R_1 C = 1 \implies f_{3dB} = \frac{1}{R_1 C}\ \text{rad/s}$$

$$\boxed{(B)\ \text{high pass filter with } f_{3dB} = \frac{1}{R_1 C}\ \text{rad/s}}$$

Key Points

- **3 dB Corner Frequency**: The frequency at which the magnitude drops to $\frac{1}{\sqrt{2}}$ of its maximum value, given by $\omega_{3dB} = \frac{1}{R_1 C}$ rad/s.

Q3.11.1 MCQ 2011 1M Ans: D ●○○

Cascaded Filter Analysis:

- **High-Pass Filter (HPF)**: Passes frequencies $f \geq 20$ Hz.
- **Low-Pass Filter (LPF)**: Passes frequencies $f \leq 30$ Hz.
- When cascaded, the overall frequency response is the intersection of both passbands:

$$20\,\text{Hz} \leq f \leq 30\,\text{Hz}$$

Conclusion: Since only a specific band of frequencies (20 Hz to 30 Hz) is allowed to pass while frequencies below 20 Hz and above 30 Hz are attenuated, the resultant system functions as a **band-pass filter**.

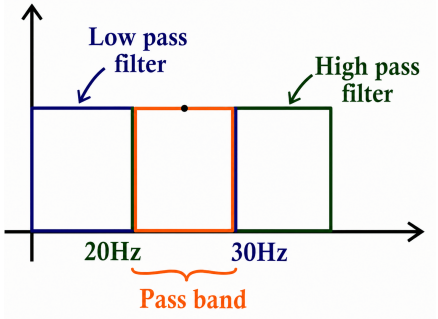


Fig. Solution Q3.11.1

(D) a band-pass filter

Mistakes to be avoided

Ensure that the HPF cut-off frequency is lower than the LPF cut-off frequency ($f_L < f_H$) to form a band-pass filter; if $f_L > f_H$, no band of frequencies is passed in cascade.

Q3.11.2 MCQ 2011 1M Ans: D ● ● ○

Stage 1: Differential Amplifier

$$V_{o1} = (V_2 - V_1) \times \left(\frac{R}{R}\right) = -V_i$$

The output of the first stage is $V_{o1} = -V_i$, giving a gain of -1.

Stage 2: Inverting Schmitt Trigger

The threshold voltages are:

$$V_{\text{UTP}} = +12 \times \left(\frac{R}{R+R}\right) = +6\,\text{V}$$

$$V_{\text{LTP}} = -12 \times \left(\frac{R}{R+R}\right) = -6\,\text{V}$$

For V_{o1}, the inverting Schmitt trigger switches to $+12$ V when V_{o1} goes below -6 V and to -12 V when V_{o1} exceeds $+6$ V.

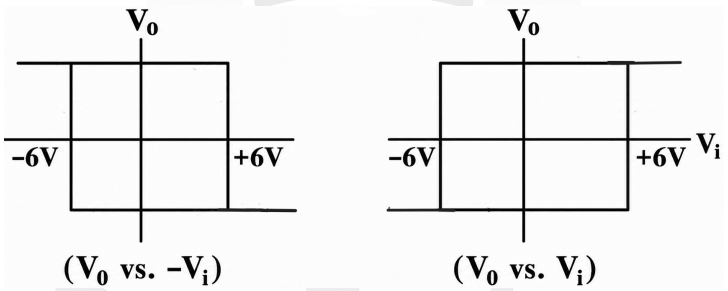


Fig. Solution Q3.11.2

(D)

Q3.10.1 MCQ 2010 1M Ans: B ● ○ ○

Method 1

By virtual short concept, the voltage at the inverting terminal is:

$$V_- = V_+ = +2\,\text{V}$$

Applying nodal analysis at the inverting terminal:

$$\frac{2-0}{R} + \frac{2-V_o}{2R} = 0$$

$$2 \times 2 + 2 - V_o = 0 \implies V_o = 6\,\text{V}$$

Method 2

The circuit acts as a standard non-inverting operational amplifier configuration with voltage gain:

$$A_v = 1 + \frac{R_f}{R_1} = 1 + \frac{2R}{R} = 3$$

Therefore, the output voltage is given by:

$$V_o = A_v \times V_{\text{in}} = (1+2) \times 2 = 6\,\text{V}$$

(B) 6 V

Q3.09.1 MCQ 2009 1M Ans: B ● ○ ○

Output Sampling Analysis:

- The feedback network is connected directly across the output node V_{out}.
- Sampling the output voltage directly corresponds to **voltage sampling** (node sampling).

Input Mixing Analysis:

- The feedback signal is fed back to the inverting terminal while input V_{in} is applied to the non-inverting terminal.
- Mixing at different nodes (series connection) corresponds to **series (voltage) mixing**.

Hence, it is a voltage-series or **voltage-voltage feedback** topology.

(B) voltage-voltage feedback

Q3.09.2 MCQ 2009 2M Ans: D ● ● ○

The current entering the non-inverting terminal from the source is:

$$i_s = \frac{V_s - V_o}{R}$$

By the virtual short concept, $V_- = V_+ = V_s$. Voltage divider at the inverting terminal yields:

$$V_s = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} V_o = \frac{V_o}{1 + j\omega CR}$$

$$\implies V_o = V_s(1 + j\omega CR)$$

Substituting V_o into the source current equation:

$$i_s = \frac{V_s - V_s(1 + j\omega CR)}{R} = -j\omega CV_s$$

Given $V_s = 10\,\text{V}$, $f = 50\,\text{Hz}$, $C = 10\,\mu\text{F}$:

$$i_s = -j(2\pi \times 50 \times 10 \times 10^{-6} \times 10)$$

$$i_s = -j10\pi\,\text{mA} = 10\pi\,\text{mA}\angle -90^\circ$$

(D) 10π mA lagging by 90°

Q3.09.3 MCQ 2009 2M Ans: D ● ● ○

This is an **inverting Schmitt trigger** circuit.
The upper and lower threshold voltages are calculated as:

$$V_{\text{UTP}} = +V_{\text{sat}} \times \frac{1}{2+1} = 6 \times \frac{1}{3} = 2\,\text{V}$$

$$V_{\text{LTP}} = -V_{\text{sat}} \times \frac{1}{1+2} = -3 \times \frac{1}{3} = -1\,\text{V}$$

Waveform Generation:

- Initially, $V_o = +6\,\text{V}$. The output remains at $+6\,\text{V}$ until V_i rises above $V_{\text{UTP}} = 2\,\text{V}$ at time t_2.
- At $t = t_2$, V_o switches to $-3\,\text{V}$ and stays there until V_i falls below $V_{\text{LTP}} = -1\,\text{V}$ at time t_4.

- At $t = t_4$, V_o switches back to +6 V.

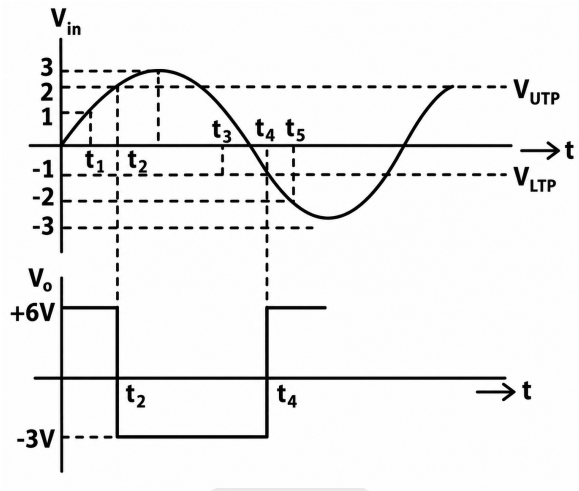


Fig. Solution Q3.09.3

(D)

Key Points

Asymmetric Saturation: Thresholds depend on saturation levels: $V_{\text{UTP}} = \dfrac{R_1}{R_1 + R_2} V_{\text{sat}}^{+}$ and $V_{\text{LTP}} = \dfrac{R_1}{R_1 + R_2} V_{\text{sat}}^{-}$.

Q3.08.1 MCQ 2008 2M Ans: B ●●○

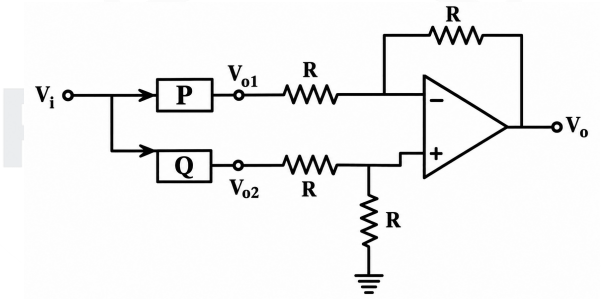


Fig. Solution Q3.08.1

For positive input cycle ($V_{\text{in}} > 0$):

$$P \rightarrow V_{01} = \text{Negative}, \quad Q \rightarrow V_{02} = 0$$

Since V_{01} is applied to the inverting terminal, V_o becomes positive due to inverting action:

$$V_o = -\left(\frac{R}{R}\right) V_{01} > 0$$

For negative input cycle ($V_{\text{in}} < 0$):

$$P \rightarrow V'_{01} = 0, \quad Q \rightarrow V'_{02} = \text{Positive}$$

Since V'_{02} is applied to the non-inverting terminal, V_o becomes positive due to non-inverting action:

$$V_o = \left(1 + \frac{R}{R}\right)\left(\frac{R}{R + R}\right) V'_{02} = V'_{02} > 0$$

Thus, the output V_o is always positive (full-wave rectified).

(B)

Q3.08.2 MCQ 2008 2M Ans: A ● ○ ○

The circuit consists of two main stages:

- **First Stage**: A Schmitt trigger acting as a square-wave generator.
- **Second Stage**: An op-amp integrator with a reference voltage v_i applied to its non-inverting terminal.

When $v_i = 0\,\text{V}$, the output at point 'P' is a symmetric triangular wave centered at 0 V with a peak-to-peak amplitude of 5 V (varying from −2.5 V to +2.5 V).

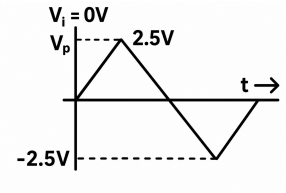


Fig. Solution Q3.08.2

Since the integrator is a linear system, applying a non-zero reference voltage $v_i = +2.5\,\text{V}$ introduces a DC offset. The entire triangular waveform shifts upward by an amount equal to v_i:

$$V_{\text{P, new}} = V_{\text{P, old}} + v_i$$

$$V_{\min} = -2.5\,\text{V} + 2.5\,\text{V} = 0\,\text{V}$$

$$V_{\max} = +2.5\,\text{V} + 2.5\,\text{V} = 5\,\text{V}$$

Thus, the waveform oscillates between 0 V and 5 V.

(A)

Key Points

Integrator DC Offset: A reference voltage at the non-inverting terminal of an op-amp integrator directly offsets the output voltage level without altering the peak-to-peak amplitude.

Q3.08.3 MCQ 2008 2M Ans: C ● ● ○

Using superposition for the circuit:
Voltage at the non-inverting terminal V_+:

$$V_+ = V_i\left(\frac{R_4}{R_3 + R_4}\right) = V_i\left(\frac{R_B}{R_B + R_B}\right) = \frac{V_i}{2}$$

Feedback impedance Z_f:

$$Z_f = R_2 \parallel \frac{1}{sC} = \frac{R_2}{1 + sCR_2}$$

Output voltage equation:

$$V_o = V_+\left(1 + \frac{Z_f}{R_1}\right) - V_i\left(\frac{Z_f}{R_1}\right)$$

$$V_o = \frac{V_i}{2}\left(1 - \frac{Z_f}{R_1}\right)$$

Substituting Z_f and $R_1 = R_2 = R_A$:

$$V_o = \frac{V_i}{2}\left(1 - \frac{1}{1 + sCR_A}\right) = \frac{V_i}{2}\left(\frac{sCR_A}{1 + sCR_A}\right)$$

Transfer function:

$$H(s) = \frac{V_o}{V_i} = \frac{1}{2}\left(\frac{sCR_A}{1 + sCR_A}\right)$$

Frequency response analysis:

- At $\omega = 0$ ($s = 0$): $|H(j0)| = 0$
- At $\omega \to \infty$ ($s \to \infty$): $|H(j\infty)| = \frac{1}{2}$

Since it attenuates low frequencies and passes high frequencies, the circuit acts as a **high pass filter**.

(C) high pass filter

Key Points

High-Pass Filter Transfer Function: Standard first-order high-pass response is given by $H(s) = \dfrac{K \cdot s/\omega_c}{1 + s/\omega_c}$.

Q3.08.4 MCQ 2008 2M Ans: D ●●○

The transfer function of the passive RC filter network is:

$$H_2(s) = \frac{V_o}{V_i} = \frac{\frac{1}{sC}}{\frac{R_A}{2} + \frac{1}{sC}} = \frac{1}{1 + sC\frac{R_A}{2}}$$

- At low frequencies ($\omega \to 0$), the capacitor is an open circuit, giving $|H_2| \to 1$.
- At high frequencies ($\omega \to \infty$), the capacitor is a short circuit, giving $|H_2| \to 0$.

Thus, this stage represents a **Low-Pass Filter**.

When cascaded with the High-Pass Filter from the previous question ($H_1(s) = \frac{1}{2} \cdot \frac{sCR_A}{1+sCR_A}$), the overall transfer function becomes:

$$H(s) = H_1(s) \cdot H_2(s) = \frac{1}{2}\left(\frac{sCR_A}{1 + sCR_A}\right)\left(\frac{1}{1 + sC\frac{R_A}{2}}\right)$$

This response attenuates both low ($\omega \to 0$) and high ($\omega \to \infty$) frequencies, peaking at an intermediate frequency. Therefore, it acts as a **Band-Pass Filter**.

(D)

Q3.07.1 MCQ 2007 1M Ans: D ●○○

The voltage at the non-inverting terminal (V_+) of the op-amp is determined by the voltage divider formed by R_1 and R_2:

$$V_+ = V\left(\frac{R_2}{R_1 + R_2}\right)$$

Due to virtual short concept under negative feedback:

$$V_- = V_+ = V\left(\frac{R_2}{R_1 + R_2}\right)$$

The current through the resistor r (which is equal to the load current I_L since no current enters the op-amp inverting terminal) is given by:

$$I_L = \frac{V_-}{r} = \frac{R_2}{R_1 + R_2} \cdot \frac{V}{r}$$

Since I_L is independent of the load resistance, the circuit acts as a **current source**.

(D) a current source with current $\dfrac{R_2}{R_1 + R_2} \cdot \dfrac{V}{r}$

Key Points

Op-Amp Voltage-Controlled Current Source: Using negative feedback, an op-amp maintains a fixed voltage across a current-sensing resistor r, forcing a constant load current independent of load impedance.

Q3.07.2 MCQ 2007 2M Ans: C ●○○

At $t = 0^-$, switch S is closed, shorting the capacitor to $-10\,\text{V}$, so $V_-(0^-) = -10\,\text{V}$.
Since $V_-(0^+) = -10\,\text{V} < V_+$, the initial output is clamped by the back-to-back Zener diodes:

$$V_{OUT}(0^+) = +5\,\text{V}$$

The non-inverting terminal voltage V_+ is:

$$V_+ = V_{OUT}\left(\frac{100\,\text{k}\Omega}{100\,\text{k}\Omega + 10\,\text{k}\Omega}\right) = \frac{50}{11}\,\text{V} \approx 4.545\,\text{V}$$

Time constant of the charging circuit:

$$\tau = RC = (1\,\text{k}\Omega)(0.01\,\mu\text{F}) = 10\,\mu\text{s}$$

Thus, V_{OUT} transitions from $+5\,\text{V}$ to $-5\,\text{V}$ at $t = 12.98\,\mu\text{s}$.

For $t > 0$, the capacitor charges towards $+10\,\text{V}$:

$$V_-(t) = V_f + (V_i - V_f)e^{-t/\tau}$$

$$= 10 + (-10 - 10)e^{-t/\tau}$$

$$V_-(t) = 10 - 20e^{-10^5 t}$$

The output V_{OUT} switches from $+5\,\text{V}$ to $-5\,\text{V}$ when $V_-(t)$ reaches V_+:

$$\frac{50}{11} = 10 - 20e^{-10^5 t}$$

$$e^{10^5 t} = \frac{220}{60} \implies 10^5 t = \ln\left(\frac{11}{3}\right) \approx 1.2993$$

$$t = 12.98\,\mu\text{s}$$

(C)

Key Points

- **Comparator Thresholding**: The transition occurs when the charging capacitor voltage at the inverting terminal crosses the non-inverting reference voltage V_+.
- **Zener Clamping**: Back-to-back Zener diodes limit the output voltage swing to $\pm V_Z$.

Q3.06.1 MCQ 2006 1M Ans: D ●○○

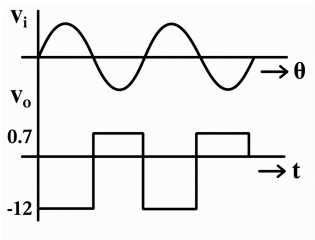
1. During positive half-cycle ($V_{in} > 0$):

- The inverting terminal potential is higher than the non-inverting terminal ($V_- > V_+$).
- The op-amp output attempts to go to negative saturation ($V_P = -V_{sat} = -12\,\text{V}$).
- With $V_P = -12\,\text{V}$, the diode connected to point P is reverse biased (OFF).
- Hence, no feedback path exists, and the output remains fixed at $V_P = -12\,\text{V}$.

Therefore, the voltage waveform at P toggles between $+0.7\,\text{V}$ and $-12\,\text{V}$.

2. During negative half-cycle ($V_{in} < 0$):

- The non-inverting terminal potential is higher than the inverting terminal ($V_+ > V_-$).
- The op-amp output goes positive, forward biasing the diode (ON).
- Negative feedback is activated, clamping the voltage at point P to the diode forward drop above virtual ground ($0.7\,\text{V}$).



(D)

Q3.06.2 MCQ 2006 2M Ans: A ● ○ ○

Point P is the non-inverting input, connected to the diode-selected resistive divider. It is not the capacitor node at the inverting input.
When $v_o = +12$ V, the lower 10 kOhm divider branch conducts. Thus

$$V_P = 12\frac{10}{10+10} = +6 \text{ V}.$$

When $v_o = -12$ V, the upper 2 kOhm branch conducts instead:

$$V_P = -12\frac{10}{2+10} = -10 \text{ V}.$$

Therefore P switches between +6 V and -10 V: the square wave in $\boxed{(A)}$. The capacitor voltage is continuous and exponential between those thresholds, but that waveform is at a different node. In each interval it charges toward the corresponding output rail (+12 or -12 V), not toward the threshold itself.

Q3.05.1 MCQ 2005 2M Ans: D ● ● ○

The voltage gain of the inverting amplifier is:

$$A_v = \frac{V_{\text{out}}}{V_{\text{in}}} = -\frac{R_F}{R_1}$$

The input resistance seen by the source is:

$$R_{\text{in}} = R_1$$

To maximize R_{in}, R_1 must be made as large as possible without violating the gain specification $-25 \le A_v \le -10$.
With $R_F \le 1\,\text{M}\Omega = 1000\,\text{k}\Omega$:

- For a gain magnitude of 25:

$$R_1 = \frac{R_F}{25} \le \frac{1000\,\text{k}\Omega}{25} = 40\,\text{k}\Omega$$

- For a gain magnitude of 10:

$$R_1 = \frac{R_F}{10} \le \frac{1000\,\text{k}\Omega}{10} = 100\,\text{k}\Omega$$

To ensure that the full required gain magnitude range up to 25 can be achieved with $R_F \le 1\,\text{M}\Omega$, R_1 cannot exceed $40\,\text{k}\Omega$.
If $R_1 = 100\,\text{k}\Omega$, the maximum achievable magnitude of gain is $\frac{1\,\text{M}\Omega}{100\,\text{k}\Omega} = 10$, failing to reach a gain of -25.
Therefore, the maximum permissible value of R_1 is $40\,\text{k}\Omega$.

$$\boxed{(D)\ 40\ \text{k}\Omega}$$

Key Points

- **Inverting Amplifier Input Resistance**: The input impedance seen by V_{in} is R_1 due to the virtual ground at the inverting node.
- **Upper Limit Constraint**: Maximize R_1 while ensuring $R_F = R_1 \times |A_v|_{\max} \le R_{F,\max}$.

Mistakes to be avoided

- Do not choose $R_1 = 100\,\text{k}\Omega$ by evaluating only the minimum gain condition, as it makes higher gains impossible under the $R_F \le 1\,\text{M}\Omega$ constraint.

Q3.05.2 MCQ 2005 2M Ans: D ● ○ ○

The op-amp operates in open-loop without negative feedback, acting as a comparator between terminals:

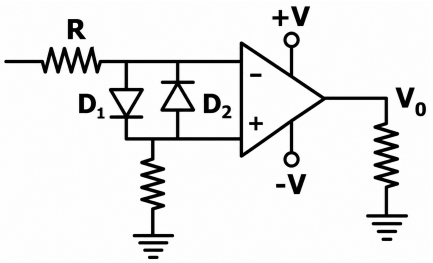


Fig. Solution Q3.05.2

1. Positive Half-Cycle ($V_{in} > 0$):

- Diode D_1 is forward biased (ON) and D_2 is reverse biased (OFF).
- D_1 holds the inverting terminal V_- higher than the non-inverting terminal V_+ by $V_D \approx 0.7\,\text{V}$ ($V_- > V_+$).
- Output saturates to negative supply:

$$V_o = -V_{sat}$$

2. Negative Half-Cycle ($V_{in} < 0$):

- Diode D_2 is forward biased (ON) and D_1 is reverse biased (OFF).
- D_2 holds V_+ higher than V_- by $V_D \approx 0.7\,\text{V}$ ($V_+ > V_-$).
- Output saturates to positive supply:

$$V_o = +V_{sat}$$

The output is a square wave inverted relative to V_{in} (a phase shift of 180°).

(D)

Q3.04.1 MCQ 2004 2M Ans: B ● ○ ○

Since no current enters the ideal op-amp terminal ($I_+ = 0\,\text{A}$), the current I_x flows entirely through resistor R_3:

$$I_x = \frac{V_x - V_y}{R_3}$$

Rearranging gives:

$$V_x - V_y = 10^6 \times I_x$$

The circuit acts as a non-inverting amplifier with input V_x at the non-inverting node:

$$V_y = V_x \left(1 + \frac{100\,\text{k}\Omega}{10\,\text{k}\Omega}\right) = 11V_x$$

Substituting $V_y = 11V_x$ into the current relation:

$$V_x - 11V_x = 10^6 I_x$$

$$-10V_x = 10^6 I_x$$

The equivalent input resistance R_{IN} is:

$$R_{IN} = \frac{V_x}{I_x} = \frac{-10^6}{10}\,\Omega = -10^5\,\Omega = -100\,\text{k}\Omega$$

(B) $-100\,\text{k}\Omega$

Q3.04.2 MCQ 2004 2M Ans: A ●●○

Warning: With the values printed in the question figure, the pole frequency is 10000 rad/s, which none of the supplied options lists.

Let v_b be the junction of the two capacitors and R2. The ideal op-amp has its inverting node at zero. Its KCL gives $v_o = -sR_1C_1v_b$. KCL at the capacitor junction gives

$$sC_1v_b + sC_2(v_b - v_o) + v_b/R_2 = 0.$$

Eliminating v_b yields the characteristic polynomial

$$s^2 + as + b = 0,$$

where

$$a = \frac{C_1 + C_2}{R_1C_1C_2},$$

$$b = \frac{1}{R_1R_2C_1C_2}.$$

Thus $\omega_0/Q = (C_1 + C_2)/(R_1C_1C_2)$. With $Q = 1$, $R_1 = 200$ kOhm and $C_1 = C_2 = 1$ nF,

$$\omega_0 = \frac{2 \times 10^{-9}}{2 \times 10^{-13}} = \boxed{10000 \text{ rad/s}}.$$

Q3.03.1 MCQ 2003 1M Ans: D ●●○

The voltage at the non-inverting terminal is obtained using the voltage divider rule:

$$V_+ = V_{\text{in}}\left(\frac{\frac{1}{sC}}{R + \frac{1}{sC}}\right) = \frac{V_{\text{in}}}{1 + sCR}$$

Using superposition, the output voltage V_{out} is:

$$V_{\text{out}} = -V_{\text{in}}\left(\frac{R_1}{R_1}\right) + V_+\left(1 + \frac{R_1}{R_1}\right)$$

$$V_{\text{out}} = -V_{\text{in}} + 2V_+ = V_{\text{in}}\left[\frac{2}{1 + sCR} - 1\right]$$

$$V_{\text{out}} = V_{\text{in}}\left[\frac{1 - sCR}{1 + sCR}\right]$$

Substituting $s = j\omega$ to determine the frequency response:

$$\frac{V_{\text{out}}}{V_{\text{in}}} = \frac{1 - j\omega CR}{1 + j\omega CR}$$

The phase shift $\phi(\omega)$ of V_{out} relative to V_{in} is:

$$\phi(\omega) = -\tan^{-1}(\omega CR) - \tan^{-1}(\omega CR) = -2\tan^{-1}(\omega CR)$$

As frequency varies from $\omega \to 0$ to $\omega \to \infty$, the maximum phase shift produced by the filter ranges across $\pm 180°$:

$$\phi_{\max} = \pm 2 \times 90° = \pm 180°$$

$$\boxed{(D) \quad \pm 180°}$$

Key Points

First-Order All-Pass Filter: Provides a constant gain of unity ($|H(j\omega)| = 1$) across all frequencies while introducing a frequency-dependent phase shift ranging from 0° to −180° (or ±180°).

Q3.03.2 MCQ 2003 2M Ans: D ● ○ ○

By virtual ground concept, the voltage at the inverting terminal is $V_- = 0\,\text{V}$. Applying KCL at the inverting node:

$$\frac{V_{in} - 0}{1\,\text{k}\Omega} + \frac{0 - V_1}{10\,\text{k}\Omega} = 0$$

$$\frac{V_{in}}{1} = \frac{V_1}{10} \implies V_1 = -10V_{in} \quad \text{— (i)}$$

Applying KCL at node 1 (junction of resistors):

$$\frac{V_1 - 0}{10\,\text{k}\Omega} + \frac{V_1}{1\,\text{k}\Omega} + \frac{V_1 - V_{out}}{10\,\text{k}\Omega} = 0$$

Multiplying the KCL equation at node 1 by 10:

$$V_1 + 10V_1 + V_1 - V_{out} = 0$$

$$12V_1 = V_{out} \quad \text{— (ii)}$$

Substituting equation (i) into equation (ii):

$$V_{out} = 12(-10V_{in}) = -120V_{in}$$

$$\frac{V_{out}}{V_{in}} = -120$$

$$\boxed{(D) \quad -120}$$

Q3.02.1 MCQ 2002 2M Ans: A ● ● ○

When the output voltage is $V_o = +15\,\text{V}$, the non-inverting terminal voltage V_+ is determined by superposition:

$$V_{UTP} = 15 \times \frac{3\,\text{k}\Omega}{10\,\text{k}\Omega + 3\,\text{k}\Omega} + 2 \times \frac{10\,\text{k}\Omega}{10\,\text{k}\Omega + 3\,\text{k}\Omega}$$

$$V_{UTP} = \frac{45 + 20}{13} = 5\,\text{V}$$

When the output voltage is $V_o = -15\,\text{V}$:

$$V_{LTP} = -15 \times \frac{3}{13} + 2 \times \frac{10}{13} = -\frac{25}{13}\,\text{V}$$

Since the input $v_{in} = 10\sin(\omega t)$ is applied to the inverting terminal, the output switches from +15 V to −15 V when v_{in} exceeds $V_{UTP} = 5\,\text{V}$.

This threshold crossing requires v_{in} to be increasing, which corresponds to the **positive slope** only.

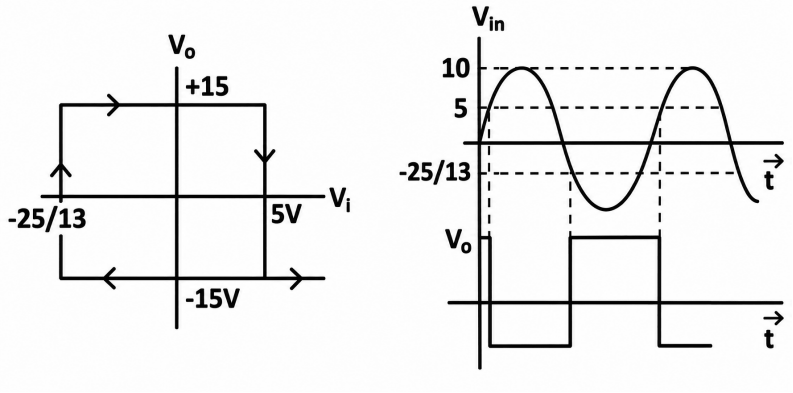


Fig. Solution Q3.02.1

(A) 5 V in the positive slope only

Key Points

Inverting Schmitt Trigger: The upper threshold V_{UTP} triggers a transition from high output ($+V_{sat}$) to low output ($-V_{sat}$) when the input rises past it with a positive slope.

Q3.02.2 Descriptive 2002 5M Ans: $2K\,\Omega, 159.15KHz$ ● ● ○

Applying Nodal Analysis at intermediate node V:

$$\frac{V-V_x}{R}+\frac{V-V_y}{\frac{1}{sC}}+\frac{V}{\frac{1}{sC}}=0 \quad -(1)$$

By potential division for V_y from V:

$$V_y=V\left(\frac{R}{R+\frac{1}{sC}}\right) \implies V=V_y\left(\frac{1+sCR}{sCR}\right)$$

Substituting V into equation (1):

$$\frac{V-V_x}{R}+(V-V_y)sC+VsC=0$$

$$\left(\frac{1+sCR}{sCR}\right)(1+2sCR)V_y=V_x+V_ysCR$$

$$\frac{V_y}{V_x}=\frac{sCR}{1+3sCR+(sCR)^2}$$

For frequency response, substitute $s=j\omega$:

$$\beta(j\omega)=\frac{j\omega CR}{1-\omega^2C^2R^2+j3\omega CR}$$

For sustained oscillations, the phase shift of $\beta(j\omega)$ must be 0°, so the imaginary part of the denominator must vanish:

$$1-\omega^2C^2R^2=0 \implies \omega=\frac{1}{RC}$$

At the frequency of oscillation ω:

$$\beta=\frac{j\omega CR}{j3\omega CR}=\frac{1}{3}$$

Applying Barkhausen criterion ($A\beta=1$):

$$A=1+\frac{R_F}{1\,\text{k}\Omega}=3 \implies R_F=2\,\text{k}\Omega$$

Frequency of oscillation f:

$$f=\frac{1}{2\pi RC}=\frac{1}{2\pi\times 10\,\text{k}\Omega\times 100\,\text{pF}}=159.15\,\text{kHz}$$

2 kΩ, 159.15 kHz

Key Points

Wien Bridge / RC Lead-Lag Oscillator: The feedback network provides a maximum feedback factor of $\beta=1/3$ with 0° phase shift at $\omega_0=1/(RC)$, requiring a non-inverting amplifier gain of $A=3$.

Q3.01.1 MCQ 2001 1M Ans: C ● ○ ○

The gain-bandwidth product (GBW) of an operational amplifier remains constant:

$$A_1 \times f_{c1} = A_2 \times f_{c2}$$

Given parameters:

- Open-loop gain, $A_1 = 10^5$
- Open-loop upper cutoff frequency, $f_{c1} = 10\,\text{Hz}$
- Closed-loop gain, $A_2 = 100$

Substituting the values:

$$10^5 \times 10\,\text{Hz} = 100 \times f_{c2}$$

$$f_{c2} = \frac{10^6}{10^2} = 10^4\,\text{Hz} = 10\,\text{kHz}$$

Due to negative feedback, the gain decreases while the bandwidth increases proportionally.

$$\boxed{(c)\ 10\,\text{kHz}}$$

Key Points

Gain-Bandwidth Product Conservation: For a single-pole op-amp, $A_{\text{OL}} \cdot f_H = \text{GBW} = \text{constant}$. Introducing negative feedback trades gain for an equal factor increase in bandwidth.

Q3.01.2 MCQ 2001 2M Ans: B ● ● ○

The non-inverting terminal voltage V_+ via potential divider is:

$$V_+ = V_\text{o} \times \frac{R}{R+R} = \frac{V_\text{o}}{2}$$

Assuming $V_\text{o} = -V_\text{sat}$, then $V_+ = -\frac{V_\text{sat}}{2}$.

When V_c drops below $-\frac{V_\text{sat}}{2}$, V_o switches to $+V_\text{sat}$, giving $V_+ = +\frac{V_\text{sat}}{2}$.

The capacitor charges towards $V_\text{c}(\infty) = +V_\text{sat}$ from $V_\text{c}(0) = -\frac{V_\text{sat}}{2}$:

$$V_\text{c}(t) = V_\text{c}(\infty) + [V_\text{c}(0) - V_\text{c}(\infty)]e^{-t/\tau}$$

$$V_\text{c}(t) = V_\text{sat} - \frac{3}{2}V_\text{sat}e^{-t/\tau}$$

The output switches state again when $V_\text{c}(t) = \frac{V_\text{sat}}{2}$:

$$\frac{V_\text{sat}}{2} = V_\text{sat} - \frac{3}{2}V_\text{sat}e^{-t/\tau}$$

$$\frac{3}{2}e^{-t/\tau} = \frac{1}{2} \implies e^{t/\tau} = 3 \implies t = \tau \ln 3$$

Since charging and discharging intervals are equal, the total time period T is:

$$T = 2t = 2\tau \ln 3$$

$$\boxed{(B)\ 2\tau \ln 3}$$

Key Points

Astable Multivibrator: For an op-amp relaxation oscillator with feedback factor $\beta = \frac{R_2}{R_1+R_2}$, the time period of oscillation is given by $T = 2\tau \ln\left(\frac{1+\beta}{1-\beta}\right)$. Here, $\beta = \frac{1}{2}$, giving $T = 2\tau \ln 3$.

Q3.01.3 MCQ 2001 2M Ans: A ●○○

To avoid distortion, the maximum rate of change of the output voltage must not exceed the slew rate (S.R.):

$$\text{S.R.} \geq \left(\frac{dV_o}{dt}\right)_{max}$$

For a sinusoidal output $V_o(t) = V_m \sin(\omega t)$:

$$\left(\frac{dV_o}{dt}\right)_{max} = V_m\omega = 2\pi f V_m$$

Given S.R. $= 62.8\ \text{V}/\mu\text{s} = 62.8 \times 10^6\ \text{V/s}$ and $V_m = 10$ V:

$$62.8 \times 10^6 \geq 2\pi f \times 10$$

$$f \leq \frac{62.8 \times 10^6}{2\pi \times 10} = 10^6\ \text{Hz} = 1\ \text{MHz}$$

Thus, the minimum frequency at which slew-rate limiting distortion sets in is 1 MHz.

(A) 1 MHz

Mistakes to be avoided

Always convert the slew rate from V/μs to V/s by multiplying by 10^6 before calculating frequency in Hz.

Q3.01.4 NAT 2001 5M Ans: 0.4 ●●○

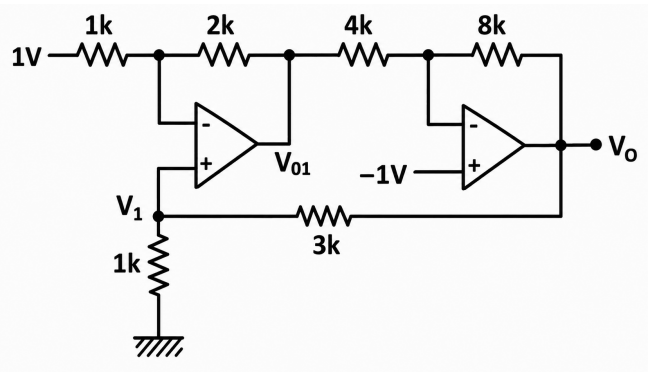


Fig. Solution Q3.01.4

Applying superposition for the 1st Op-Amp:

$$V_{o1} = V_1\left(\frac{2}{1}+1\right) + 1\times\left(-\frac{2}{1}\right) = 3V_1 - 2$$

From the voltage divider network at the non-inverting terminal:

$$V_1 = V_o \times \left(\frac{1}{1+3}\right) = \frac{V_o}{4}$$

Substituting V_1 into V_{o1}:

$$V_{o1} = 3\left(\frac{V_o}{4}\right) - 2 = \frac{3}{4}V_o - 2$$

Applying superposition for the 2nd Op-Amp:

$$V_o = V_{o1}\left(-\frac{8}{4}\right) + (-1)\left(1+\frac{8}{4}\right) = -2V_{o1} - 3$$

Substituting V_{o1} into the equation for V_o:

$$V_o = -2\left(\frac{3}{4}V_o - 2\right) - 3$$

$$V_o = -\frac{3}{2}V_o + 4 - 3 \implies \left(1+\frac{3}{2}\right)V_o = 1$$

$$\frac{5}{2}V_o = 1 \implies V_o = \frac{2}{5} = 0.4\,\text{V}$$

$$\boxed{0.4}$$

Q3.01.5 Descriptive | 2001 | 5M | Ans: $100K\Omega, 1M\Omega, 1.5915nF$ | ●●○

Applying nodal analysis at the inverting terminal:

$$\frac{0-V_i}{R_1+\frac{1}{sC}} + \frac{0-V_o}{R_2} = 0$$

The transfer function is:

$$H(s) = \frac{V_o}{V_i} = \frac{-R_2}{R_1+\frac{1}{sC}} = \frac{-sCR_2}{1+sCR_1}$$

For $s = j\omega$, the magnitude response $|A(\omega)|$ is:

$$|A(\omega)| = \left|\frac{V_o}{V_i}\right| = \frac{\omega CR_2}{\sqrt{1+(\omega CR_1)^2}}$$

At $\omega = 0$, $A(0) = 0$; and at $\omega \to \infty$, $A(\infty) = \frac{R_2}{R_1}$. Thus, the circuit acts as a **High-Pass Filter**.

At high frequencies, C behaves as a short circuit, so input resistance is $R_1 = 100\,\text{k}\Omega$.
Given 3-dB cutoff frequency $f_c = 1\,\text{kHz} = 10^3\,\text{Hz}$:

$$f_c = \frac{1}{2\pi R_1 C} = 10^3$$

$$C = \frac{1}{2\pi\times 100\times 10^3\times 10^3} \approx 1.5915\,\text{nF}$$

High frequency gain magnitude $|A(\infty)| = \left|-\frac{R_2}{R_1}\right| = 10$:

$$R_2 = 10R_1 = 10\times 100\,\text{k}\Omega = 1\,\text{M}\Omega$$

$$\boxed{R_1 = 100\,\text{k}\Omega, R_2 = 1\,\text{M}\Omega, C = 1.5915\,\text{nF}}$$

Key Points

Active High-Pass Filter: The 3-dB corner frequency depends on the input RC network $\omega_c = \frac{1}{R_1C}$, while the high-frequency passband gain is governed by $A_v = -\frac{R_2}{R_1}$.

Q3.00.1 MCQ | 2000 | 1M | Ans: D | ●○○

The feedback factor β is defined as the ratio of the feedback voltage V_f (voltage at the non-inverting terminal V_+) to the output voltage V_o.
Using the potential divider rule across the 10 kΩ and 90 kΩ resistors:

$$V_f = V_+ = V_o\left(\frac{10\,\text{k}\Omega}{10\,\text{k}\Omega + 90\,\text{k}\Omega}\right)$$

Simplifying the expression for β:

$$\beta = \frac{V_f}{V_o} = \frac{10}{100} = \frac{1}{10}$$

$$\boxed{\text{(D) } \frac{1}{10}}$$

Q3.00.2 MCQ 2000 1M Ans: D ● ○ ○

Due to virtual ground concept:

$$V_a = V_+ = 0\,\text{V}$$

Since no current enters the ideal op-amp terminal, the entire 1 mA current flows through the 1 kΩ feedback resistor toward the output. The output voltage V_o is given by:

$$V_o = V_a - (1\,\text{mA} \times 1\,\text{k}\Omega)$$

$$V_o = 0\,\text{V} - 1\,\text{V} = -1\,\text{V}$$

$$\boxed{(D)\ \text{-1 V}}$$

Q3.00.3 Descriptive 2000 5M Ans: 10^3 rad/sec, −50, 10 ● ● ○

The transfer function of the filter is:

$$T(s) = \frac{-\frac{s}{R_1 C}}{s^2 + \frac{2s}{R_3 C} + \frac{1}{RR_3C^2}}$$

Comparing denominator with $s^2 + 2\xi\omega_0 s + \omega_0^2$:

$$R = R_1 \parallel R_2 = \frac{2 \times \frac{2}{3}}{2 + \frac{2}{3}} = 0.5\,\text{k}\Omega$$

$$\omega_0 = \frac{1}{\sqrt{RR_3C^2}} = \frac{1}{C\sqrt{RR_3}}$$

$$\omega_0 = \frac{1}{10^{-7}\sqrt{0.5 \times 10^3 \times 200 \times 10^3}} = 10^3\ \text{rad/s}$$

At resonance $s = j\omega_0$, center frequency gain A_0 is:

$$A_0 = T(j\omega_0) = \frac{-\frac{j\omega_0}{R_1 C}}{\frac{2j\omega_0}{R_3 C}} = -\frac{R_3}{2R_1}$$

$$A_0 = -\frac{200\,\text{k}\Omega}{2 \times 2\,\text{k}\Omega} = -50$$

Comparing damping term $2\xi\omega_0 = \frac{2}{R_3 C}$:

$$Q = \frac{1}{2\xi} = \frac{\omega_0 R_3 C}{2} = \frac{10^3 \times 200 \times 10^3 \times 0.1 \times 10^{-6}}{2} = 10$$

$$\boxed{\omega_0 = 10^3\ \text{rad/s},\ A_0 = -50,\ Q = 10}$$

Q3.99.1 NAT 1999 5M Ans: * ● ● ○

Case 1: Diode OFF When $V_i > 0$, the inverting gain gives:

$$V_{o1} = -V_i\left(\frac{5\,\text{k}\Omega}{1\,\text{k}\Omega}\right) = -5V_i < 0$$

Since cathode voltage $V_1 = 0\,\text{V}$ and anode voltage $V_{o1} < 0$, the diode remains reverse-biased (OFF). Thus:

$$V_1 = 0\,\text{V} \quad \text{for } V_i > 0$$

Case 2: Diode ON When $V_i < 0$, V_{o1} tries to go positive, forward-biasing the diode (ON). The equivalent feedback resistance is $5\,\text{k}\Omega \parallel 10\,\text{k}\Omega = \frac{10}{3}\,\text{k}\Omega$.

$$V_1 = V_{o1} = -V_i\left(\frac{\frac{10}{3}\,\text{k}\Omega}{1\,\text{k}\Omega}\right) = -3.33V_i$$

Peak value of V_1 occurs when $V_i = -1\,\text{V}$:

$$V_{1,\text{peak}} = 3.33\,\text{V}$$

The low-pass RC filter extracts the average (DC) value V_0 of the half-wave rectified voltage waveform V_1:

$$V_0 = \frac{1}{2\pi}\int_{\pi}^{2\pi} 3.33\sin(\theta)\,d\theta = \frac{3.33}{\pi} \approx 1.06\,\text{V}$$

$$V_{1,\text{peak}} = 3.33\,\text{V},\ V_0 = 1.06\,\text{V}$$

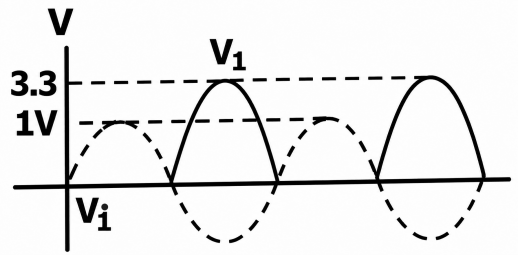


Fig. Solution Q3.99.1

Key Points

- **Precision Half-Wave Rectifier**: The diode in the feedback loop conducts only during negative half-cycles of V_i, producing non-zero V_1 only when $V_i < 0$.
- **Average DC Output**: The RC network operates as an integrator/low-pass filter to output $V_0 = V_{1,\text{avg}} = \frac{V_{1,\text{peak}}}{\pi}$.

Q3.98.1 Descriptive | 1998 | 2M | Ans: A-R, B-S, C-P | ● ○ ○

Circuit A: No feedback is provided between output and input terminals. Hence, the op-amp operates in open loop and acts as a **Comparator (R)**.

Circuit B: At $\omega = 0$ (DC), capacitors act as open circuits:

$$\text{Gain}_{\omega=0} = -\frac{R_f}{R_1} \neq 0$$

At $\omega = \infty$ (high frequency), capacitors act as short circuits, bypassing the feedback resistor:

$$V_o = 0 \implies \text{Gain}_{\omega=\infty} = 0$$

Passes low frequencies and attenuates high frequencies, acting as a **Low-pass filter (S)**.

Circuit C: At $\omega = 0$ (DC), input series capacitors act as open circuits:

$$\text{Gain}_{\omega=0} = 0$$

At $\omega = \infty$ (high frequency), capacitors act as short circuits:

$$\text{Gain}_{\omega=\infty} \neq 0$$

Passes high frequencies and blocks low frequencies, acting as a **High-pass filter (P)**.

$$\boxed{\text{A} \rightarrow \text{R},\ \text{B} \rightarrow \text{S},\ \text{C} \rightarrow \text{P}}$$

Q3.98.2 Descriptive | 1998 | 5M | Ans: $10, 1, 10^{10}\Omega$ | ● ● ○

Calculation of V_s and Gain $\frac{V_0}{V_s}$:

Given: $R_i = 100\,\text{k}\Omega$, $R_0 = 50\,\Omega$, $R_\text{L} = 5\,\text{k}\Omega$, $R_\text{s} = 2\,\text{k}\Omega$, $A = 10^5$, $V_0 = 10\,\text{V}$.
Applying KCL at the output node:

$$\frac{V_0}{R_\text{L}} + \frac{V_0 - AV_i}{R_0} - I_i = 0$$

Since $V_i = I_i R_i$:

$$V_0\left(\frac{1}{R_\text{L}} + \frac{1}{R_0}\right) = I_i\left(\frac{AR_i}{R_0} + 1\right)$$

Substituting values ($V_0 = 10\,\text{V}$):

$$10\left(\frac{1}{5000} + \frac{1}{50}\right) = I_i\left(\frac{10^5 \times 10^5}{50} + 1\right)$$

$$I_i \approx 1.01 \times 10^{-9}\,\text{A}$$

Now, using input loop equation:

$$V_\text{s} = I_i(R_\text{s} + R_i) + V_0 \approx 10\,\text{V}$$

$$\text{Voltage Gain } \frac{V_0}{V_\text{s}} = \frac{10}{10} = 1$$

Calculation of Input Resistance Z_in:

Using equivalent model at input node:

$$V_\text{in} = V_i + V_0 = I_i R_i + V_0$$

Expressing V_0 in terms of I_i:

$$V_0 = I_i \frac{\left(\frac{AR_i}{R_0} + 1\right)}{\left(\frac{1}{R_\text{L}} + \frac{1}{R_0}\right)}$$

Substituting into V_in:

$$Z_\text{in} = \frac{V_\text{in}}{I_i} = R_i + \frac{\left(\frac{AR_i}{R_0} + 1\right)}{\left(\frac{1}{R_\text{L}} + \frac{1}{R_0}\right)}$$

Substituting numerical values:

$$Z_\text{in} = 100\,\text{k}\Omega + \frac{\left(\frac{10^5 \times 10^5}{50} + 1\right)}{\left(\frac{1}{5000} + \frac{1}{50}\right)} \approx 10^{10}\,\Omega$$

$$\boxed{V_\text{s} = 10\,\text{V},\ \frac{V_0}{V_\text{s}} = 1,\ Z_\text{in} = 10^{10}\,\Omega}$$

Key Points

- **Non-inverting Voltage Follower**: Series-shunt feedback increases input impedance by factor of $(1 + A\beta)$ and provides unity voltage gain.
- **Input Resistance Enhancement**: $Z_\text{in} = R_i(1 + A\beta)$, yielding extremely high input resistance ($\approx 10^{10}\,\Omega$).

Q3.98.3 NAT 1998 5M Ans: * ●○○

Feedback Factor (β) Derivation:
Let Z_p be the parallel combination of R and C, and Z_s be the series combination of R and C:

$$Y_\text{p} = j\omega C + \frac{1}{R}, \quad Z_\text{s} = R + \frac{1}{j\omega C}$$

The feedback attenuation factor $\beta = \frac{V_\text{f}}{V_\text{o}}$ is given by:

$$\beta = \frac{Z_\text{p}}{Z_\text{p} + Z_\text{s}} = \frac{1}{1 + Y_\text{p} Z_\text{s}}$$

Substituting Y_p and Z_s:

$$\beta = \frac{1}{1 + \left(j\omega C + \frac{1}{R}\right)\left(R + \frac{1}{j\omega C}\right)} = \frac{1}{3 + j\left(\omega CR - \frac{1}{\omega CR}\right)}$$

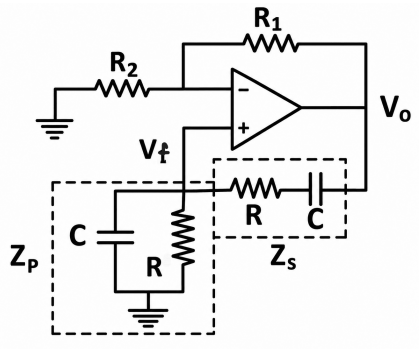


Fig. Solution Q3.98.3

Barkhausen Criteria & Oscillation Conditions:
For sustained oscillations, $A\beta = 1\angle 0°$. Thus, the imaginary part of β must be zero:

$$\omega CR - \frac{1}{\omega CR} = 0 \implies \omega = \frac{1}{RC}$$

$$f = \frac{1}{2\pi RC}$$

At this resonant frequency $\omega = \frac{1}{RC}$, $\beta = \frac{1}{3}$.
For $|A\beta| = 1$:

$$|A| \cdot \left(\frac{1}{3}\right) = 1 \implies A = 3$$

Since $A = 1 + \frac{R_1}{R_2}$:

$$1 + \frac{R_1}{R_2} = 3 \implies R_1 = 2R_2$$

$$\boxed{f = \frac{1}{2\pi RC},\ R_1 = 2R_2}$$

Key Points

Wien Bridge Oscillator: Requires zero phase shift in the feedback network at resonance, forcing the imaginary part to zero.

Q3.97.1 MCQ 1997 1M Ans: B ● ● ○

A major advantage of active filters is that they can be realized without using **inductors**.

In conventional passive RC/RLC filters, inductors are bulky, expensive, non-ideal at low frequencies, and difficult to fabricate on Integrated Circuits (ICs).

Active Filter Realization:
By using active elements such as operational amplifiers along with resistors and capacitors (RC networks), inductors can be eliminated or simulated (e.g., using gyrators).

(B) inductors

Q3.97.2 Descriptive 1997 2M Ans: adder, -3 ● ● ○

Circuit Identification:
The given circuit acts as an **inverting summer / adder**.
Using the ideal op-amp virtual ground concept ($V_- = V_+ = 0\,\text{V}$), the output voltage V_0 is given by:

$$V_0 = -\left(V_1\frac{R_f}{R_1} + V_2\frac{R_f}{R_2}\right)$$

Output Voltage Calculation:
Given: $V_1 = 1\,\text{V}$, $V_2 = 2\,\text{V}$, $R_1 = 1\,\text{k}\Omega$, $R_2 = 1\,\text{k}\Omega$, $R_f = 1\,\text{k}\Omega$.
Substituting the given values:

$$V_0 = -\left(1 \times \frac{1}{1} + 2 \times \frac{1}{1}\right) = -3\,\text{V}$$

$$\boxed{-3\,\text{V}}$$

Q3.97.3 NAT 1997 2M Ans: * ● ● ○

Positive Half Cycle ($V_{in} > 0$):
During the positive half cycle of the input signal $4 \sin 10t$:

- The voltage at the inverting terminal is greater than at the non-inverting terminal ($V_- > V_+$).
- The op-amp output goes to negative saturation ($-V_{sat}$).
- The buffer transmits this negative voltage to the FET gate, turning the FET **OFF**.
- No current flows through resistance R, resulting in:

$$V = 0\,\text{V}$$

Negative Half Cycle ($V_{in} < 0$):
During the negative half cycle of the input signal:

- The voltage at the inverting terminal is less than at the non-inverting terminal ($V_- < V_+$).
- The op-amp output goes to positive saturation ($+V_{sat}$).
- This turns the FET **ON**, creating a low-resistance path between drain and source.
- The voltage across resistance R connected between +12 V and −12 V becomes:

$$V = 12\,\text{V} - (-12\,\text{V}) = 24\,\text{V}$$

$$\boxed{V = 0\,\text{V (Positive Half Cycle)}, \quad V = 24\,\text{V (Negative Half Cycle)}}$$

Q3.97.4 NAT 1997 5M Ans: 33.863 ● ○ ○

Oscillator Identification:
The given circuit is a **Wien-Bridge Oscillator**.
The frequency of oscillation for a Wien-Bridge oscillator is given by the formula:

$$f = \frac{1}{2\pi RC}$$

Given values:

$$R = 1\,\text{k}\Omega = 10^3\,\Omega, \quad C = 4.7\,\mu\text{F} = 4.7 \times 10^{-6}\,\text{F}$$

Substituting the values into the formula:

$$f = \frac{1}{2\pi \times (4.7 \times 10^{-6}) \times 10^3} = \frac{1}{2\pi \times 4.7 \times 10^{-3}}$$

$$f \approx 33.863\,\text{Hz}$$

$$\boxed{33.863}$$

Q3.96.1 MCQ 1996 1M Ans: A ● ○ ○

Non-Inverting Terminal Voltage (V_+):
Using the superposition theorem and potential divider rule:

$$V_+ = -2 \times \frac{R}{R+R} + (2 + \sin 100t) \times \frac{R}{R+R}$$

$$V_+ = -1 + 1 + \frac{1}{2}\sin 100t = \frac{1}{2}\sin 100t$$

Output Voltage (V_0):
For a non-inverting op-amp configuration with feedback resistance $2R$ and input resistance R:

$$V_0 = \left(1 + \frac{2R}{R}\right) \times V_+ = 3V_+$$

$$V_0 = 3 \times \frac{1}{2}\sin 100t = \frac{3}{2}\sin 100t$$

$$\boxed{(A) \quad (3/2)\sin(100t)}$$

Q3.96.2 MCQ 1996 1M Ans: A ● ○ ○

Case 1: Switch S_1 Open

When switch S_1 is open, the total feedback resistance R_f consists of two series resistors of value R:

$$R_f = R + R = 2R$$

The magnitude of the voltage gain is:

$$x = -\frac{2R}{R} = -2$$

Case 2: Switch S_1 Closed

When switch S_1 is closed, one resistor R is short-circuited, making the effective feedback resistance $R_f' = R$:

The new gain x' is given by:

$$x' = -\frac{R}{R} = -1$$

Expressing x' in terms of x:

$$x' = \frac{x}{2}$$

(A) x / 2

Q3.95.1 Descriptive 1995 1M Ans: V_0 ● ○ ○

Common Mode Voltage Formula:

The common-mode input voltage is the average of the two input terminal voltages:

$$V_{cm} = \frac{1}{2}(V_+ + V_-) = \frac{1}{2}(V_i + V_0)$$

Voltage Follower Relation: For an ideal voltage follower buffer circuit:

$$V_i = V_0$$

Substituting $V_i = V_0$ into the expression:

$$V_{cm} = \frac{1}{2}(V_0 + V_0) = V_0$$

V_0

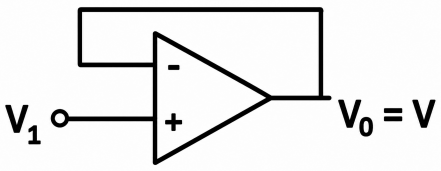


Fig. Solution Q3.95.1

Q3.95.2 NAT 1995 5M Ans: 0.1 ● ○ ○

Transfer Function Derivation:

Using superposition theorem:

$$V_{o1} = V_i\left(-\frac{R_1}{R_1}\right) = -V_i$$

Non-inverting terminal voltage V_+ by voltage divider:

$$V_+ = V_i\left(\frac{1/j\omega C}{R + 1/j\omega C}\right) = V_i\left(\frac{1}{1 + j\omega RC}\right)$$

Output due to non-inverting terminal:

$$V_{o2} = V_+\left(1 + \frac{R_1}{R_1}\right) = 2V_i\left(\frac{1}{1 + j\omega RC}\right)$$

Net output voltage V_0:

$$V_0 = V_{o1} + V_{o2} = V_i\left(\frac{1 - j\omega RC}{1 + j\omega RC}\right)$$

$$\frac{V_0(s)}{V_i(s)} = \frac{1 - sRC}{1 + sRC}$$

Phase Shift & Calculation of C:

Magnitude and Phase Response:

$$\left|\frac{V_0}{V_i}\right| = 1, \quad \phi = -2\tan^{-1}(\omega RC)$$

Given $\phi = 270^\circ \equiv -90^\circ$:

$$-2\tan^{-1}(\omega RC) = -90^\circ \implies \tan^{-1}(\omega RC) = 45^\circ$$

$$\omega RC = 1 \implies C = \frac{1}{\omega R}$$

Given $\omega = 1000\,\text{rad/s}$ and $R = 10\,\text{k}\Omega = 10^4\,\Omega$:

$$C = \frac{1}{1000 \times 10^4} = 10^{-7}\,\text{F} = 0.1\,\mu\text{F}$$

$\boxed{0.1\,\mu\text{F}}$

Key Points

- **All-Pass Filter**: The given circuit acts as a first-order active phase shifter (all-pass filter) with a constant unity gain magnitude.
- **Phase Angle Formula**: $\phi = -2\tan^{-1}(\omega RC)$.

Q3.95.3 Descriptive 1995 5M Ans: R_1 ●●○

Test Source Analysis:

To find the input impedance Z_i, apply a test voltage source V_t at the input terminal and let I_t be the drawn test current.

At the non-inverting terminal:

$$V_+ = V_t$$

Since no current enters the ideal op-amp input terminals, no current flows through the $R - L$ branch. Thus:

$$V_- = V_0$$

By virtual short concept:

$$V_- = V_+ \implies V_0 = V_t$$

Input Impedance Calculation:

Applying KCL at the non-inverting terminal node:

$$\frac{V_t - 0}{R_1} - I_t + \frac{V_t - V_0}{R_1} = 0$$

Since $V_0 = V_t$, the term $\frac{V_t - V_0}{R_1} = 0$:

$$\frac{V_t}{R_1} - I_t = 0 \implies \frac{V_t}{I_t} = R_1$$

Therefore, the input impedance Z_i is:

$$Z_i = \frac{V_t}{I_t} = R_1$$

$\boxed{R_1}$

Q3.94.1 Descriptive 1994 1M Ans: True ●○○

Analog Comparator Operation:

An analog comparator compares two voltage levels applied to its non-inverting (V_+) and inverting (V_-) terminals.

Because of its extremely high open-loop gain ($A_{OL} \to \infty$) and open-loop operation (no negative feedback): **Output Voltage States:**

- If $V_+ > V_-$, then $V_0 = +V_{sat}$
- If $V_+ < V_-$, then $V_0 = -V_{sat}$

Thus, the given statement is true.

True

Q3.94.2 Descriptive 1994 1M Ans: a-S, b-R, c-T, d-Q ● ○ ○

Fault Analysis (Parts a & b):

- **(a) C_1 is open:** The input AC signal V_s is disconnected from transistor TR_1. The DC biasing remains unaffected. Thus, all DC voltages are normal and $V_0 = 0$. → **(S)**
- **(b) C_3 is open:** C_3 acts as a bypass capacitor for R_{e2}. When C_3 opens, the emitter resistance R_{e2} is no longer bypassed for AC, introducing local negative feedback. This decreases the 2nd stage gain and thus decreases V_0, while DC voltages remain normal. → **(R)**

Fault Analysis (Parts c & d):

- **(c) C_4 is open:** C_4 blocks DC in the feedback path R_F. If C_4 is open, global negative feedback is disabled. Overall voltage gain increases, leading to an increased V_0, while DC conditions stay normal. → **(T)**
- **(d) R_{c2} is shorted:** Shorting R_{c2} connects the collector of TR_2 directly to V_{CC} ($V_{C2} = V_{CC}$). Consequently, no AC signal voltage is developed at the collector, resulting in $V_0 = 0$. → **(Q)**

a-S, b-R, c-T, d-Q

Key Points

- **Coupling Capacitors**: Block DC bias while allowing AC signals to pass.
- **Bypass Capacitors**: Increase AC voltage gain by eliminating AC degeneration across emitter resistors.
- **Feedback Loop**: Removal of negative feedback loop increases overall amplifier gain.

Q3.94.3 NAT 1994 1M Ans: 1 V ● ○ ○

Non-Inverting Voltage (V_+) Calculation:

Using superposition theorem at non-inverting terminal:

1. Due to 2 V source (with −1 V source grounded):

$$V_{+1} = 2 \times \frac{R/2}{R + R/2} = \frac{2}{3}\,\text{V}$$

2. Due to −1 V source (with 2 V source grounded):

$$V_{+2} = -1 \times \frac{R/2}{R + R/2} = -\frac{1}{3}\,\text{V}$$

Total non-inverting terminal voltage:

$$V_+ = V_{+1} + V_{+2} = \frac{2}{3} - \frac{1}{3} = \frac{1}{3}\,\text{V}$$

Output Voltage (V_0) Calculation:

The amplifier configuration is non-inverting with non-inverting gain:

$$A_v = 1 + \frac{R_\text{f}}{R_1} = 1 + \frac{2R}{R} = 3$$

Therefore, the output voltage is:

$$V_0 = \left(1 + \frac{2R}{R}\right) \times V_+ = 3 \times \frac{1}{3} = 1\,\text{V}$$

1 V

Q3.92.1 MCQ 1992 1M Ans: C ● ○ ○

Virtual Ground Concept:

The output voltage of an op-amp is given by:

$$V_0 = A_{OL}(V_+ - V_-)$$

where A_{OL} is the open-loop voltage gain.

Rearranging the expression for differential input voltage:

$$V_+ - V_- = \frac{V_0}{A_{OL}}$$

Derivation & Conclusion:

Since the output voltage V_0 is always bounded between $+V_{sat}$ and $-V_{sat}$, as $A_{OL} \to \infty$:

$$V_+ - V_- = 0 \implies V_+ = V_-$$

This phenomenon is known as the **virtual ground** (or virtual short) principle, which occurs due to infinite open-loop gain.

(C) The open loop gain of the OPAMP is infinity

Q3.92.2 NAT 1992 2M Ans: * ● ○ ○

Circuit Operation & Charging Phase:

- Initially, the capacitor is uncharged so $V_0 = 0\,\text{V}$.
- When $V_i > 0$, diode D_1 conducts (ON) and charges the capacitor following V_i. Thus, $V_0 = V_i$.
- Diode D_2 remains OFF as its anode is grounded (0 V) and cathode is at positive voltage $V_0 > 0$.
- At $t = 0.5\,\text{s}$, V_i reaches its peak of 1 V, charging the capacitor to $V_0 = 1\,\text{V}$.

Peak Hold Phase & Final Waveform:

- For $t > 0.5\,\text{s}$, V_i decreases ($V_i < 1\,\text{V}$) while $V_- = V_0 = 1\,\text{V}$.
- Since $V_+ < V_-$, the op-amp output switches to $-V_{sat}$, reverse-biasing D_1 (OFF).
- With D_1 OFF, the capacitor has no discharge path and holds its peak voltage of 1 V indefinitely.

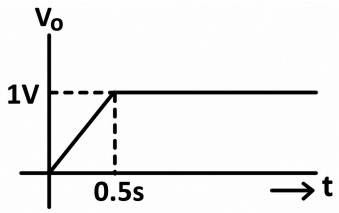


Fig. Solution Q3.92.2

Key Points

- **Precision Peak Detector**: The circuit acts as a positive peak detector, tracking V_i until it peaks and holding the maximum value.
- **Op-Amp Buffer**: Eliminates the diode forward voltage drop V_γ inside the feedback loop.

Q3.92.3 NAT 1992 2M Ans: * ●●●

Integrator Stage Analysis:
For the upper two integrator op-amps:

$$V_2 = -\frac{1}{RC}\int V_1\,dt \implies V_1 = -RC\frac{dV_2}{dt} \quad -(1)$$

$$V = -\frac{1}{RC}\int V_2\,dt \implies V_2 = -RC\frac{dV}{dt} \quad -(2)$$

Substituting (2) into (1):

$$V_1 = R^2C^2\frac{d^2V}{dt^2} \quad -(3)$$

Lower Op-Amp Analysis:
At the non-inverting terminal, using superposition:

$$V_+ = \frac{1}{2}(V_2 + e^t)$$

By virtual short concept, $V_- = V_+ = \frac{1}{2}(V_2 + e^t)$. Applying KCL at the inverting terminal node:

$$C\frac{d}{dt}(V_+ - V_1) = \frac{V_+}{R} \implies RC\frac{d}{dt}(V_+ - V_1) = V_+$$

Integrating both sides:

$$RC(V_+ - V_1) = \int V_+\,dt = \frac{1}{2}\int e^t\,dt + \frac{1}{2}\int V_2\,dt$$

Differential Equation Realization:
Substitute V_+ into the integrated expression:

$$RC\left[\frac{V_2 + e^t}{2} - V_1\right] = \frac{1}{2}e^t + \frac{1}{2}\int V_2\,dt$$

Substituting $V_2 = -RC\frac{dV}{dt}$, $V_1 = R^2C^2\frac{d^2V}{dt^2}$, and $\int V_2\,dt = -RC\,V$:

$$RC\left[\frac{-RC\frac{dV}{dt} + e^t}{2} - R^2C^2\frac{d^2V}{dt^2}\right] = \frac{1}{2}e^t - \frac{1}{2}RC\,V$$

Dividing both sides by $-R^3C^3$:

$$\frac{d^2V}{dt^2} + \frac{1}{2RC}\frac{dV}{dt} + \frac{1}{2R^2C^2}V = \frac{1}{2R^2C^2}e^t\left(1 + \frac{1}{RC}\right)$$

Comparing with $\frac{d^2v}{dt^2} + a\frac{dv}{dt} + bv = f(t)$:

$$a = \frac{1}{2RC}, \quad b = \frac{1}{2R^2C^2}, \quad f(t) = \frac{e^t}{2R^2C^2}\left(1 + \frac{1}{RC}\right)$$

$$\boxed{a = \frac{1}{2RC},\ b = \frac{1}{2R^2C^2},\ f(t) = \frac{e^t}{2R^2C^2}\left(1 + \frac{1}{RC}\right)}$$

Key Points

- **Op-Amp Integrator**: $V_o = -\frac{1}{RC}\int V_{in}\,dt$.
- **Superposition and KCL**: Node analysis combined with virtual ground condition allows modeling active filter systems into standard second-order linear differential equations.

Q3.91.1 NAT 1991 2M Ans: * ●●○

First Stage Analysis (Integrator):
The first op-amp acts as an inverting integrator. Its output V_1' is given by:

$$V_1' = -\frac{1}{RC}\int V_1\,dt$$

Given $R = 10\,\text{k}\Omega = 10^4\,\Omega$ and $C = 100\,\mu\text{F} = 100 \times 10^{-6}\,\text{F}$:

$$RC = 10^4 \times 10^{-4} = 1\,\text{s}$$

$$V_1' = -\frac{1}{1}\int V_1\,dt = -\int V_1\,dt$$

Second Stage Analysis (Inverting Summer):
The second op-amp acts as an inverting summing amplifier with inputs V_1', V_2, and V_3:

$$V_0 = -\left(V_1'\frac{R_f}{R_1} + V_2\frac{R_f}{R_2} + V_3\frac{R_f}{R_3}\right)$$

Substituting $R_f = 20\,\text{k}\Omega$, $R_1 = 10\,\text{k}\Omega$, $R_2 = 5\,\text{k}\Omega$, and $R_3 = 4\,\text{k}\Omega$:

$$V_0 = -\left(V_1'\frac{20}{10} + V_2\frac{20}{5} + V_3\frac{20}{4}\right)$$

$$V_0 = -2V_1' - 4V_2 - 5V_3$$

Substituting $V_1' = -\int V_1\,dt$:

$$V_0 = 2\int V_1\,dt - 4V_2 - 5V_3$$

$$\boxed{2\int V_1 dt - 4V_2 - 5V_3}$$

Key Points

Inverting Summer: $V_o = -\sum\left(\frac{R_f}{R_i}V_i\right)$.

Q3.91.2 NAT 1991 2M Ans: *

(i) When $V_i \le 0$:
The Zener diode becomes forward-biased and acts as a short circuit (ideal diode).

- Equivalent input resistance:

$$R_{\text{in}} = 1\,\text{k}\Omega \parallel 1\,\text{k}\Omega = 0.5\,\text{k}\Omega$$

- Output voltage gain:

$$\text{Gain} = \frac{V_0}{V_i} = -\frac{R_\text{f}}{R_{\text{in}}} = -\frac{1\,\text{k}\Omega}{0.5\,\text{k}\Omega} = -2$$

(ii) When $0 < V_i \le 5$ V:
The Zener diode is reverse-biased and OFF (open circuit).

- Equivalent input resistance: $R_{\text{in}} = 1\,\text{k}\Omega$
- Output voltage gain:

$$\text{Gain} = \frac{V_0}{V_i} = -\frac{1\,\text{k}\Omega}{1\,\text{k}\Omega} = -1$$

(iii) When $V_i > 5$ V:
The Zener diode enters breakdown mode and maintains a constant voltage of $V_Z = 5$ V.
Using superposition theorem:

$$V_0 = (V_i - 5)\left(-\frac{1\,\text{k}\Omega}{1\,\text{k}\Omega}\right) + V_i\left(-\frac{1\,\text{k}\Omega}{1\,\text{k}\Omega}\right)$$

$$V_0 = -(V_i - 5) - V_i = 5 - 2V_i$$

The incremental gain in this region is:

$$\text{Gain} = \frac{dV_0}{dV_i} = -2$$

$$\boxed{\text{Gain} = -2\ (V_i \le 0),\ -1\ (0 < V_i \le 5\,\text{V}),\ -2\ (V_i > 5\,\text{V})}$$

Key Points

- **Forward-Biased Zener**: Acts as an open/short depending on ideal model.
- **Breakdown Region**: Holds fixed voltage $V_Z = 5\,\text{V}$ across its terminals for $V_i > V_Z$.

Q3.91.3 NAT | 1991 | 2M | Ans: * | ● ○ ○

Transfer Characteristic Curves (V_0 vs V_i):
Combining all three operational regions:

- **Region I ($V_i \leq 0$):**

$$V_0 = -2V_i \quad (\text{Slope} = -2)$$

- **Region II ($0 < V_i \leq 5\,\text{V}$):**

$$V_0 = -V_i \quad (\text{Slope} = -1)$$

At $V_i = 5\,\text{V}$, $V_0 = -5\,\text{V}$.

- **Region III ($V_i > 5\,\text{V}$):**

$$V_0 = 5 - 2V_i \quad (\text{Slope} = -2)$$

Salient Points Summary:

- Origin point $(0, 0)$ is continuous.
- Breakpoint occurs at $V_i = 5\,\text{V}$, where $V_0 = -5\,\text{V}$.
- Slope changes from -1 to -2 at $V_i = 5\,\text{V}$.

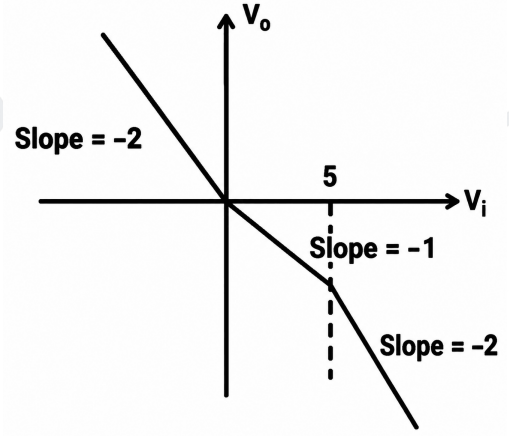


Fig. Solution Q3.91.3

Key Points

Piecewise Linearity: Sudden slope change occurs at Zener breakdown voltage $V_i = 5\,\text{V}$.

SOLUTIONS

IV

BJTs

Topics

1. BJT Physics and Biasing
2. Current Mirror Circuit Using BJT
3. Small Signal Analysis of BJT Amplifiers
4. Differential Amplifiers and Feedback Topologies Using BJTs

Unit Snapshot*

Stream: EC	**Questions:** 28	**MCQ:** 21	**MSQ:** 0	**NAT:** 7
1M: 4	**2M:** 24	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2016)–2M(2024)

**See preface for how Avg Weightage and Range are calculated.*

Q4.24.1 MCQ | 2024 | 2M | Ans: A | ● ○ ○

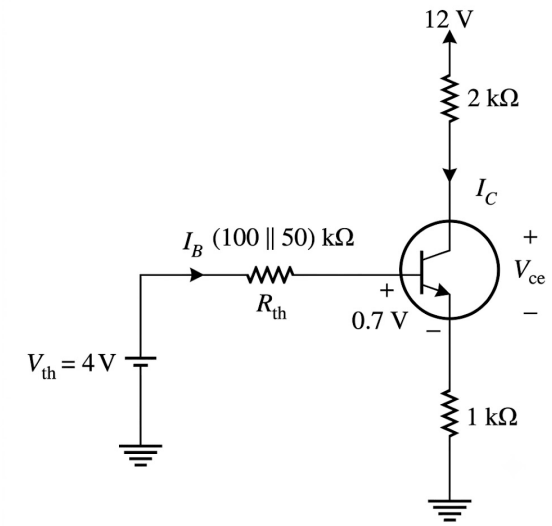


Fig. Solution Q4.24.1

Calculating Thevenin equivalent voltage V_{th} and resistance R_{th} across the base terminal:

$$V_{th} = \frac{12 \times 50}{100 + 50} = 4\,\text{V}$$

$$R_{th} = 50\,\text{k}\Omega \parallel 100\,\text{k}\Omega = 33.33\,\text{k}\Omega$$

Applying KVL in the base-emitter loop:

$$-4 + I_B(33.33\,\text{k}\Omega) + 0.7 + (1+\beta)I_B \times 1\,\text{k}\Omega = 0$$

$$I_B = \frac{3.3}{33.33\,\text{k}\Omega + 101\,\text{k}\Omega} = 24.56\,\mu\text{A}$$

$$I_C = \beta I_B = 100 \times 24.56 \times 10^{-6} = 2.46\,\text{mA}$$

$$I_E = (1+\beta)I_B = 2.47\,\text{mA}$$

Applying KVL in the collector-emitter loop:

$$-12 + (2\,\text{k}\Omega)I_C + V_{CE} + I_E R_E = 0$$

$$12 = 2\,\text{k}\Omega \times I_C + V_{CE} + I_E \times 1\,\text{k}\Omega$$

$$V_{CE} = 12 - (2\,\text{k}\Omega)(2.46\,\text{mA}) - (2.47\,\text{mA})(1\,\text{k}\Omega)$$

$$V_{CE} = 12 - 7.39 = 4.61\,\text{V} \approx 4.6\,\text{V}$$

(A) 4.6 V and 2.46 mA

Key Points

- Apply Thevenin theorem across the base voltage divider circuit.
- Use $I_E = (1+\beta)I_B$ and $I_C = \beta I_B$ to link loop currents.

Q4.21.1 NAT | 2021 | 2M | Ans: 16.70 to 17.70 | ● ● ○

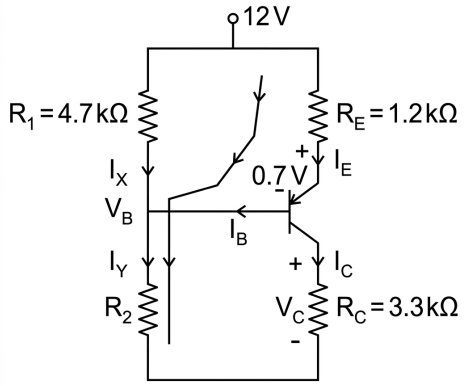


Fig. Solution Q4.21.1

Given:
(i) $\beta = 100$
(ii) $V_{BE} = -0.7\,\text{V}$
(iii) $V_s = 12\,\text{V}$
From the circuit diagram:

$$I_C = \frac{V_C}{R_C}$$

$$I_C = \frac{5}{3.3} = 1.515\,\text{mA}$$

Since,

$$I_C = \beta I_B$$

$$I_B = \frac{1.515}{100} = 0.01515\,\text{mA}$$

Also,

$$I_E = (1+\beta) I_B$$

$$I_E = 1.5303\,\text{mA}$$

Applying KVL around the emitter loop:

$$-12 + 1.2 \times 1.5303 + 0.7 + V_B = 0$$

$$V_B = 9.4636\,\text{V}$$

The current through R_1 is

$$I_X = \frac{12 - V_B}{4.7} = \frac{12 - 9.4636}{4.7} = 0.5397\,\text{mA}$$

Applying KCL at the base node:

$$I_Y = I_X + I_B$$

$$I_Y = (0.5397 + 0.01515)\,\text{mA} = 0.5548\,\text{mA}$$

Therefore,

$$R_2 = \frac{V_B}{I_Y} = \frac{9.4636}{0.5548 \times 10^{-3}} = 17.0576\,\text{k}\Omega$$

$$\boxed{17.06}$$

Key Points

- For PNP transistors, $V_{EB} = 0.7\,\text{V}$ (or $V_{BE} = -0.7\,\text{V}$).
- Calculate node voltages sequentially using KVL and standard transistor current relations ($I_C = \beta I_B$, $I_E = (1+\beta)I_B$).

Q4.18.1 NAT | 2018 | 2M | Ans: 90 | ● ● ○

Given:

(i) $V_{BE} = 0.7\,\text{V}$

(ii) DC current gain, $\beta = 100$

Thevenin's equivalent resistance across terminals a and b may be obtained as:

$$R_{TH} = \frac{V_{TH}}{I_N} = \frac{V_{OC}}{I_{SC}} \quad \ldots(\text{i})$$

To find V_{OC} or V_{TH} :

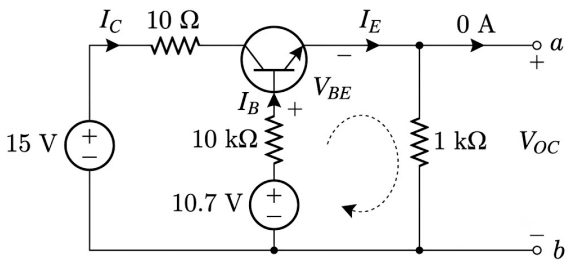


Fig. Solution Q4.18.1

Applying KVL in the loop shown:

$$-10.7 + 10I_B + V_{BE} + I_E \times 1 = 0$$

$$10I_B + (1+\beta) \times 1I_B = 10.7 - V_{BE}$$

$$10I_B + 101I_B = 10.7 - 0.7$$

$$I_B = \frac{10}{111}\,\text{mA} = 0.09\,\text{mA}$$

$$I_E = (1+\beta)I_B = 101 \times 0.09 = 9.09\,\text{mA}$$

Hence, V_{OC} or $V_{TH} = I_E \times 1 = 9.09\,\text{V}$ $\ldots(\text{ii})$

To find I_{SC} or I_N :

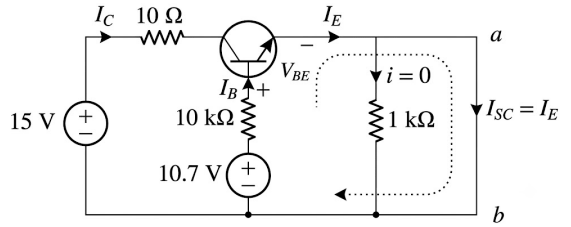


Fig. Solution Q4.18.1

The 1 kΩ resistor is in parallel with the short-circuited path, hence current through it will be zero. Therefore, $I_E = I_{SC}$.

Applying KVL in the loop shown:

$$-10.7 + 10I_B + V_{BE} = 0$$

$$I_B = \frac{10.7 - 0.7}{10} = 1\,\text{mA}$$

Therefore,

$$I_{SC} = I_E = (1+\beta)I_B = 101 \times 1 = 101\,\text{mA} \quad \ldots\text{(iii)}$$

Substituting equations (ii) and (iii) in equation (i):

$$R_{TH} = \frac{V_{OC}}{I_{SC}} = \frac{9.09}{101}\,\text{k}\Omega = 90\,\Omega$$

90

Key Points

- Find open-circuit voltage V_{OC} and short-circuit current I_{SC} across terminals a-b.
- R_{TH} is calculated as $R_{TH} = \frac{V_{OC}}{I_{SC}}$.

Q4.17.1 NAT | 2017 | 2M | Ans: 172.7 ● ● ○

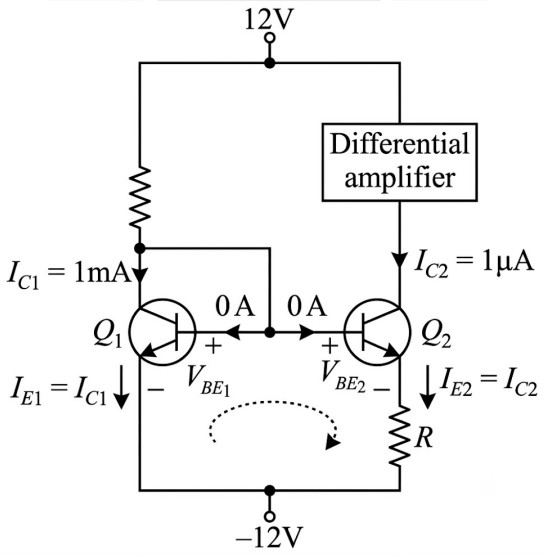


Fig. Solution Q4.17.1

Given:

(i) Thermal voltage, $V_T = 25\,\text{mV}$

(ii) $V_{BE_1} \neq V_{BE_2}$ because emitter terminals of both transistors are not at the same potential.

(iii) $I_{B_1} = 0 = I_{B_2}$ (negligible)

(iv) Transistors are matched, but $I_{C_1} \neq I_{C_2}$

(v) Bias current of differential amplifier is $1\,\mu\text{A}$

From the circuit diagram, $I_{C_1} = 1\,\text{mA}$ and $I_{C_2} = 1\,\mu\text{A}$.
Since base currents are negligible, $I_{B_1} = I_{B_2} = 0$.

Q_1 is a diode-connected transistor, so collector current is given by:

$$I_{C_1} = I_0 e^{V_{BE_1}/V_T}$$

$$V_{BE_1} = V_T \ln\left(\frac{I_{C_1}}{I_0}\right) \quad \ldots (i)$$

where I_0 = reverse saturation current, V_T = thermal voltage.
Collector current of transistor Q_2 is given by:

$$I_{E_2} = I_{C_2} = I_0 e^{V_{BE_2}/V_T}$$

$$V_{BE_2} = V_T \ln\left(\frac{I_{C_2}}{I_0}\right) \quad \ldots (ii)$$

Applying KVL in the loop shown:

$$-V_{BE_1} + V_{BE_2} + I_{C_2}R = 0$$

$$I_{C_2}R = V_{BE_1} - V_{BE_2}$$

From equations (i) and (ii):

$$I_{C_2}R = V_T \ln\left(\frac{I_{C_1}}{I_0}\right) - V_T \ln\left(\frac{I_{C_2}}{I_0}\right)$$

$$I_{C_2}R = V_T \left[\ln\frac{I_{C_1}}{I_0} - \ln\frac{I_{C_2}}{I_0}\right]$$

$$I_{C_2}R = V_T \ln\left(\frac{I_{C_1}/I_0}{I_{C_2}/I_0}\right) = V_T \ln\left(\frac{I_{C_1}}{I_{C_2}}\right)$$

$$R = \frac{V_T}{I_{C_2}} \ln\left(\frac{I_{C_1}}{I_{C_2}}\right) = \frac{25 \times 10^{-3}}{10^{-6}} \ln\left(\frac{10^{-3}}{10^{-6}}\right)$$

$$R = 25 \times 10^3 \ln(1000) = 172.7\,\text{k}\Omega$$

$$\boxed{172.7}$$

Key Points

- **Widlar Current Mirror Circuit:** For a Widlar current source, $V_{BE_1} \neq V_{BE_2}$ and the emitter resistance is calculated using:

$$R_E = \frac{V_T}{I_{\text{out}}} \ln\left(\frac{I_{\text{ref}}}{I_{\text{out}}}\right)$$

Q4.17.2 MCQ | 2017 | 2M | Ans: D | ● ● ○

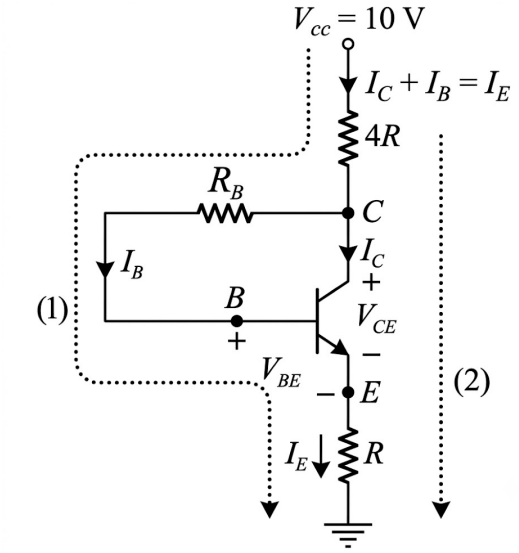


Fig. Solution Q4.17.2

Given:

(i) $V_{BE} = 0.7\,\text{V}$

(ii) $V_{CE} = \dfrac{V_{CC}}{2} = 5\,\text{V}$

(iii) DC current gain, $\beta = 29$

Applying KVL in loop (1):

$$-V_{CC} + 4RI_E + I_BR_B + V_{BE} + RI_E = 0$$

$$-10 + (\beta + 1)I_B \times 4R + I_BR_B + 0.7 + (\beta + 1)I_B \times R = 0$$

$$-10 + 30 \times 4RI_B + I_BR_B + 0.7 + 30RI_B = 0$$

$$-9.3 + 120RI_B + I_BR_B + 30RI_B = 0$$

$$I_B(150R + R_B) = 9.3$$

$$I_B = \frac{9.3}{150R + R_B} \quad \ldots(\text{i})$$

Applying KVL in loop (2):

$$-10 + (1 + \beta)I_B \times 4R + V_{CE} + (1 + \beta)I_B \times R = 0$$

$$-10 + 150I_BR + 5 = 0$$

$$I_B = \frac{5}{150R} = \frac{1}{30R} \quad \ldots(\text{ii})$$

From equations (i) and (ii):

$$30R \times 9.3 = 150R + R_B$$

$$279R = 150R + R_B$$

$$R_B = 129R \implies \frac{R_B}{R} = 129$$

D) 129

Key Points

- Relate base and emitter currents via $I_E = (1 + \beta)I_B$.
- Express I_B by writing KVL for two independent loops and equate them to find the ratio $\frac{R_B}{R}$.

Q4.16.1 NAT 2016 2M Ans: 19 ●●○

Given:

(i) Zener voltage, $V_Z = 5.3\,\text{V}$

(ii) $V_{BE} = 0.6\,\text{V}$

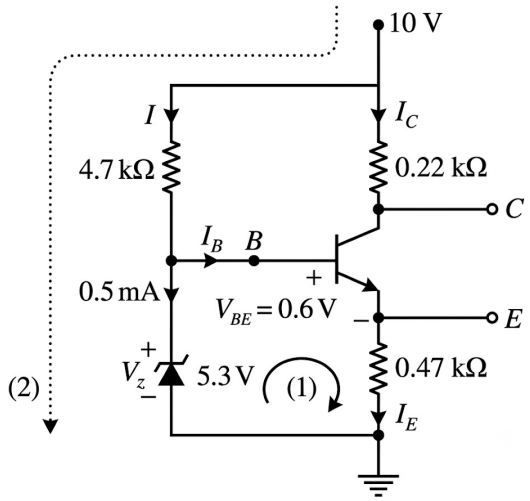


Fig. Solution Q4.16.1

Due to the DC biasing of 10 V, the Zener diode is in the breakdown region. Therefore, it is replaced by a constant voltage source $V_Z = 5.3\,\text{V}$.

Applying KVL in loop (1):

$$-5.3 + 0.6 + I_E \times 0.47 = 0$$

$$I_E = 10\,\text{mA}$$

Applying KVL in loop (2):

$$-10 + I \times 4.7 + 5.3 = 0$$

$$I = 1\,\text{mA}$$

From figure,

$$I_B = I - 0.5\,\text{mA} = 1 - 0.5\,\text{mA} = 0.5\,\text{mA}$$

Collector current, $I_C = I_E - I_B$

$$I_C = 10 - 0.5 = 9.5\,\text{mA}$$

Current gain, $\beta = \frac{I_C}{I_B} = \frac{9.5}{0.5} = 19$

19

Key Points

- When a Zener diode operates in the breakdown region, model it as a constant DC voltage source equal to V_Z.
- Apply KVL across loops to find node currents, then use $\beta = \frac{I_C}{I_B}$.

Q4.15.1 NAT | 2015 | 1M | Ans: 105.13 | ● ○ ○

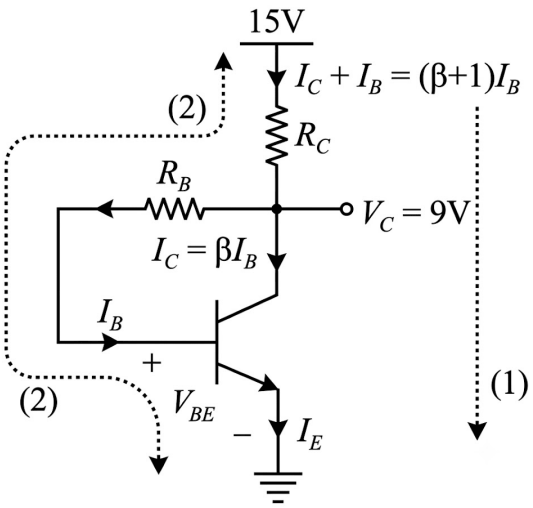


Fig. Solution Q4.15.1

Given: $\beta = 75$, $V_C = 9\,\text{V}$
Here we take $V_{BE} = 0.7\,\text{V}$.
Applying KVL in loop (1),

$$-15 + I_B(\beta + 1)R_C + 9 = 0$$

$$I_B(\beta + 1)R_C = 6\,\text{V} \quad \ldots(\text{i})$$

Applying KVL in loop (2),

$$-15 + I_B(\beta + 1)R_C + I_B R_B + 0.7 = 0$$

From equation (i),

$$-15 + 6 + I_B R_B + 0.7 = 0$$

$$I_B R_B = 8.3\,\text{V} \quad \ldots(\text{ii})$$

From equation (i) and (ii),

$$\frac{I_B R_B}{I_B(\beta + 1)R_C} = \frac{8.3}{6}$$

$$\frac{R_B}{R_C} = \frac{8.3}{6} \times (75 + 1) = 105.13$$

$$\boxed{105.13}$$

Key Points

- In a collector-to-base feedback bias circuit, the total current through R_C is $I_C + I_B = (\beta + 1)I_B$.
- Expressing the loop equations allows solving for the ratio $\frac{R_B}{R_C}$ directly without determining I_B individually.

Q4.15.2 NAT | 2015 | 2M | Ans: 0.75 | ● ● ○

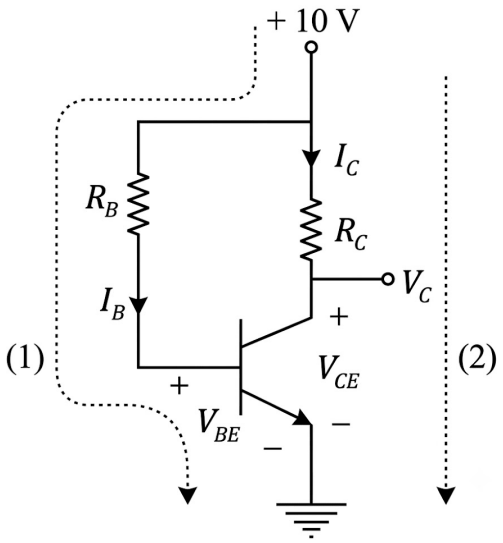


Fig. Solution Q4.15.2

Given fixed bias circuit is shown below,
Given transistor is in active region because $V_{CE} > 0.2\,\text{V}$ when $V_C = 2\,\text{V}$ and $V_C = 4\,\text{V}$.
Applying KVL in loop (1),

$$-10 + R_B I_B + V_{BE} = 0$$

$$I_B = \frac{10 - V_{BE}}{R_B}$$

Collector current is given by,

$$I_C = \beta I_B$$

Since, change in R_C does not affect the base current I_B as well as I_C.
Thus, collector current I_C = constant.
Applying KVL in loop (2),

$$-10 + I_C R_C + V_C = 0$$

$$I_C = \frac{10 - V_C}{R_C} \quad \ldots(\text{i})$$

For $V_C = 2\,\text{V}$,
From equation (i),

$$I_C = \frac{10 - 2}{R_C} = \frac{8}{R_C} \quad \ldots(\text{ii})$$

For $V_C = 4\,\text{V}$,

$$I_C = \frac{10 - V_C}{R'_C} = \frac{10 - 4}{R'_C} = \frac{6}{R'_C} \quad \ldots(\text{iii})$$

From equation (ii) and (iii),

$$\frac{R'_C}{R_C} = \frac{6}{8} = 0.75$$

$$\boxed{0.75}$$

Key Points

- In a fixed bias circuit operating in the active region, base current I_B and collector current I_C remain independent of collector resistance R_C.
- Equating expressions for I_C under two different collector voltages yields the required resistance ratio directly.

Q4.15.3 MCQ 2015 1M Ans: C ●○○

When CB and BE junctions are both in forward bias region then transistor is said to be in saturation region.

C) Both the CB and BE junctions are forward biased

Key Points

Operating Region	Emitter-base junction	Collector-base junction	Acting device
Active	Forward bias	Reverse bias	Amplifier
Saturation	Forward bias	Forward bias	ON switch
Cut-off	Reverse bias	Reverse bias	OFF switch
Inverse Active	Reverse bias	Forward bias	Attenuator

Q4.14.1 NAT 2014 2M Ans: 22.30 ●●○

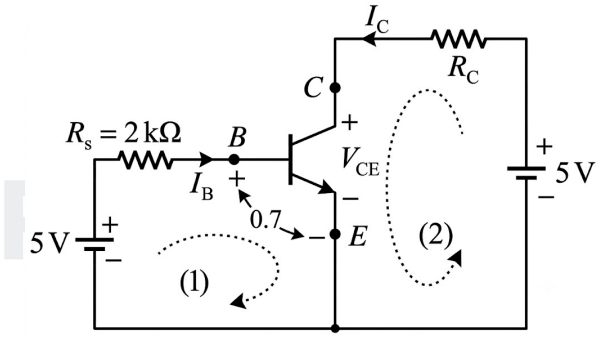


Fig. Solution Q4.14.1

Given:

(i) DC current gain, $\beta = 100$

(ii) According to question, Transistor is always working in active region.

(iii) $V_{CE(sat)} = 0.2\,V$, $V_{BE} = 0.7\,V$

Applying KVL in loop (1),

$$-5 + 2I_B + 0.7 = 0$$

$$I_B = \frac{5 - 0.7}{2} = \frac{4.3}{2} = 2.15\,mA$$

Collector current is given by,

$$I_C = \beta I_B = 100 \times 2.15\,mA = 215\,mA$$

Applying KVL in loop (2),

$$-5 + I_C R_C + V_{CE} = 0$$

$$V_{CE} = 5 - 215R_C$$

Condition for *n-p-n* transistor to operate in active region,

$$V_{CE} > V_{CE(\text{sat})}$$

$$V_{CE} > 0.2\,\text{V}$$

$$(5 - 215 \times R_C) > 0.2$$

$$R_C < \frac{5 - 0.2}{215} < 22.32\,\Omega$$

So, we choose $R_{C(\max)}$ is $22.30\,\Omega$.

$$\boxed{22.30}$$

Key Points

- For an *n-p-n* transistor to stay in active mode, V_{CE} must be strictly greater than $V_{CE(\text{sat})}$.
- Maximum collector resistance $R_{C(\max)}$ occurs at the edge of saturation where $V_{CE} = V_{CE(\text{sat})}$.

Q4.11.1 MCQ 2011 1M Ans: D ●●○

Given data:

(i) DC current gain, $\beta = 30$

(ii) Collector to base reverse saturation current, $I_{CB0} \approx 0$ A

(iii) Diode forward voltage, $V_\gamma = 0.7$ V

(iv) $V_{BE} = 0.7$ V, $V_{CE(\text{sat})} = 0.2$ V

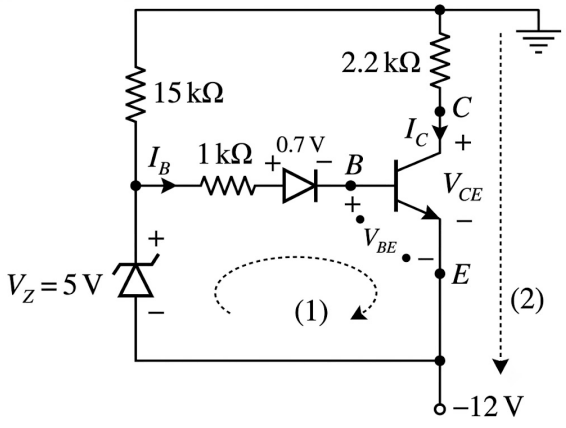


Fig. Solution Q4.11.1

From the figure, the Zener diode is in the reverse breakdown region due to the −12 V voltage source. So, the Zener diode works as a 5 V battery as shown below:

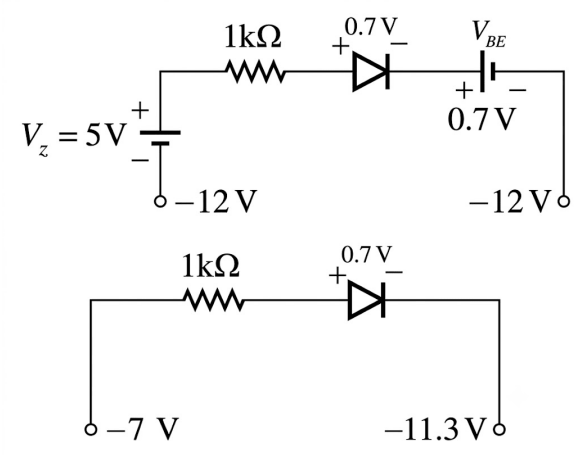


Fig. Solution Q4.11.1

Here, the P-terminal of the diode is at a higher potential than the N-terminal of the diode. Therefore, the diode and emitter junction J_E will be forward biased due to −12 V.
Hence, the BJT will be either in the active region or in the saturation region.

Method 1

Assuming the transistor is in the active region,

$$V_{BE(\text{active})} = 0.7 \text{ V}$$

Applying KVL in loop (1):

$$-5 + I_B \times 1 + 0.7 + 0.7 = 0$$

$$I_B = 5 - 1.4 = 3.6 \text{ mA}$$

Collector current is given by:

$$I_C = \beta I_B = 30 \times 3.6 = 108 \text{ mA}$$

Applying KVL in loop (2):

$$0 + 2.2 I_C + V_{CE} - 12 = 0$$

$$V_{CE} = 12 - 2.2 \times 108 = -225.6 \text{ V}$$

$$V_{CB} = V_{CE} - V_{BE} = -225.6 - 0.7 = -226.3 \text{ V}$$

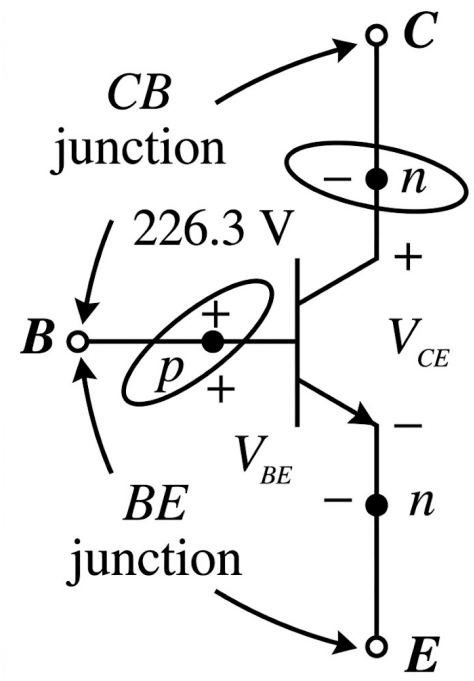


Fig. Solution Q4.11.1

From the above figure, it is clear that the collector of the given npn BJT is negatively biased and the base is positively biased; therefore, the collector-to-base junction is in forward bias. But in active mode, the collector-to-base junction must be reverse biased. Hence, our assumption is wrong.
Therefore, the transistor is in the saturation region.
Applying KVL in loop (2):

$$2.2I_{C(\text{sat})} + V_{CE(\text{sat})} - 12 = 0$$

Assume $V_{CE(\text{sat})} = 0.2$ V:

$$2.2I_{C(\text{sat})} + 0.2 - 12 = 0$$

$$I_{C(\text{sat})} = 5.36 \text{ mA}$$

Hence, the correct option is (D).

Method 2

Assuming the transistor is in the saturation region:

$$V_{BE(\text{sat})} = 0.7 \text{ V}, \quad V_{CE(\text{sat})} = 0.2 \text{ V}$$

For a transistor to be in the saturation region:

$$I_{B(\text{min})} \le I_B \implies \frac{I_{C(\text{sat})}}{\beta} \le I_B \quad \text{— (i)}$$

Applying KVL in loop (1):

$$-5 + I_B \times 1 + 0.7 + 0.7 = 0$$

$$I_B = 5 - 1.4 = 3.6 \text{ mA} \quad \text{— (ii)}$$

Applying KVL in loop (2):

$$0 + 2.2I_{C(\text{sat})} + V_{CE(\text{sat})} - 12 = 0$$

$$I_{C(\text{sat})} = \frac{12 - V_{CE(\text{sat})}}{2.2} = \frac{12 - 0.2}{2.2} = 5.36 \text{ mA}$$

Hence,

$$I_{B(\min)} = \frac{I_{C(\text{sat})}}{\beta} = \frac{5.36 \times 10^{-3}}{30} = 1.78 \text{ mA} \quad \text{— (iii)}$$

From equations (ii) and (iii):

$$I_{B(\min)} < I_B \quad \text{— (iv)}$$

From equations (i) and (iv), the transistor satisfies the condition of saturation. Hence, our assumption is correct and the transistor is working in the saturation region.
Therefore, current through collector is given by:

$$\boxed{I_{C(\text{sat})} = 5.36 \text{ mA}}$$

Key Points

- Transistor operating region can be determined by checking the bias conditions of both the base-emitter and collector-base junctions.
- For saturation, the base current must be greater than or equal to $I_{B(\min)} = \frac{I_{C(\text{sat})}}{\beta}$.

Q4.10.1 MCQ | 2010 | 2M | Ans: A

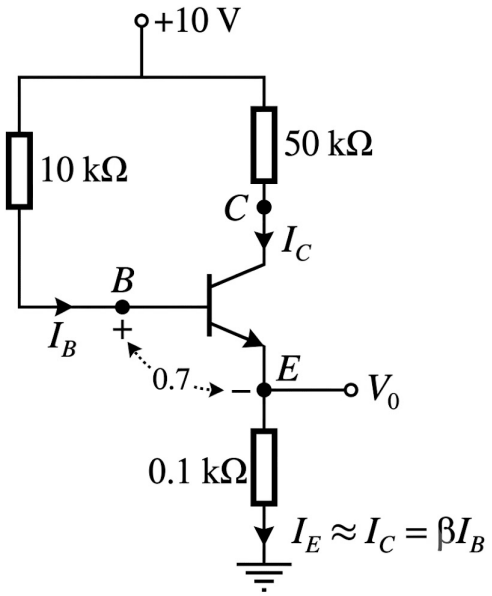


Fig. Solution Q4.10.1

Given:

(i) $V_{BE} = 0.7 \text{ V}$

(ii) $I_C \approx I_E$

(iii) DC current gain, $\beta = \frac{I_C}{I_B} = 100$

Applying KVL in loop shown,

$$-10 + I_B \times 10 + 0.7 + 0.1 \times I_B \times 100 = 0$$

$$I_B = 0.465 \text{ mA}$$

From figure,

$$V_0 = I_E \times 0.1 = \beta I_B \times 0.1$$

$$V_0 = 100 \times 0.465 \times 0.1 = 4.65\,\text{V}$$

(A) 4.65 V

Key Points

- Writing KVL through the base-emitter loop allows solving for I_B by expressing $I_E \approx \beta I_B$.
- The emitter output voltage V_0 is given by the voltage drop across R_E: $V_0 = I_E R_E$.

Q4.08.1 MCQ 2008 2M Ans: B ●●○

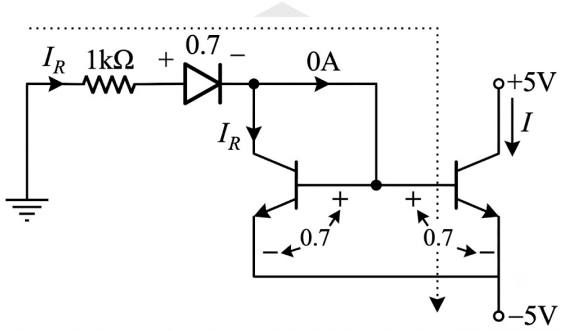


Fig. Solution Q4.08.1

Given:

(i) Both BJT are perfectly matched i.e. $V_{BE_1} = V_{BE_2}$, $V_{BE} = 0.7\,\text{V}$

(ii) β is large. Assume $\beta = \infty$, so $I_B = 0\,\text{A}$ and $I_C = I_E$.

(iii) Forward voltage drop in diode = 0.7 V

Diode and J_E junction will be forward biased due to −5 V.
Applying KVL in the loop shown,

$$I_R \times 1 + 0.7 + 0.7 - 5 = 0$$

$$I_R = 3.6\,\text{mA}$$

From the concept of current mirror circuit,

$$I = I_R = 3.6\,\text{mA}$$

(B) 3.6 mA

Key Points

- In a BJT current mirror circuit with matched transistors and large β, the output current I mirrors the reference current I_R.
- In circuit calculations, keep unit consistency: resistances in kΩ, currents in mA, and voltages in V.

Q4.07.1 MCQ 2007 1M Ans: D ● ● ●

The original GATE 2007 EE Q1 circuit has a PNP transistor: the arrow points into its base at the upper emitter branch. Assuming forward-active operation with $V_{EB} = 0.7$ V and $\beta = 100$, the emitter/base loop gives (currents in mA, resistances in kOhm)

$$10 = 101I_B + 0.7 + 270I_B.$$

Hence $I_B = 9.3/371 = 0.0250674$ mA and $I_C = 100I_B = 2.50674$ mA. The node voltages are

$$V_B = 270I_B = 6.7682 \text{ V},$$
$$V_E = V_B + 0.7 = 7.4682 \text{ V},$$
$$V_C = I_C(1) = 2.5067 \text{ V}.$$

The emitter-base junction is forward biased, and $V_C < V_B$ makes the collector-base junction reverse biased for a PNP device. This validates forward active, option (D). No saturation assumption is needed.

Q4.07.2 MCQ 2007 2M Ans: B ● ● ○

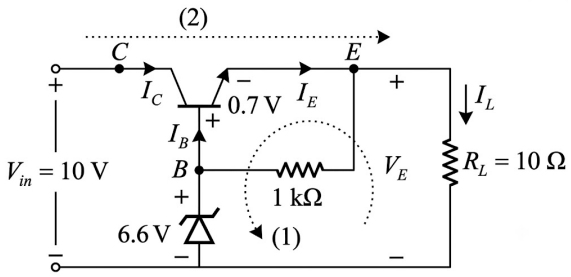


Fig. Solution Q4.07.2

Given circuit is shown below,
Applying KVL in loop (1),

$$-V_E - 0.7 + 6.6 = 0$$
$$V_E = 5.9 \text{ V}$$

and

$$I_L = \frac{V_E}{R_L} = \frac{5.9}{10} = 0.59 \text{ A}$$

Applying KVL in loop (2),

$$-V_{\text{in}} + V_{CE} + V_E = 0$$
$$V_{CE} = V_{\text{in}} - V_E = 10 - 5.9 = 4.1 \text{ V}$$

Applying KCL at node E,

$$I_E = \frac{5.9 - 6.6}{1000} + \frac{5.9}{10}$$
$$I_E \approx 0.59 \text{ A}$$

Assuming β to be high,

$$I_C \approx I_E \approx 0.59 \text{ A}$$

Therefore, power dissipated in the transistor is given by,

$$P = V_{CE} \times I_C = 4.1 \times 0.59 = 2.419 \text{ W}$$

(B) 2.419 W

Key Points

- Power dissipation in a BJT operating in active region is evaluated using $P \approx V_{CE} \times I_C$.
- Zener diode establishes a fixed base voltage, regulating emitter voltage via base-emitter voltage drop $V_E = V_Z - V_{BE}$.

Q4.06.1 MCQ | 2006 | 2M | Ans: B | ● ● ●

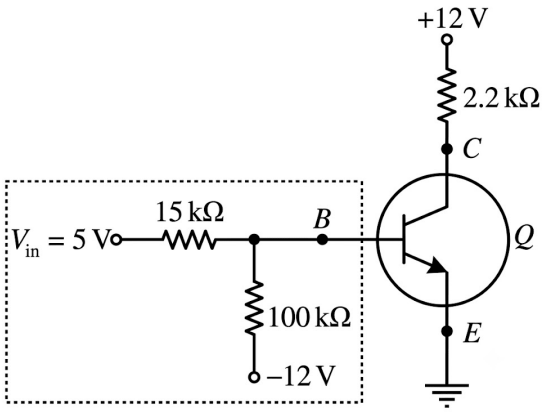


Fig. Solution Q4.06.1

Given:

(i) DC current gain, $\beta = 30$

(ii) Input voltage, $V_{in} = 5\,\text{V}$

(iii) Collector to base reverse saturation current, $I_{CBO} = 20\,\text{nA}$

Replace dotted box shown above by Thevenin equivalent circuit as shown below:

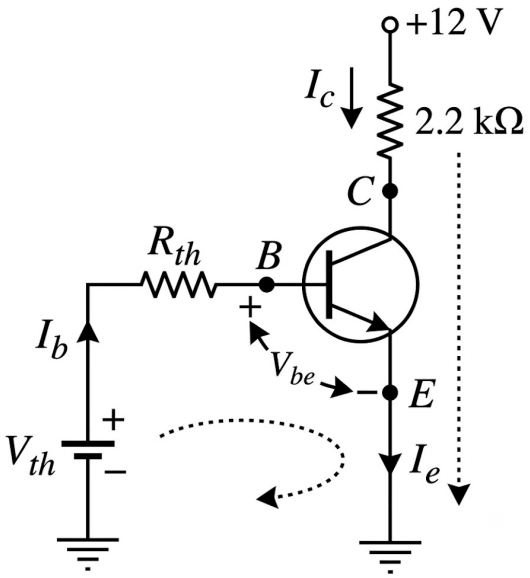


Fig. Solution Q4.06.1

Calculation of V_{TH} using superposition theorem:

(i) Consider −12 V only:

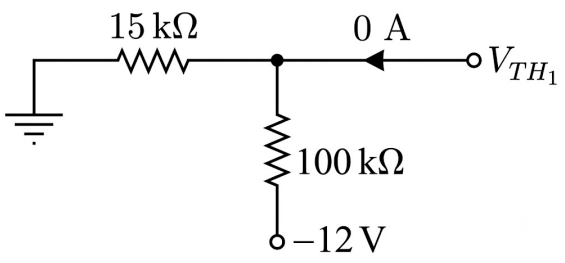


Fig. Solution Q4.06.1

$$V_{\text{TH}_1} = \frac{-12 \times 15}{15 + 100}\ \text{V}$$

(ii) Consider 5 V only:

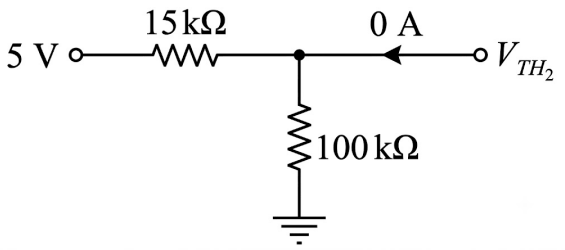


Fig. Solution Q4.06.1

$$V_{\text{TH}_2} = \frac{5 \times 100}{100 + 15}\ \text{V}$$

Hence,

$$V_{\text{TH}} = V_{\text{TH}_1} + V_{\text{TH}_2} = \frac{5 \times 100}{100 + 15} - \frac{12 \times 15}{100 + 15} = 2.78\ \text{V}$$

and

$$R_{\text{TH}} = 15 \parallel 100 = 13.04\ \text{k}\Omega$$

OR

We can also find V_{TH} and R_{TH} as follows:

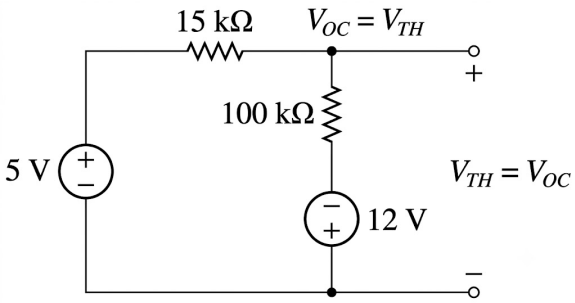


Fig. Solution Q4.06.1

$$\frac{V_{\text{TH}} - 5}{15} + \frac{V_{\text{TH}} + 12}{100} = 0$$

$$V_{TH} = 2.782\,\text{V}$$

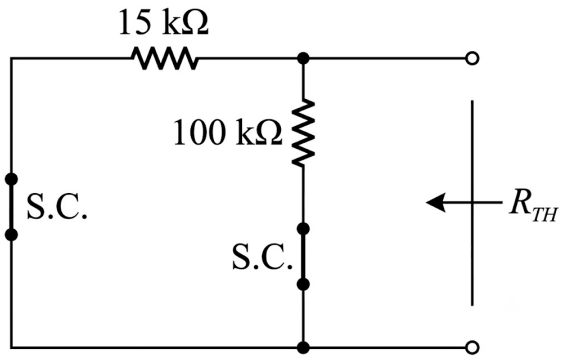


Fig. Solution Q4.06.1

$$R_{TH} = 15 \parallel 100 = 13.04\,\text{k}\Omega$$

Since, input circuit is forward biased due to V_{TH} ($V_{TH} > 0.7\,\text{V}$), hence BJT will be either in active region or in saturation region.

Method 1

Assuming transistor Q is in active region,

$$V_{BE(\text{active})} = 0.7\,\text{V}$$

Applying KVL in input loop,

$$-V_{TH} + R_{TH}I_B + V_{BE(\text{active})} = 0$$

$$-2.78 + 13.04 \times I_B + 0.7 = 0$$

$$I_B = 0.159\,\text{mA}$$

Collector current is given by,

$$I_C = \beta I_B = 30 \times 0.159 = 4.78\,\text{mA}$$

Applying KVL in output loop,

$$-12 + I_C \times 2.2 + V_{CE} = 0$$

$$V_{CE} = 1.484\,\text{V}$$

$$V_{CB} = V_{CE} - V_{BE} = 1.484 - 0.7 = 0.784\,\text{V}$$

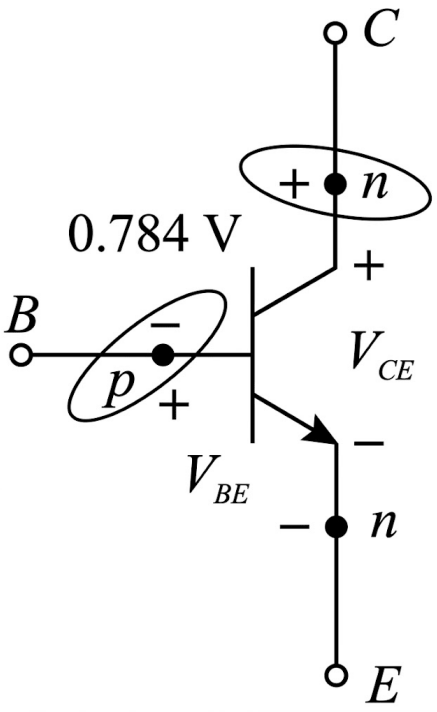


Fig. Solution Q4.06.1

From above figure, it is clear that, collector side of *n-p-n* transistor is connected to +ve voltage and base side of *n-p-n* transistor is −ve biased, therefore collector to base junction is in reverse bias.
Therefore, the transistor is in active region. Hence, the correct option is (B).

Method 2

Assuming transistor Q is in saturation region,

$$V_{BE(\text{sat})} = 0.8\,\text{V}, \quad V_{CE(\text{sat})} = 0.2\,\text{V}$$

For transistor to be in saturation region,

$$I_{B(\min)} \le I_B \implies \frac{I_{C(\text{sat})}}{\beta} \le I_B \quad \ldots(\text{i})$$

Applying KVL in input loop,

$$-V_{\text{TH}} + R_{\text{TH}} I_B + V_{BE(\text{sat})} = 0$$
$$-2.78 + 13.04 \times I_B + 0.8 = 0$$
$$I_B = 0.151\,\text{mA} \quad \ldots(\text{ii})$$

Applying KVL in output loop,

$$-12 + I_{C(\text{sat})} \times 2.2 + V_{CE(\text{sat})} = 0$$
$$I_{C(\text{sat})} = \frac{12 - 0.2}{2.2} = 5.36\,\text{mA}$$
$$I_{B(\min)} = \frac{I_{C(\text{sat})}}{\beta} = \frac{5.36}{30} = 0.178\,\text{mA} \quad \ldots(\text{iii})$$

From equation (ii) and (iii),

$$I_{B(\min)} > I_B \quad \ldots(\text{iv})$$

From equation (i) and (iv), the transistor does not satisfy the condition of saturation. Hence, our assumption is wrong.
Therefore, the transistor is in active region.

B) Active

Key Points

- Convert complex base biasing network into its equivalent Thevenin voltage V_{TH} and resistance R_{TH}.
- Test for saturation mode by checking whether actual base current $I_B \geq I_{C(sat)}/\beta$; if not, the transistor operates in the active region.

Q4.05.1 MCQ | 2005 | 2M | Ans: B ● ● ○

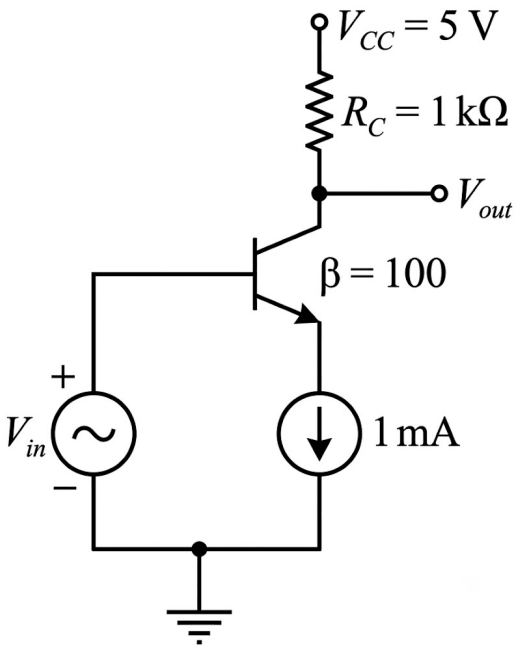


Fig. Solution Q4.05.1

Given circuit is shown above.
DC equivalent circuit:

(i) All capacitors are open circuited.

(ii) AC voltage source is replaced by short circuit.

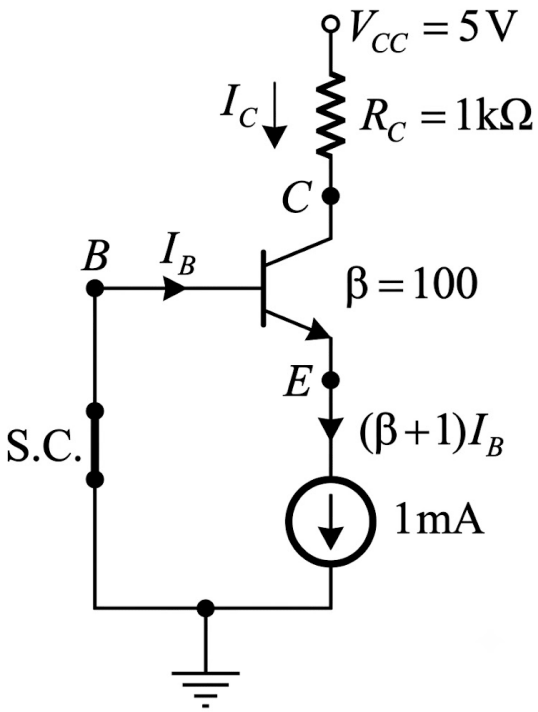


Fig. Solution Q4.05.1

From figure,

$$I_E = (\beta + 1)I_B$$

$$I_B = \frac{I_E}{1+\beta} = \frac{1}{1+100} = \frac{1}{101}\,\text{mA}$$

$$I_B \approx 10\,\mu\text{A}$$

B) $10\,\mu$A

Key Points

- For DC analysis, deactivate AC voltage sources by short-circuiting them.
- The relationship between emitter current and base current is given by $I_E = (1 + \beta)I_B$.

Q4.04.1 MCQ | 2004 | 2M | Ans: C | ● ● ○

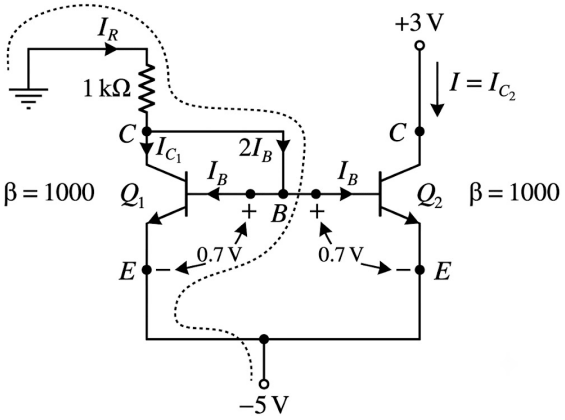


Fig. Solution Q4.04.1

Given current mirror circuit is shown above.
Since both transistors Q_1 and Q_2 are perfectly matched,

$$V_{BE_1} = V_{BE_2} = V_{BE} = 0.7\,\text{V}$$

$$I_{C_1} = I_{C_2} = I_C$$

Applying KVL in the loop shown,

$$I_R \times 1 + 0.7 - 5 = 0$$

$$I_R = 5 - 0.7 = 4.3\,\text{mA}$$

Applying KCL at collector terminal of Q_1,

$$I_C + 2I_B - I_R = 0$$

$$I_C + 2\frac{I_C}{\beta} - I_R = 0$$

$$I_R = \frac{\beta + 2}{\beta} I_C \implies I_C = \left(\frac{\beta}{\beta + 2}\right) I_R$$

Since $\beta = 1000 \gg 2$, $I_C \approx I_R$:

$$I_{C_2} = I_C = I = I_R = 4.3\,\text{mA}$$

C) 4.3 mA

Key Points

- In a standard two-transistor current mirror with finite β, $I_{\text{out}} = \left(\frac{\beta}{\beta+2}\right) I_{\text{ref}}$.
- For very large β, the base currents become negligible ($I_B \approx 0$), making $I_{\text{out}} \approx I_{\text{ref}}$.

Q4.04.2 MCQ 2004 2M Ans: D ●○○

When a BJT is in the saturation region, both the base-emitter and collector-base junctions are forward biased.

D) Saturation

Key Points

- **Active Region:** Emitter-base junction forward biased, collector-base junction reverse biased (acts as an amplifier).

- **Saturation Region:** Emitter-base junction forward biased, collector-base junction forward biased (acts as an ON switch).
- **Cut-off Region:** Both junctions reverse biased (acts as an OFF switch).
- **Inverse Active Region:** Emitter-base junction reverse biased, collector-base junction forward biased.

Q4.04.3 MCQ | 2004 | 2M | Ans: B

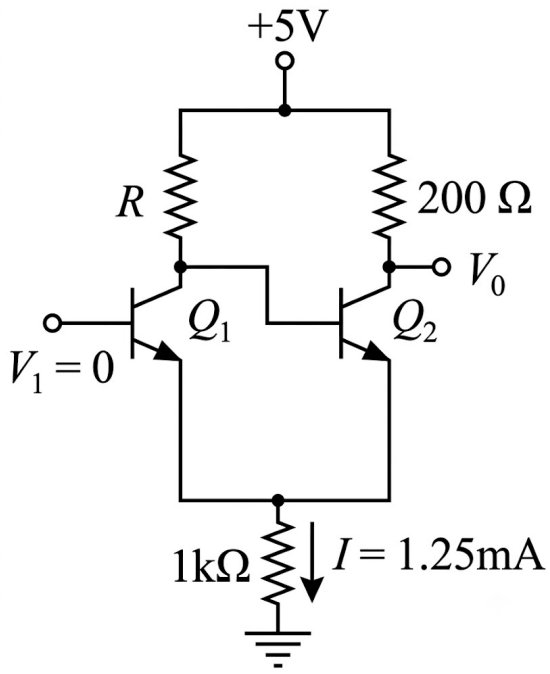


Fig. Solution Q4.04.3

PrepFusion

Given circuit is shown above.
Applying KVL in above shown loop,

$$-V_i + V_{BE} + 1.25 \times 1 = 0$$

$$V_{BE} = -1.25\ \text{V}$$

Since $V_{BE} < 0$, transistor Q_1 is in cut-off region.
Current through resistor R will drive Q_2 into saturation region. Due to which output V_0 will act as logic low level.
If transistor Q_2 was in cut-off region, then output V_0 would have been at logic high level.

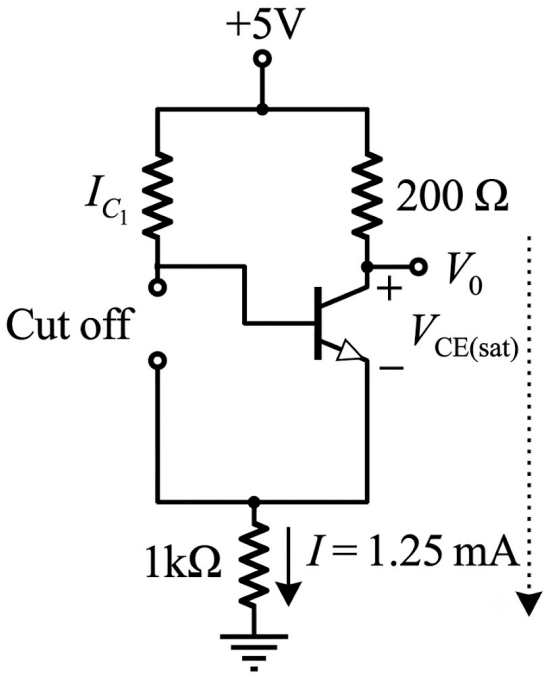


Fig. Solution Q4.04.3

Applying KVL in loop shown,

$$V_0 = V_{CE(\text{sat})} + 1.25 \times 10^{-3} \times 1 \times 10^3$$

$$V_0 = 0.1 + 1.25 = 1.35 \text{ V}$$

B) 1.35 V

Key Points

- When $V_{BE} < 0$, the BJT is reverse biased and enters the cut-off region (acting as an open switch).
- When Q_2 operates in saturation, $V_{CE} = V_{CE(\text{sat})}$, and output node voltage is the sum of $V_{CE(\text{sat})}$ and the voltage drop across R_E.

Q4.03.1 MCQ | 2003 | 2M | Ans: B | ● ● ○

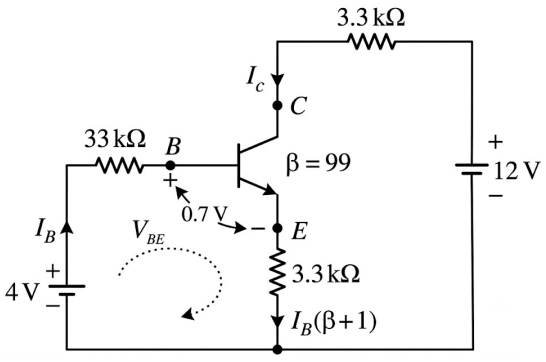


Fig. Solution Q4.03.1

Given:

(i) $h_{FE} = \beta = 99$

(ii) $V_{BE} = 0.7\,\text{V}$

Applying KVL in the loop shown by dotted line,

$$-4 + 33 \times I_B + 0.7 + 100 \times I_B \times 3.3 = 0$$

$$(33 + 330)I_B = 3.3$$

$$I_B = \frac{3.3}{33 + 330}\,\text{mA}$$

Collector current (I_C) is given by,

$$I_C = \beta I_B$$

$$I_C = \frac{99 \times 3.3}{33 + 330} \approx \frac{100 \times 3.3}{33 + 330}$$

$$I_C \approx \frac{3.3}{3.3 + 0.33}\,\text{mA}$$

$$\boxed{\text{B)}\ \frac{3.3}{3.3 + 0.33}\,\text{mA}}$$

Key Points

- In standard BJT calculations, keep consistency with units: resistances in kΩ, currents in mA, and voltages in V.
- Express I_E in terms of I_B using $I_E = (1 + \beta)I_B$ when solving the base-emitter loop.

Q4.03.2 MCQ 2003 2M Ans: C ●●○

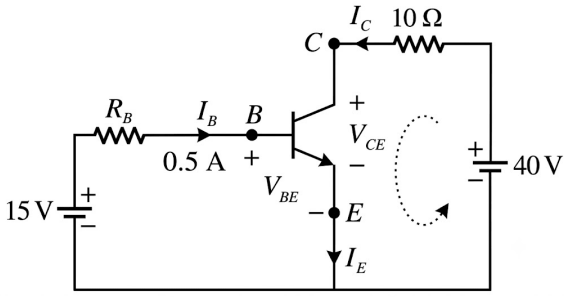


Fig. Solution Q4.03.2

Given: $\beta = 10$

Assuming BJT is in active region,

$$V_{BE(\text{active})} = 0.7\,\text{V}$$

Collector current is given by,

$$I_C = \beta I_B = 10 \times 0.5 = 5\,\text{A}$$

Applying KVL in above shown loop,

$$-40 + 10I_C + V_{CE} = 0$$

$$V_{CE} = 40 - 10 \times 5 = -10\,\text{V}$$

$$V_{CB} = V_{CE} - V_{BE(\text{active})} = -10 - 0.7 = -10.7\,\text{V}$$

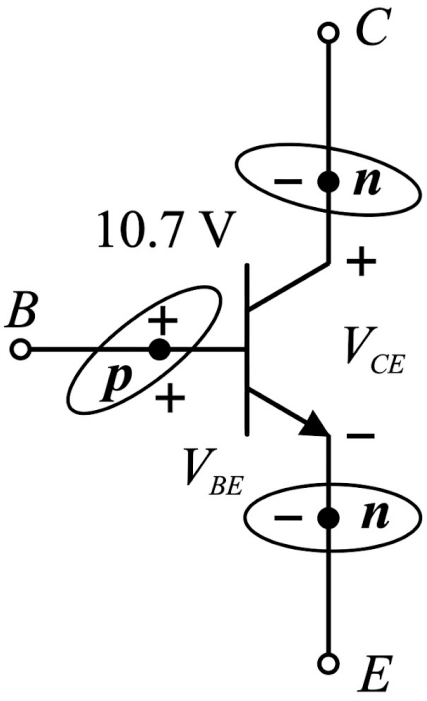


Fig. Solution Q4.03.2

From above figure, it is clear that collector terminal of *n*-*p*-*n* transistor is −vely biased and base terminal of *n*-*p*-*n* transistor is +vely biased. So, collector to base junction of given BJT is in forward bias. But in case of active region, collector to base junction should be reverse biased. Therefore, our assumption is wrong.
Hence, we can say the transistor is in saturation region.
For an ideal transistor in saturation region,

$$V_{CE(\text{sat})} \approx 0\,\text{V}$$

So, taking only outer part of the given circuit,

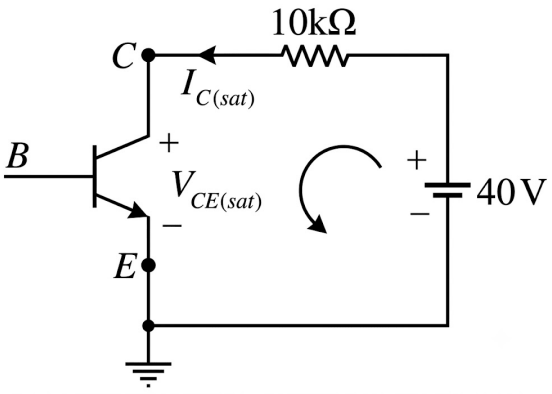


Fig. Solution Q4.03.2

Applying KVL in loop shown,

$$-40 + 10I_{C(\text{sat})} + V_{CE(\text{sat})} = 0$$

$$-40 + 10I_{C(\text{sat})} = 0$$

$$I_{C(\text{sat})} = \frac{40}{10} = 4\,\text{A}$$

Thus, operating point $[V_{CE(\text{sat})}, I_{C(\text{sat})}]$ for given transistor is $[0\,\text{V}, 4\,\text{A}]$.

C) [0 V, 4 A]

Key Points

- If active region assumption yields $V_{CB} < 0$ (forward biased collector-base junction for an n-p-n), the transistor operates in saturation.
- For an ideal transistor operating in saturation, $V_{CE(\text{sat})} \approx 0\,\text{V}$.

Q4.02.1 MCQ 2002 2M Ans: B

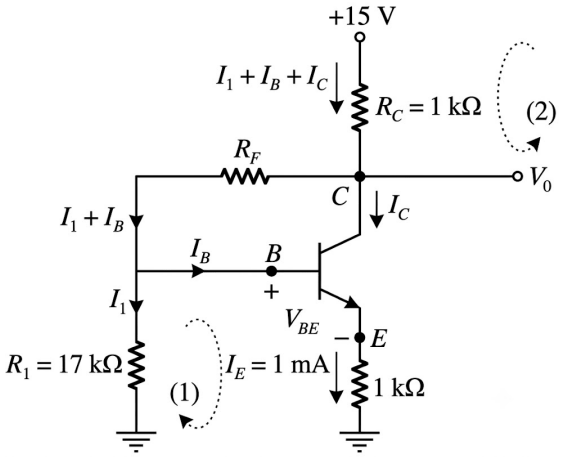


Fig. Solution Q4.02.1

Given:

(i) $I_E = 1\,\text{mA},\ \beta = 99$

(ii) $V_{BE} = 0.7\,\text{V}$

Feedback bias circuit is shown above.
From figure, $I_E = 1\,\text{mA}$.
Applying KVL in loop (1),

$$-I_1 \times 17 + V_{BE} + 1 \times 1 = 0$$

$$-I_1 \times 17 + 0.7 + 1 = 0$$

$$I_1 = \frac{1.7}{17} = 0.1\,\text{mA}$$

From figure, current through R_C is,

$$I_{R_C} = I_1 + I_B + I_C$$

Since $I_E = I_B + I_C = 1\,\text{mA}$,

$$I_{R_C} = I_1 + I_E = 0.1 + 1 = 1.1\,\text{mA}$$

B) 1.1 mA

Key Points

- Total collector branch current is the sum of feedback bias current $I_1 + I_B$ and collector current I_C.
- Applying $I_E = I_B + I_C$ simplifies $I_{R_C} = I_1 + I_E$.

Q4.02.2 MCQ 2002 2M Ans: C ● ○ ○

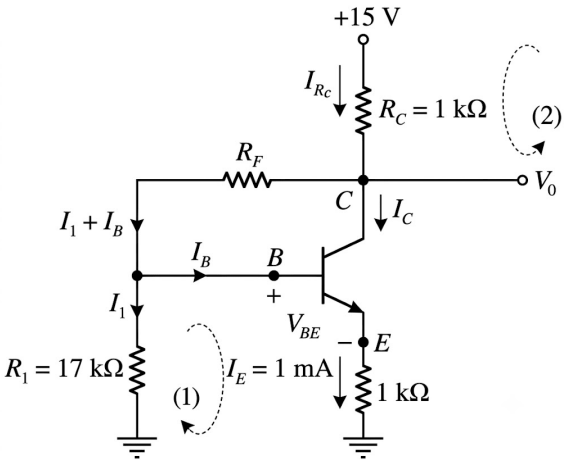


Fig. Solution Q4.02.2

From the previous question,

$$I_{R_C} = 1.1\ \text{mA}$$

Applying KVL in loop (2),

$$-15 + R_C(I_{R_C}) + V_0 = 0$$

$$V_0 = 15 - R_C(I_{R_C})$$

$$V_0 = 15 - 1 \times 1.1 = 13.9\ \text{V}$$

C) 13.9 V

Key Points

- Node voltage V_0 at the collector terminal equals $V_{CC} - I_{R_C}R_C$.

Q4.02.3 MCQ 2002 2M Ans: A ● ○ ○

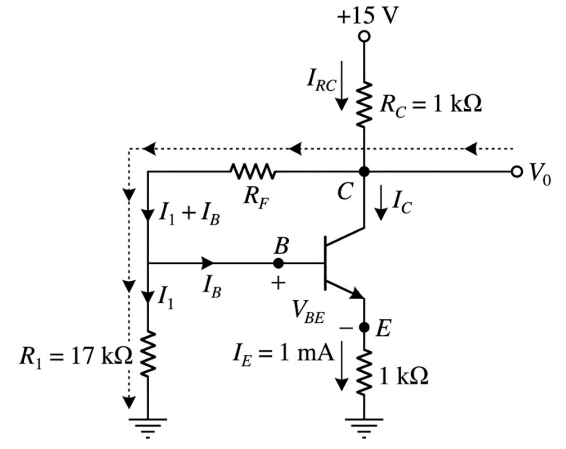


Fig. Solution Q4.02.3

From previous calculations,

$$I_{R_C} = 1.1\,\text{mA}, \quad I_E = 1\,\text{mA}, \quad I_1 = 0.1\,\text{mA}$$

Base current (I_B) is given by,

$$I_B = \frac{I_E}{1+\beta} = \frac{1}{1+99} = 0.01\,\text{mA}$$

Applying KVL from collector to ground (along dotted line),

$$-V_0 + R_F(I_1 + I_B) + 17I_1 = 0$$

$$-13.9 + R_F(0.1 + 0.01) + 17 \times 0.1 = 0$$

$$R_F \times 0.11 = 12.2$$

$$R_F = 110.9\,\text{k}\Omega$$

A) 110.9 kΩ

Key Points

- Derive I_B using $I_E = (1+\beta)I_B$.
- Write KVL across the feedback branch path containing R_F and R_1 to determine R_F.

Q4.00.1 MCQ | 2000 | 2M | Ans: B

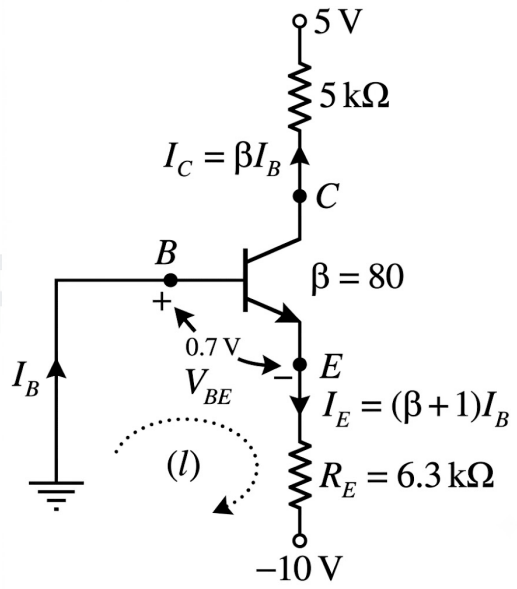


Fig. Solution Q4.00.1

Given circuit is shown above.
Applying KVL in input loop (1),

$$+0.7 + (\beta+1)I_B \times 6.3 - 10 = 0$$

$$+0.7 + (80+1)I_B \times 6.3 - 10 = 0$$

$$81 \times 6.3 \times I_B = 9.3$$

$$I_B = 0.0182\,\text{mA} = 18.2\,\mu\text{A}$$

B) 18.2 μA

Key Points

- Express I_E in terms of base current using $I_E = (1 + \beta)I_B$.
- Apply KVL starting from ground through the base-emitter junction and R_E to the $-10\,\text{V}$ supply.

Q4.94.1 MCQ | 1994 | 2M | Ans: B | ● ● ○

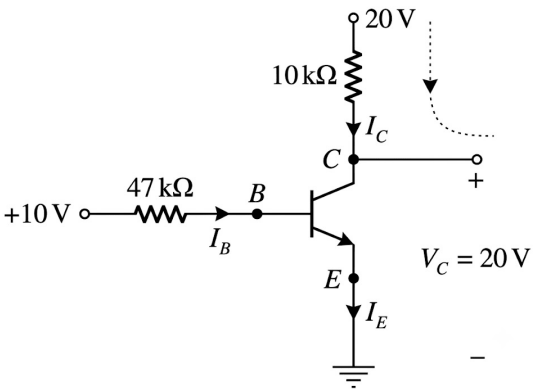


Fig. Solution Q4.94.1

Given:
Collector to ground voltage V_C is $+20\,\text{V}$.
Applying KVL in loop shown by dotted line,

$$-20 + I_C \times 10 + V_C = 0$$

$$-20 + I_C \times 10 + 20 = 0$$

$$I_C = 0\,\text{A}$$

Collector current I_C is given by,

$$I_C = \beta I_B$$

$$I_B = \frac{I_C}{\beta} = 0$$

So, both I_B and I_C are zero.
So that, $I_E = I_C + I_B = 0\,\text{A}$.
It shows, emitter current will be zero only when emitter to ground connection is open as shown below,

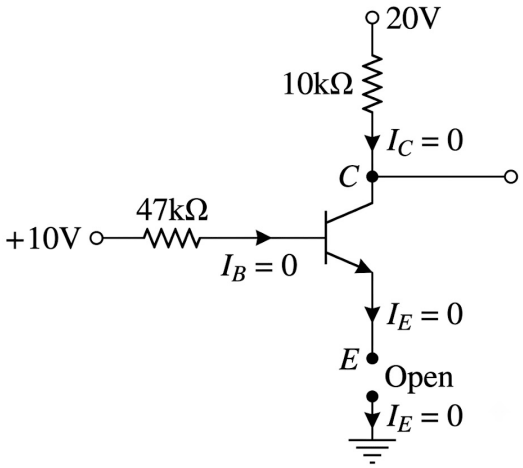


Fig. Solution Q4.94.1

$$I_C = I_B = I_E = 0$$

B) Emitter terminal is open

Key Points

- If V_C equals the supply voltage V_{CC}, no current flows through the collector resistor ($I_C = 0$).
- Since $I_C = 0$ and $I_B = 0$, the emitter current $I_E = 0$, which indicates an open-circuit fault at the emitter terminal.

Q4.91.1 MCQ 1991 2M Ans: C ● ● ○

Given:

(i) Common emitter amplifier

(ii) $\beta = 100$

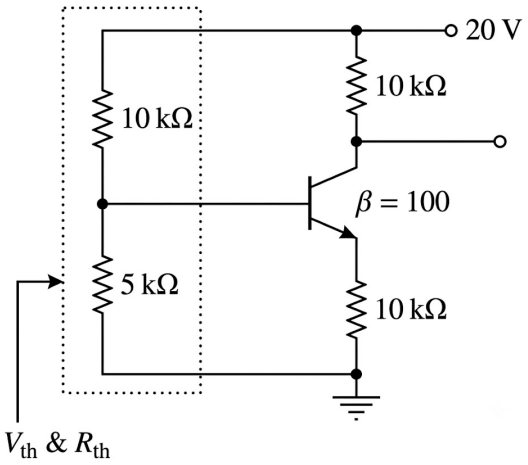


Fig. Solution Q4.91.1

Thevenin equivalent circuit of given self-bias circuit: Left side of the above circuit (shown by dotted line) is converted into Thevenin equivalent circuit as shown below,

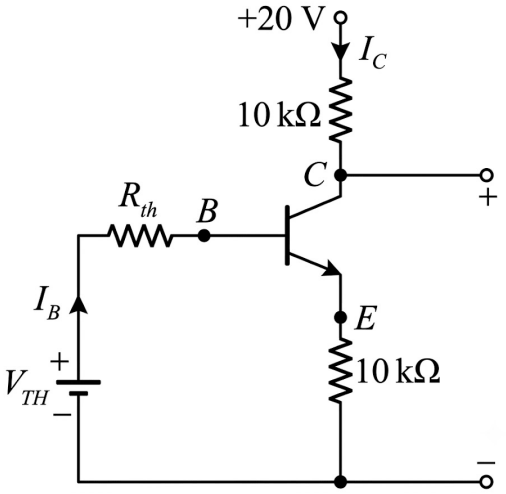


Fig. Solution Q4.91.1

$$R_{TH} = (5 \parallel 10) = \frac{5 \times 10}{5 + 10} = \frac{10}{3}\,\text{k}\Omega$$

$$V_{TH} = 20 \times \frac{5}{5 + 10} = \frac{20}{3}\,\text{V} \quad \text{[By VDR]}$$

So, given self-bias circuit becomes as,

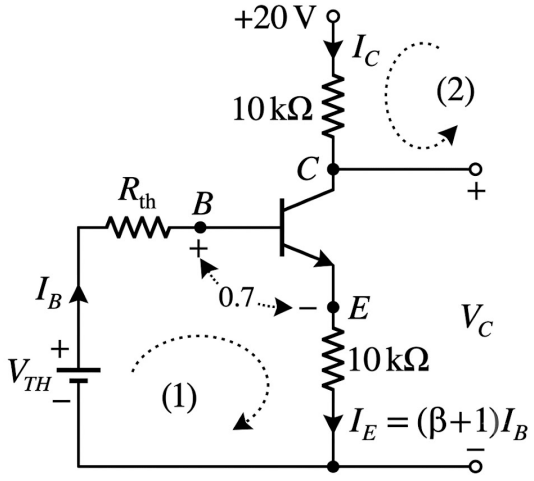


Fig. Solution Q4.91.1

Applying KVL in loop (1),

$$-V_{TH} + R_{TH} \times I_B + 0.7 + 10I_E = 0$$

$$-\frac{20}{3} + \frac{10}{3} \times I_B + 0.7 + (\beta + 1)I_B \times 10 = 0$$

$$I_B = \frac{\frac{20}{3} - 0.7}{\frac{10}{3} + (\beta + 1) \times 10} = \frac{\frac{20}{3} - 0.7}{\frac{10}{3} + 101 \times 10}$$

$$I_B = 5.84 \times 10^{-3}\,\text{mA}$$

So, base current I_B is 5.84×10^{-3} mA.

Collector current I_C is given by,

$$I_C = \beta I_B$$

$$I_C = 100 \times 5.84 \times 10^{-3} = 0.584\,\text{mA}$$

Applying KVL in loop (2), for calculating collector voltage V_C,

$$-20 + 10I_C + V_C = 0$$

$$V_C = 20 - I_C \times 10 = 20 - 0.584 \times 10$$

$$V_C = 14.1\,\text{V} \approx 14\,\text{V}$$

C) 14 V

Key Points

- Convert the base divider network into its Thevenin equivalent (V_{TH}, R_{TH}) to analyze self-biased BJT circuits.
- Express emitter current as $I_E = (\beta + 1)I_B$ when solving the input KVL loop.

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SOLUTIONS

V

Miscellaneous Topics

Topics

1. CMOS Inverter

Unit Snapshot*

Stream: EC	**Questions:** 1	**MCQ:** 1	**MSQ:** 0	**NAT:** 0
1M: 0	**2M:** 1	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 2M(2026)–2M(2026)

**See preface for how Avg Weightage and Range are calculated.*

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Q5.26.1 MCQ | 2026 | 2M | Ans: D

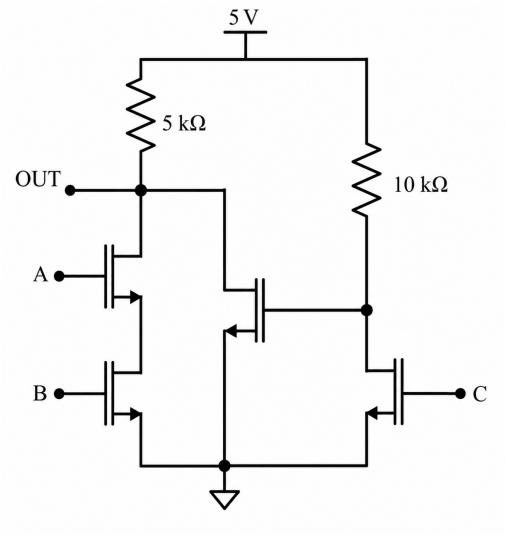

Fig. Solution Q5.26.1

The two NMOS transistors controlled by A and B are connected in series. Hence, they form an AND operation (AB).
The NMOS transistor on the right is controlled by C. Since it drives the gate of the middle NMOS, the middle NMOS conducts when $C = 0$.
Hence, this branch represents $\overline{C}$.
The two pull-down branches are connected in parallel. Therefore, they are ORed together.

$$AB + \overline{C}.$$

Since this is an NMOS pull-down network, the output is the complement of the pull-down logic.

$$\text{OUT} = \overline{AB + \overline{C}}$$

Using De Morgan's theorem,

$$\text{OUT} = (\overline{AB})\, C$$

Maximum Power Consumption
Power dissipated in the 5 kΩ resistor is

$$P_{5k} = \frac{5^2}{5 \times 10^3} = 5 \text{ mW}.$$

This occurs whenever the output is pulled LOW.
Power dissipated in the 10 kΩ resistor is

$$P_{10k} = \frac{5^2}{10 \times 10^3} = 2.5 \text{ mW},$$

which occurs when $C = 1$.
The maximum power occurs for

$$A = 1, \qquad B = 1, \qquad C = 1,$$

where

- the A-B pull-down path conducts, dissipating 5 mW,
- transistor C conducts, dissipating 2.5 mW in the 10 kΩ resistor.

Hence,

$$P_{max} = 5 + 2.5 = 7.5 \text{ mW}.$$

Therefore,

D. OUT = $\overline{ABC}$, $P_{max} = 7.5$ mW

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Also in this Series

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NexusX	Network Theory, Analog Electronics, Digital Electronics, Control Systems and Signals & Systems — EC, EE, IN
SiliconX	Electronic Devices, Electromagnetic Field Theory and Communication Systems — EC
PowerX	Electrical Machines, Power Systems, Power Electronics, Measurements and Electromagnetic Field Theory — EE, IN
Engineering Mathematics	Linear Algebra, Calculus, Differential Equations, Complex Variables, Probability & Statistics, Numerical Methods, Transform Theory and Vector Calculus — all branches
General Aptitude	Verbal and quantitative aptitude, arranged year-wise and session-wise — all branches

Spotted a wrong answer key, a typo or a question we have mis-tagged? Report it on the PYQ portal at `pyq.prepfusion.in` — corrections go into the next printing, and you can request a video solution for any question from the same place.

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